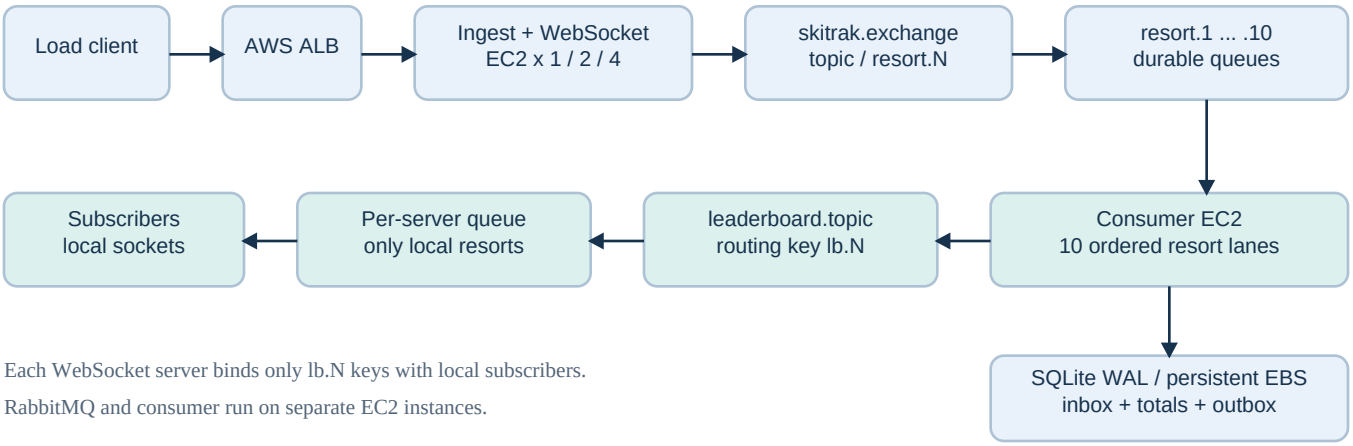
SkiTrak | Assignment 2

Architecture 1/2 - deployment, topology and connection locality



Each WebSocket server binds only lb.N keys with local subscribers.

RabbitMQ and consumer run on separate EC2 instances.

Routing and state ownership

Java 11/Tomcat 9 ingest servers publish persistent swipe events through one reused producer connection and a bounded, exclusive channel pool. The dedicated RabbitMQ EC2 routes skitrak.exchange topic keys resort.1 through resort.10 to ten durable queues. A separate Consumer EC2 owns one FIFO aggregation lane and one SQLite database per resort.

Subscriber-aware topic backplane

The Consumer publishes complete versioned Top-20 snapshots to leaderboard.topic using lb.{resortID}. Each WebSocket node owns an exclusive generated queue and binds only the resorts with local subscribers. This bridges connection locality without requiring resort co-location. It reduces uninterested traffic but adds binding, bootstrap and reconnect coordination.

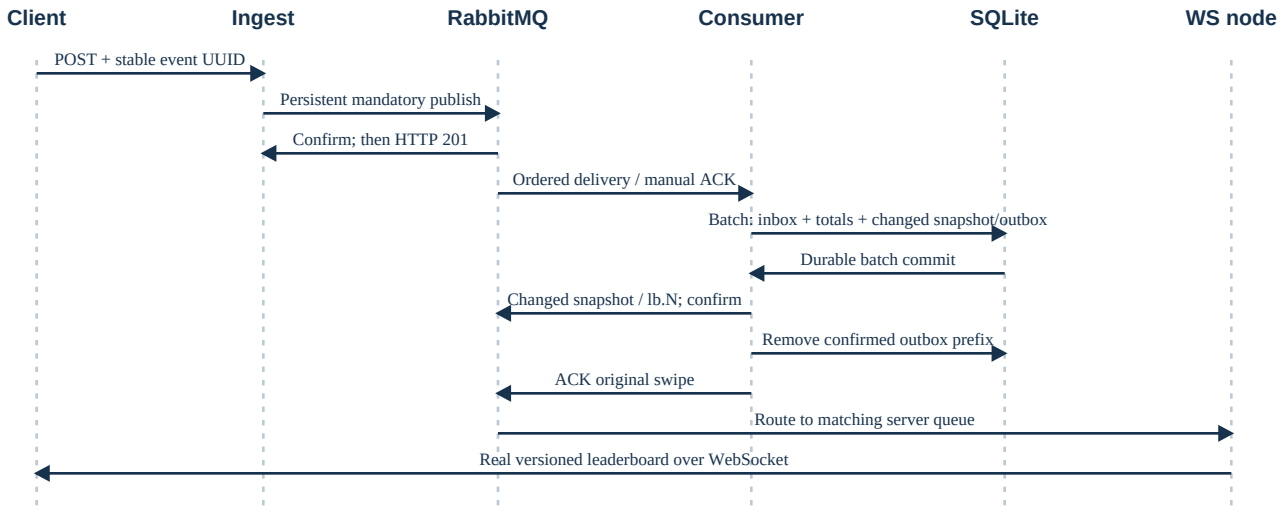
First subscriber binds before bootstrap from the authoritative Consumer snapshot. Live updates and bootstrap responses reconcile by resort/day/version. Last unsubscribe removes the binding. Broker reconnect recreates the private queue and restores bindings; periodic snapshot refresh repairs transient notification loss.

ALB and threading

ALB forwards HTTP and WebSocket upgrades to ROOT-deployed ingest apps. Health path /health: 30 s interval, 5 s timeout,healthy 2 / unhealthy 3. Idle timeout 120 s; generated-cookie stickiness 300 s. Each WebSocket stays on the node that accepted it; after failure a new connection bootstraps on another healthy node. Configurable worker pools process fair serial resort lanes; ten resorts cap useful aggregation parallelism at ten. Dynamic scaling is supported but disabled during fixed-worker comparisons.

Delivery and failure handling

Architecture 2/2 - swipe-to-leaderboard sequence



Durable aggregation and acknowledgement

A stable event UUID is reused through Idempotency-Key on retries. Ingest validates the official lift mapping and returns 201 only after mandatory routing and publisher confirmation. Uncertain outcomes return 503. Per-resort SQLite WAL/FULL transactions atomically persist dedupe inbox, skier totals and any changed snapshot/version/outbox. Batches hold at most 64 events with a 10 ms partial-batch flush trigger. Top-N changes are retained individually and published in order. Grouped broadcast confirms precede source ACKs; the confirmed outbox prefix is deleted. Prefetch 128 and ACK batch 16 are fixed for the fresh comparisons.

Recovery and boundaries

One recovery owner rebuilds a failed producer connection/channel pool; stale leases cannot re-enter its replacement. Consumer failures retry in the same lane before reconnect/redelivery. Inbox state prevents double-counting and persistent outbox state permits notification replay. Conflicting/poison events pause a resort instead of silently reordering it. systemd/Docker restart processes; persistent directories survive process restart. Source queues default to 24 h TTL, capacity 100000 and reject-publish overflow; expired events reach dead-letter queues and are not counted as successful aggregation.

The backplane is transient notification transport; the Consumer is the authoritative bootstrap source. Publisher confirm is broker acceptance, not proof that every WebSocket client received an update. Host/disk loss and multi-host aggregation failover require additional replication and ownership design. Fresh report measurements, separate from this architecture handout, disclose sampled queue peaks, clock-dependent lag means and missed performance targets.

SkiTrak | Assignment 2

Fresh performance tests and validation

Fengying Zeng | CS6650 | Session 20261108T135433Z

Repository: https://github.com/Anne-Zeng/cs6650-assignment1.git

Repository URL is the configured project remote. Publication of the local A2 changes has not been verified; this report does not claim that they have been pushed.

English performance and validation report

17 of 17 planned benchmark runs collected. This report contains only the new session. Earlier report values and tuning experiments are excluded.

The repository evidence directories preserve complete client output, CSVs, final WebSocket snapshots, monitor JSONL, actual RabbitMQ UI screenshots and provenance hashes. This PDF summarizes the metrics; it does not replace the original records.

Measurement method

Client on the Mac; ALB stages with 1/2/4 responding ingest nodes; dedicated RabbitMQ and Consumer EC2 instances. Ten resorts,Top-N = 20,seed 6650. Main comparisons use 10 fixed Consumer workers, prefetch 128, maximum database batch 64,10 ms flush trigger, ACK batch 16 and 64 publisher channels per ingest. A separate fixed-worker sweep uses 5/10/15/20. SQLite FULL synchronization remains enabled.

Persistent data is retained between runs; database growth and cache state are not reset. Tests are single observations, not statistical estimates. Queue peaks/means are sample statistics over the client window including warmup and drain. Consumer lag depends on host clocks and is calculated per run from cumulative sums/counts; it is not a per-event upper bound.

Console and application evidence

RabbitMQ screenshots are captured from the actual Management UI in UTC. Overview covers all queues; numerical queue metrics cover the ten raw resort queues only. Ten-minute console history can contain earlier tests, so use each run timestamp. Application counter distribution is separate from ALB CloudWatch/access-log evidence. The user supplies AWS console screenshots for the same UTC windows.

Original session directory: results/a2-fresh/20261108T135433Z

Results and limits

All 17 planned runs completed: 9,000,000 events sent, 9,000,000 additional events processed, zero final failed requests, and zero additional duplicates in benchmark counters. There were 269 application retries. All recorded final Ready, in-flight and pending-outbox counts were zero.

At 512 client threads and 500K events, four ingests increased main HTTP throughput by 25.80% relative to one ingest, but end-to-end throughput decreased by 5.16%. Load balancing therefore increased admission capacity without removing the downstream completion limit.

The four-ingest 1M stress run reached 6,551.14 main HTTP events/s and 3,323.33 end-to-end events/s. Ready peaked at 377,327, mean Consumer lag was 49,377.24 ms, drain/verification took 110.256 s, and the client retried 77 times. This run failed the queue-depth and lag targets despite eventual completion.

Consumer mean I/O wait in the 1M run was 41.39%, with mean OS-observed writes of 4,842.87 IOPS; broker mean CPU was 54.74%. These support downstream storage/processing pressure, but do not establish EBS IOPS as the only cause. SQL work, transaction scheduling, confirmations and broker pressure remain plausible contributors. OS IOPS are not identical to EBS CloudWatch IOPS.

For the fixed-worker sweep, mean lag stayed below 100 ms, but 15 workers had a sampled Ready peak of 1,059. Ten ordered resort lanes cap useful parallelism at ten; 15/20 configured workers did not improve throughput over ten. Dynamic scaling exists in the implementation but was disabled and not validated by this fixed-worker experiment.

Target interpretation

High-concurrency runs did not consistently satisfy queue depth < 1,000 or Consumer lag < 100 ms. Low-concurrency sampled peaks and mean lag are observations, not a proof that every message met a latency bound. Count reconciliation plus WebSocket final-state checks supports no observed loss in these finite runs, not a general guarantee against host/disk failure.

Deployment and measurement settings

Role	EC2 type	Private IP
Ingest 1	t3.micro	172.31.45.254
Ingest 2	t3.micro	172.31.24.108
Ingest 3	t3.micro	172.31.45.75
Ingest 4	t3.micro	172.31.44.196
Consumer	t3.medium	172.31.36.151
RabbitMQ	t3.medium	172.31.32.233

Instructor clarification: use Top-N = 20 and bracket the ten-resort concurrency cap with 5/10/15/20 Consumer workers. The original handout lists 10/20/40/80; this report follows the instructor-endorsed clarification provided by the student.

The supplied official lift_vertical.json is used with demoLifts=false. For example, lift 21 adds 267 ft. The measured configuration uses a maximum store batch of 64 (the earlier 32-event tuning configuration is excluded), a roughly 10 ms partial-batch flush trigger, prefetch 128, ACK batch 16, and one producer connection with 64 pooled channels per ingest.

The recorded gp3 configuration was 6,000 provisioned IOPS and 125 MiB/s; the lab policy denied further volume modification. This restriction is context, not an explanation that turns overload runs into passes. The fresh session does not contain a controlled IOPS-only comparison.

ALB: HTTP 80 to HTTP 8080, /health every 30 s, timeout 5 s, healthy threshold 2, unhealthy threshold 3; idle timeout 120 s, generated-cookie stickiness 300 s. Four instances are registered in stages; no EC2 recreation is required between stages.

Clock synchronization was checked on EC2 at a point in time. The Mac clock was not calibrated by that check. Consequently, WebSocket snapshot-to-receipt wall-clock differences are not presented as calibrated one-way latency.

CloudWatch captures use Sum for RequestCount and Average for TargetResponseTime, period 60 s. LoadBalancer-wide metrics aggregate traffic and do not establish per-instance distribution. The latter is supported by application counters, routing probes and the independent cookie/distribution verification. Minute buckets can straddle run boundaries.

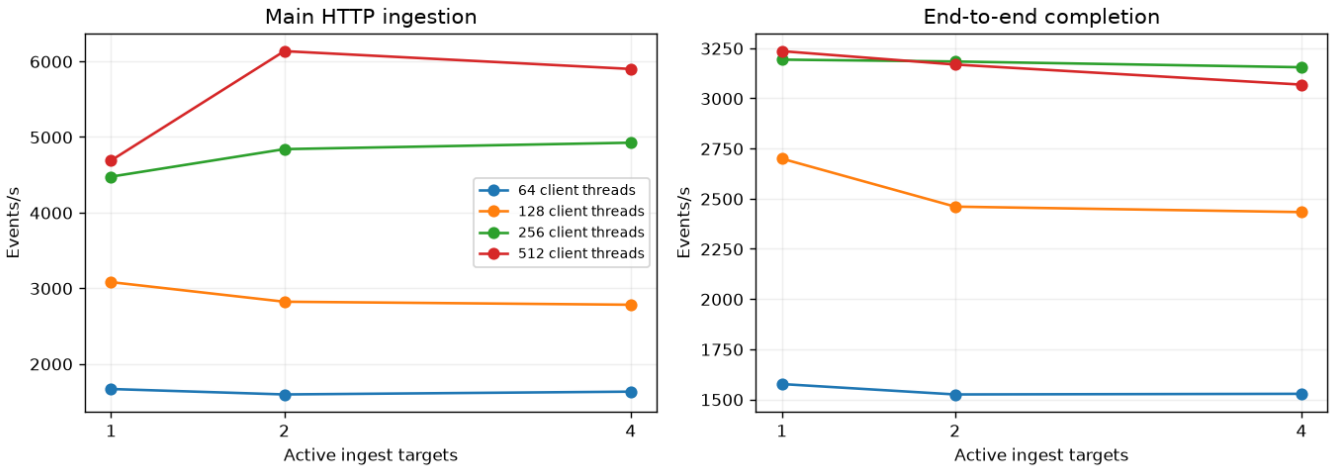
Benchmark matrix

Nodes	Events	Client	Workers	HTTP main/s	E2E/s	Retries	Ready peak	Mean lag ms
1	500000	64	10	1,666.74	1,576.41	0	0	19.66
1	500000	128	10	3,080.16	2,699.50	0	0	25.18
1	500000	256	10	4,475.42	3,192.84	6	55,463	5,177.43
1	500000	512	10	4,689.51	3,235.22	131	59,967	7,135.63
2	500000	64	10	1,594.86	1,524.64	0	0	18.22
2	500000	128	10	2,821.24	2,460.44	0	29	29.39
2	500000	256	10	4,839.15	3,183.52	0	68,120	9,306.91
2	500000	512	10	6,135.42	3,168.37	55	145,164	16,665.74
4	500000	64	10	1,631.83	1,527.74	0	0	18.87
4	500000	128	10	2,781.22	2,433.20	0	26	24.43
4	500000	256	10	4,924.34	3,154.89	0	80,309	7,641.57
4	500000	512	10	5,899.50	3,068.38	0	141,495	13,798.40
4	1000000	512	10	6,551.14	3,323.33	77	377,327	49,377.24
4	500000	128	5	2,604.84	2,275.86	0	138	42.40
4	500000	128	10	3,155.08	2,650.72	0	446	34.96
4	500000	128	15	3,032.48	2,643.07	0	1,059	38.50
4	500000	128	20	3,036.31	2,620.63	0	111	39.38

Read the metrics together: faster HTTP ingestion is not faster pipeline completion when queue backlog and drain increase. Final success after retries is not the absence of failed HTTP attempts. A run is not claimed to meet every target solely because its client exit code is 0.

All old runs remain archived and excluded. Per-run configurations and errors below must be considered before making scaling claims.

Measured scaling changes



Client threads	Ingests	Main HTTP change vs1	End-to-end change vs1
64	2	-4.31%	-3.28%
64	4	-2.09%	-3.09%
128	2	-8.41%	-8.86%
128	4	-9.71%	-9.86%
256	2	+8.13%	-0.29%
256	4	+10.03%	-1.19%
512	2	+30.83%	-2.07%
512	4	+25.80%	-5.16%

Changes describe single observations with matching application tuning. Database size/cache state and test time differ, so they are not statistically isolated causal estimates. Faster ingestion with little completion improvement and rising backlog identifies a downstream capacity limit. Low-concurrency runs may be limited by admitted workload instead of server capacity.

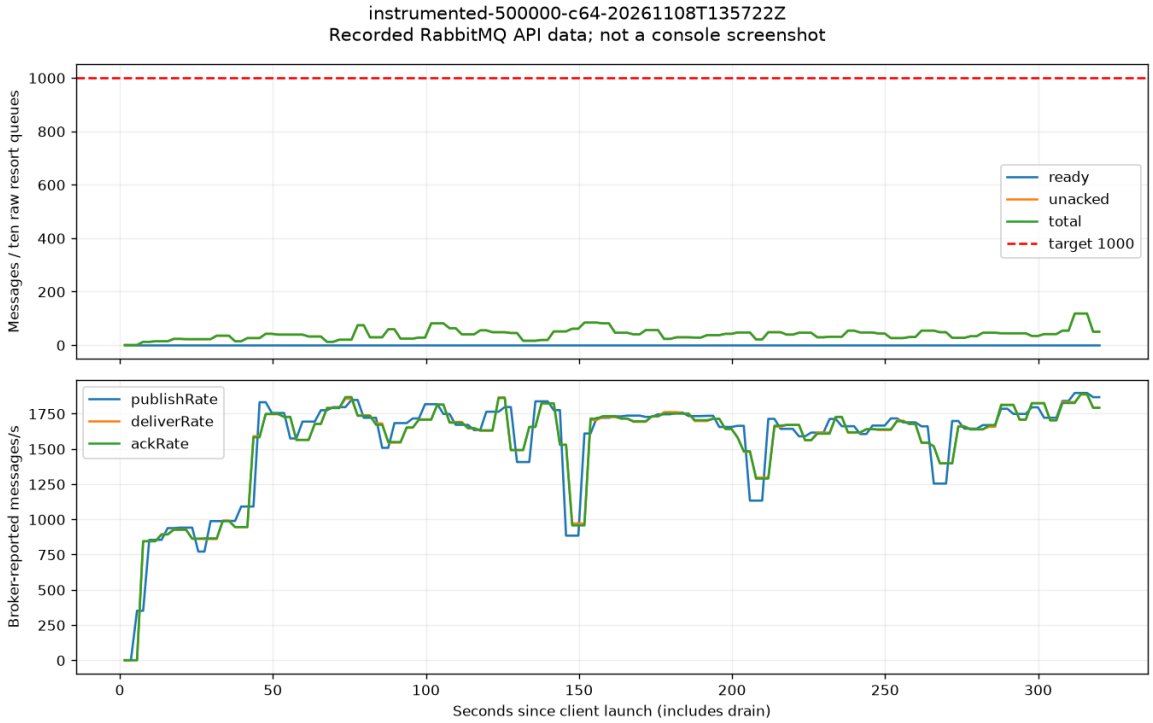
A finite overload run can accumulate a large queue and then drain after publishing stops. Oscillating unacknowledged counts alone do not prove sustained Ready backlog. Use the Ready curve, publish/delivery rates, run-specific mean lag and final count reconciliation together.

Run 01 | 1 ingest(s), 500,000 events

Client threads=64; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

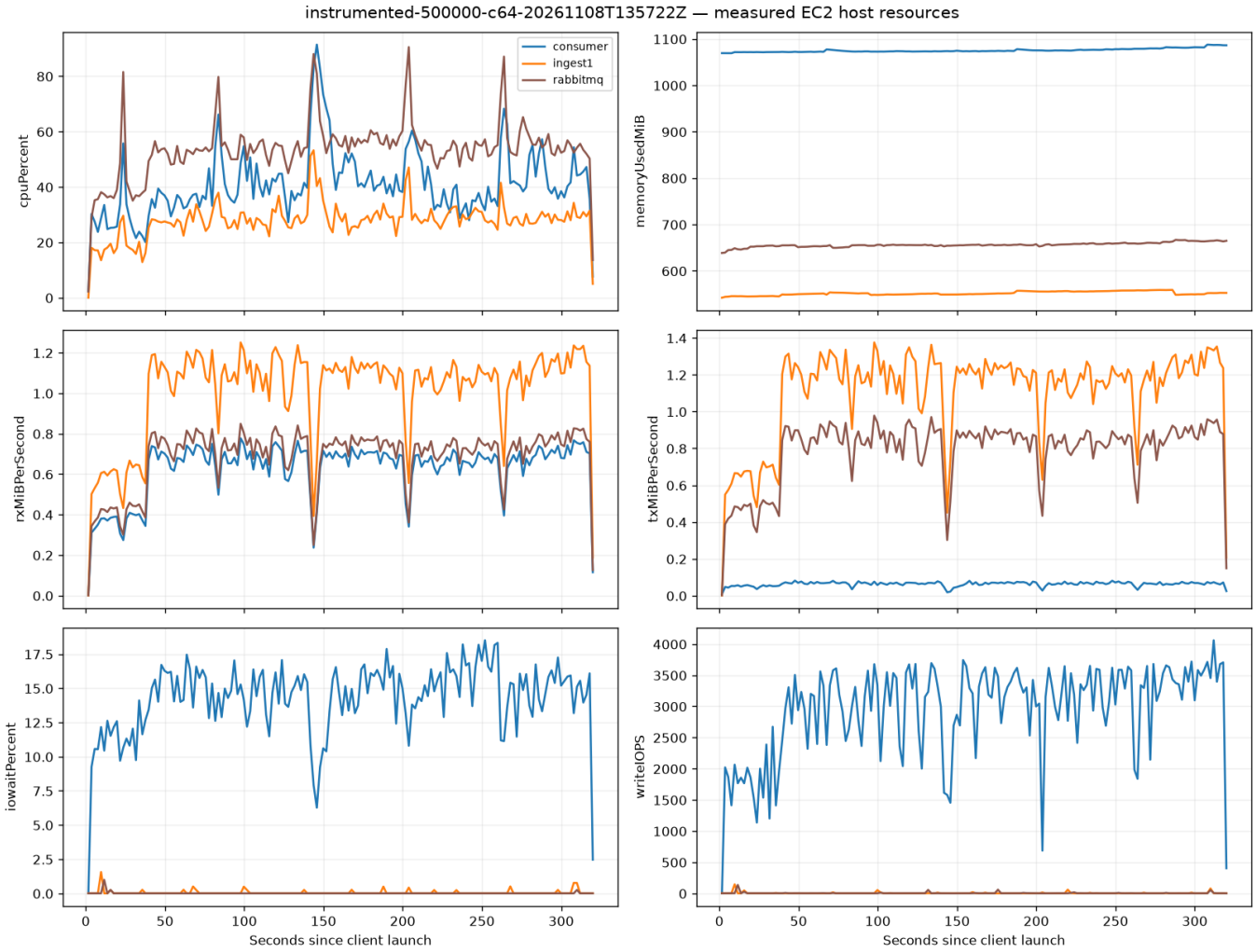
UTC 2026-11-08T13:57:39.954246+00:00 to 2026-11-08T14:03:01.243264+00:00

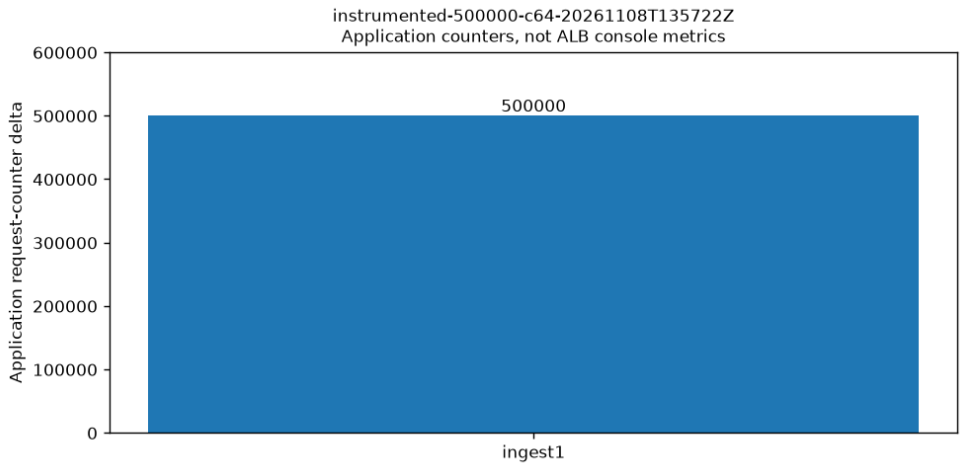
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	1,666.74 / 1,580.80 / 1,576.41
Ingestion / program runtime / drain seconds	316.295 / 319.491 / 0.789
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	38.18 / 51.73 / 93.47
Ready sampled mean / maximum	0.00 / 0
Unacked / Ready+Unacked maximum	118 / 118
Broker publish / delivery / ACK sampled mean per second	1,552.24 / 1,541.85 / 1,541.61
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	19.66
WebSocket updates / reconnections	911 / 0



Source: instrumented-500000-c64-20261108T135722Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 01 | resource use and distribution

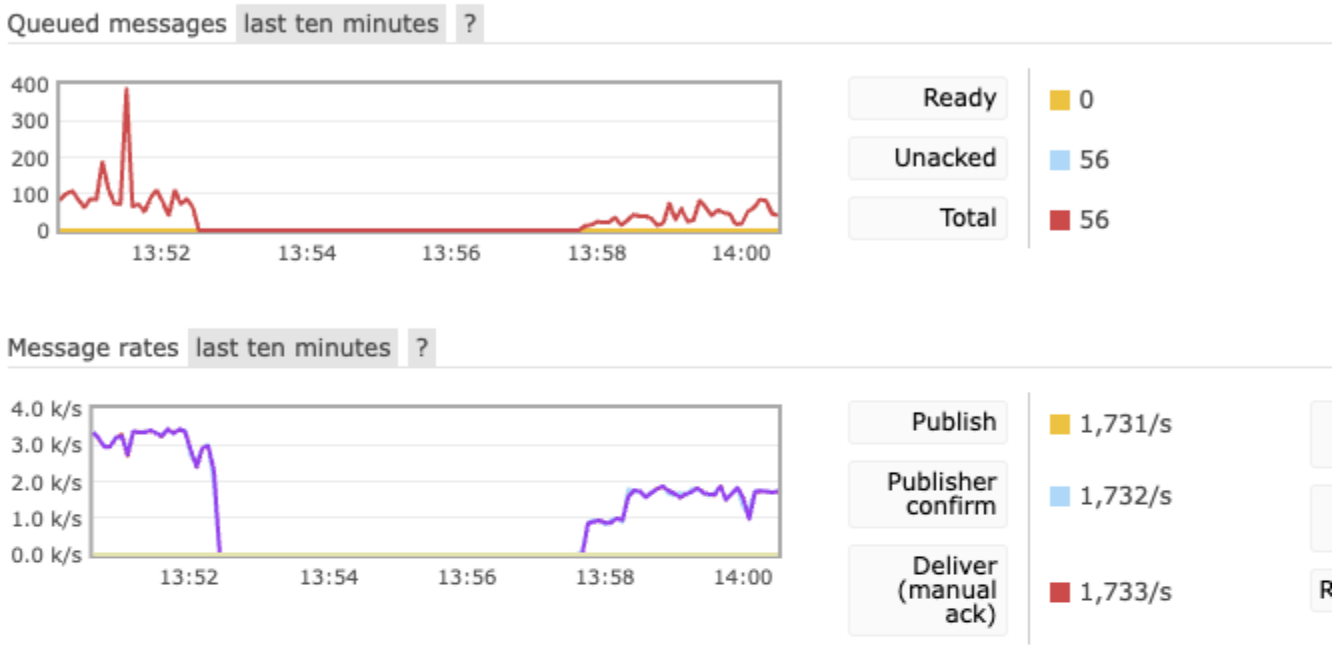




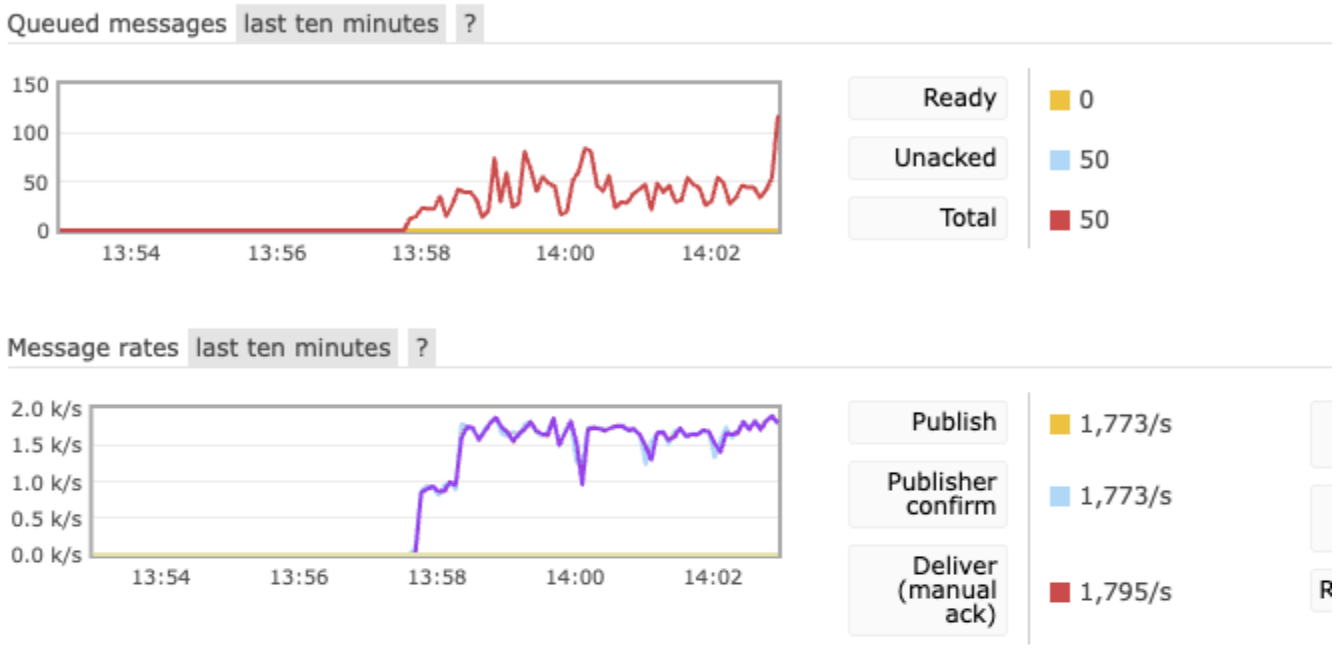
Collection gaps: None detected.

Run 01 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T13:57:39.954246+00:00 to 2026-11-08T14:03:01.243264+00:00.

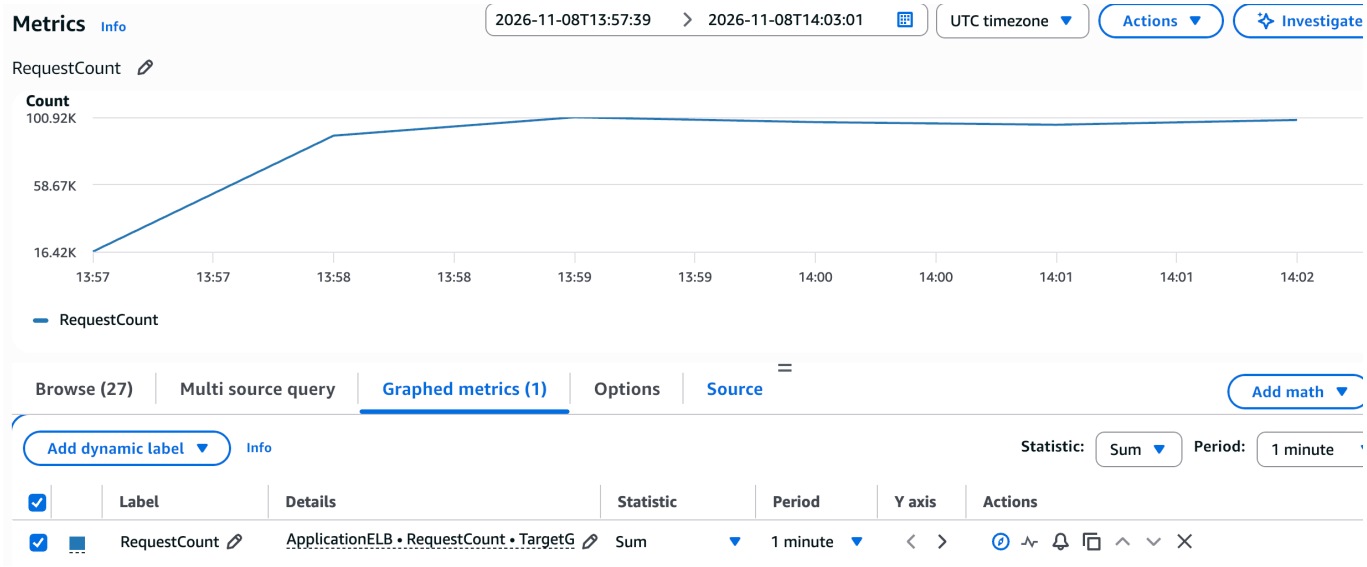


rabbitmq-20261108T140041357168Z.png | Original screenshot retained unchanged.

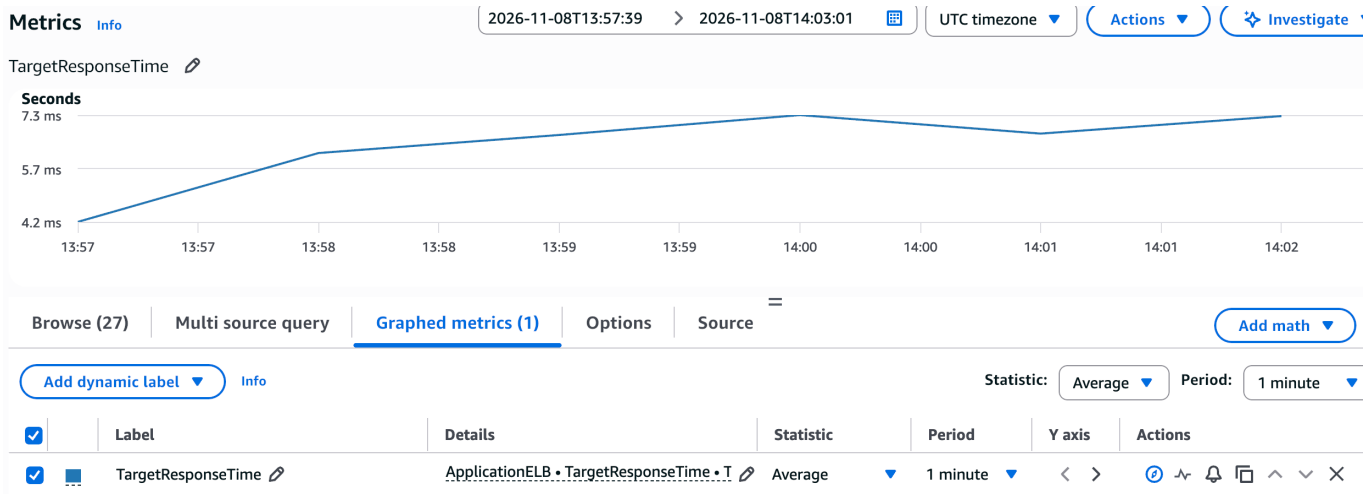


rabbitmq-20261108T140307393941Z-final.png | Original screenshot retained unchanged.

Run 01 | supplied AWS CloudWatch captures



alb-request-count.png



alb-target-response-time.png

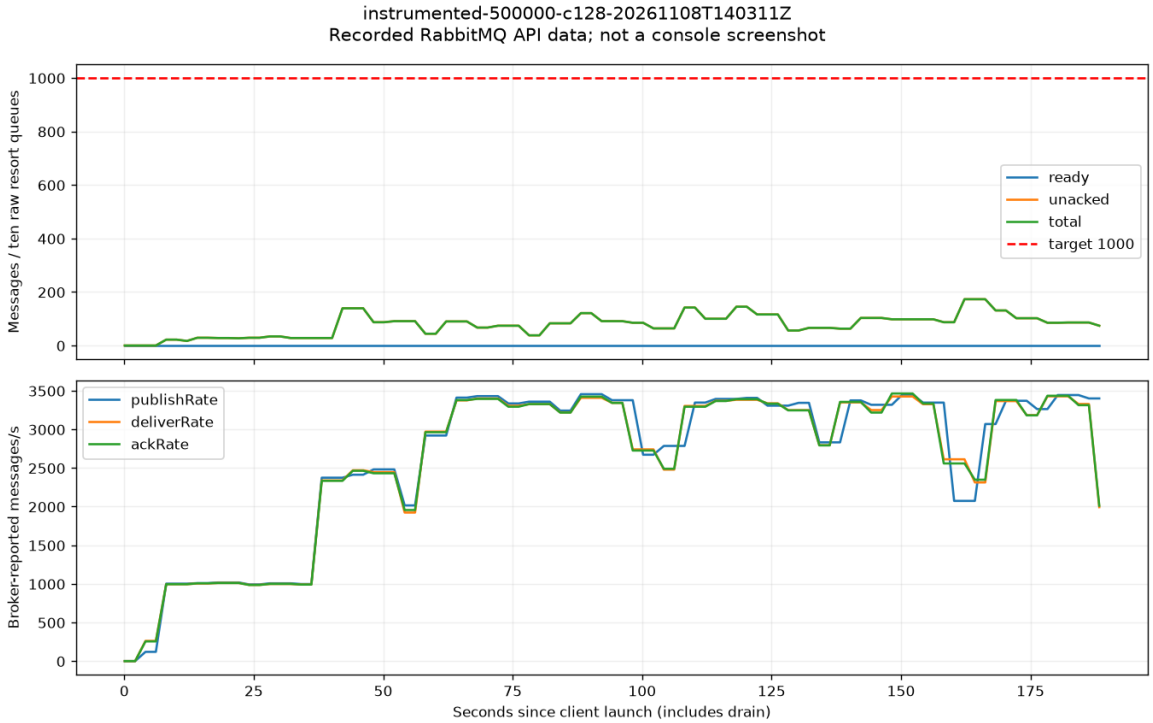
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 02 | 1 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

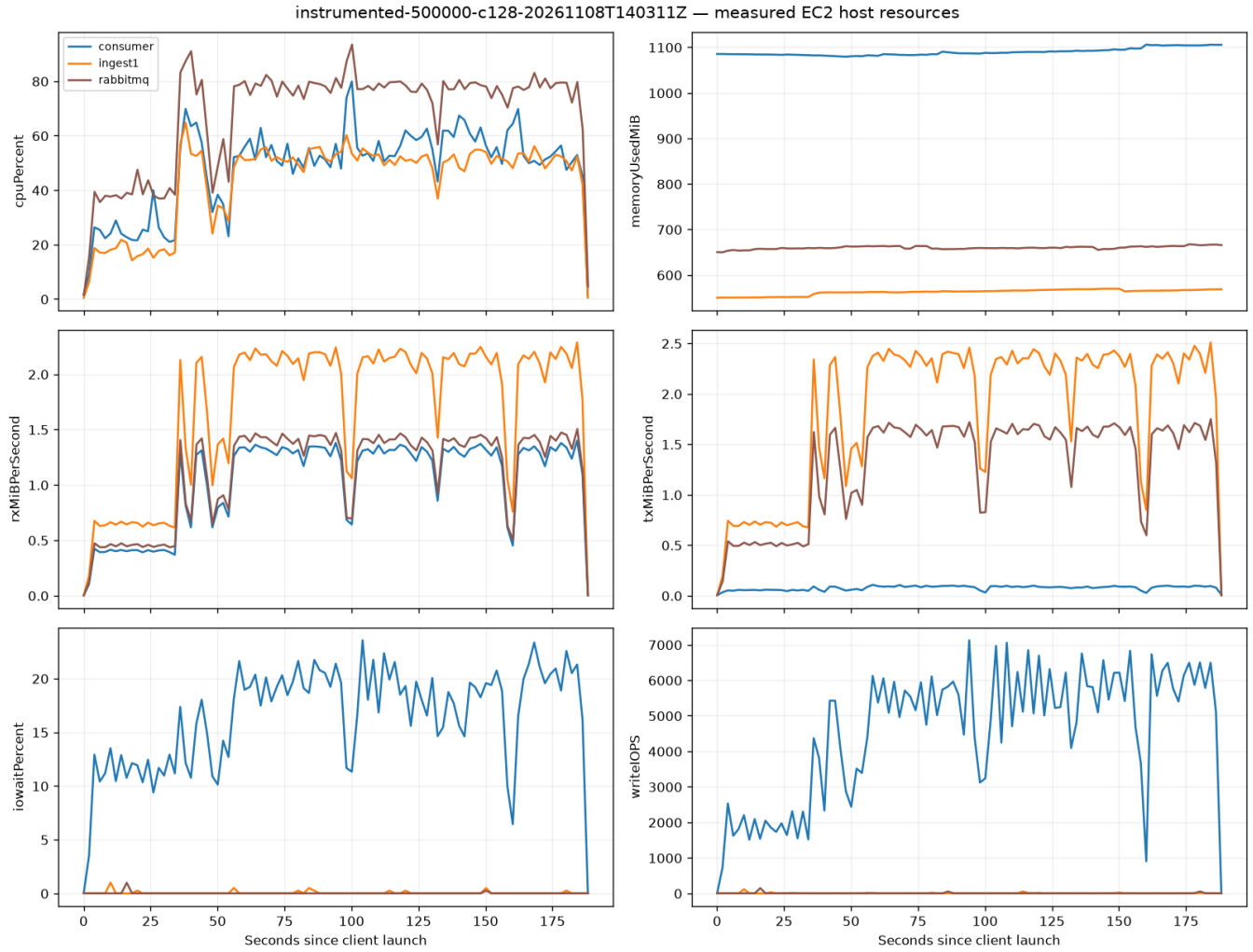
UTC 2026-11-08T14:03:24.169506+00:00 to 2026-11-08T14:06:32.887825+00:00

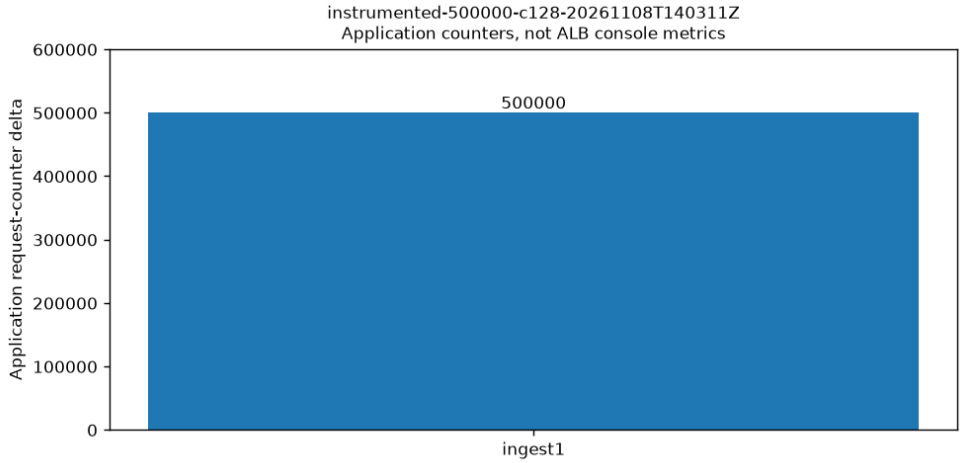
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	3,080.16 / 2,711.06 / 2,699.50
Ingestion / program runtime / drain seconds	184.429 / 187.473 / 0.766
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	40.94 / 56.09 / 134.59
Ready sampled mean / maximum	0.00 / 0
Unacked / Ready+Unacked maximum	173 / 173
Broker publish / delivery / ACK sampled mean per second	2,637.75 / 2,607.96 / 2,607.50
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	25.18
WebSocket updates / reconnections	911 / 0



Source: instrumented-500000-c128-20261108T140311Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 02 | resource use and distribution

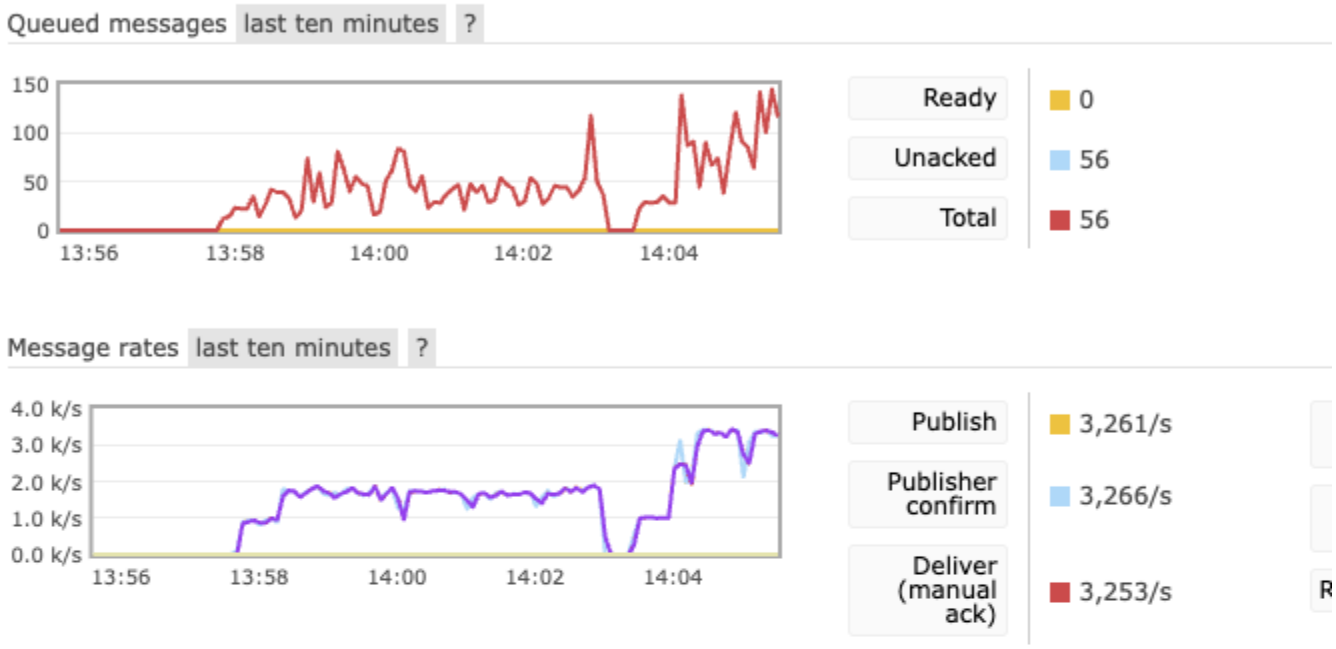




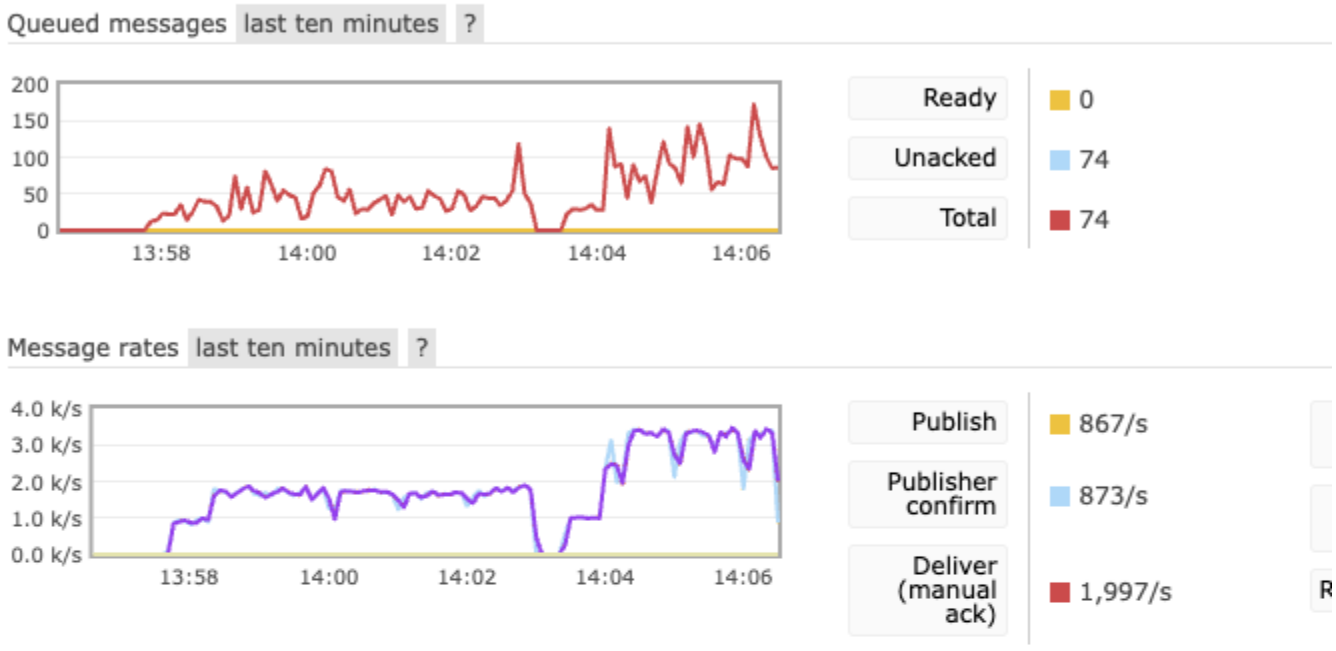
Collection gaps: None detected.

Run 02 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:03:24.169506+00:00 to 2026-11-08T14:06:32.887825+00:00.

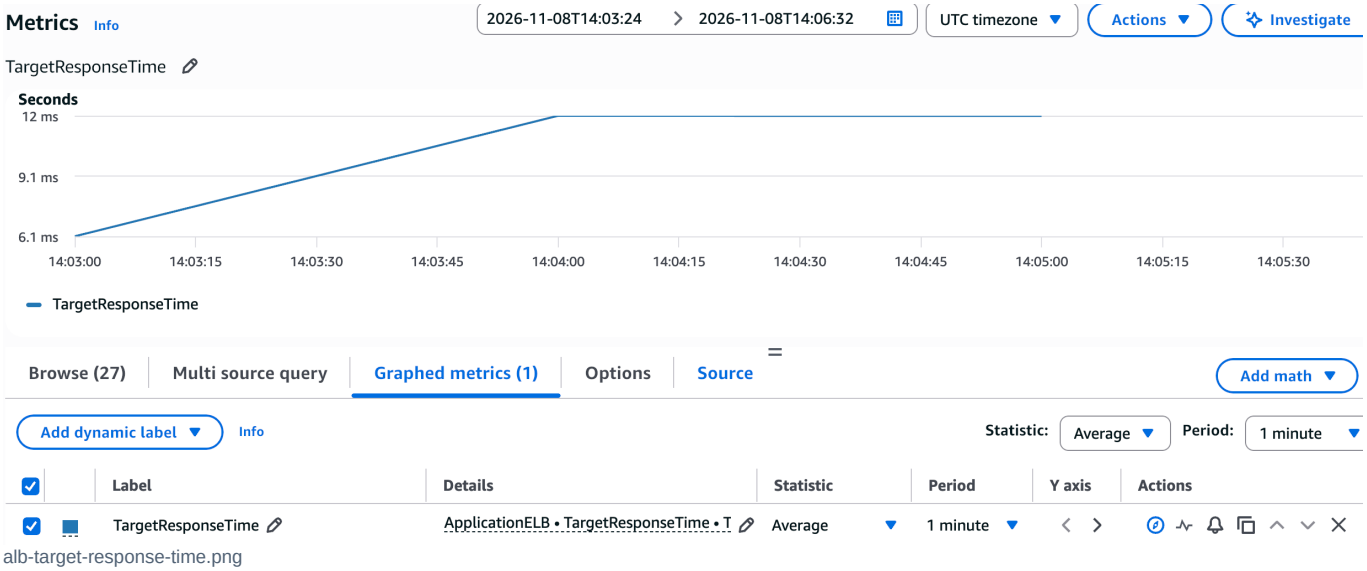


rabbitmq-20261108T140540312026Z.png | Original screenshot retained unchanged.

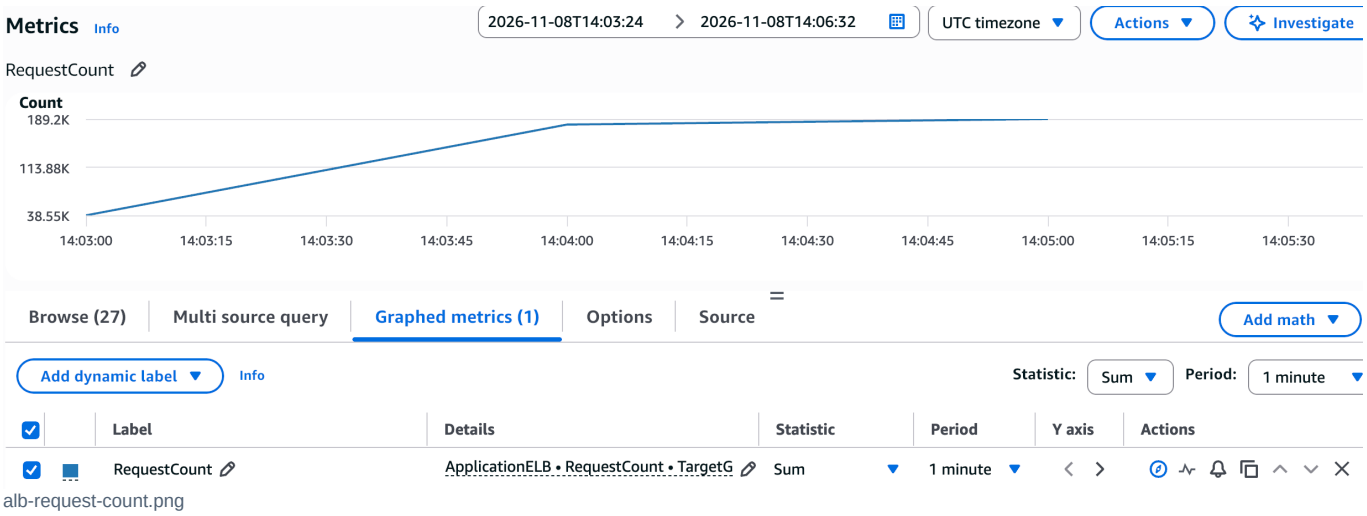


rabbitmq-20261108T140639050709Z-final.png | Original screenshot retained unchanged.

Run 02 | supplied AWS CloudWatch captures



alb-target-response-time.png



alb-request-count.png

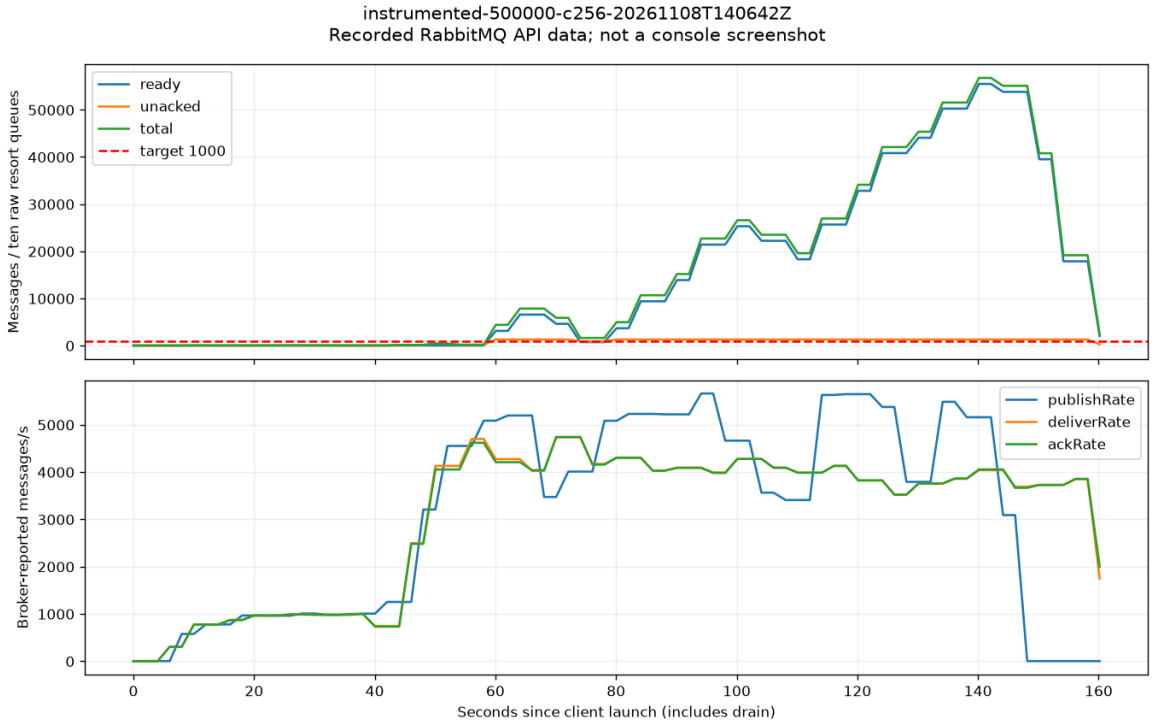
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 03 | 1 ingest(s), 500,000 events

Client threads=256; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

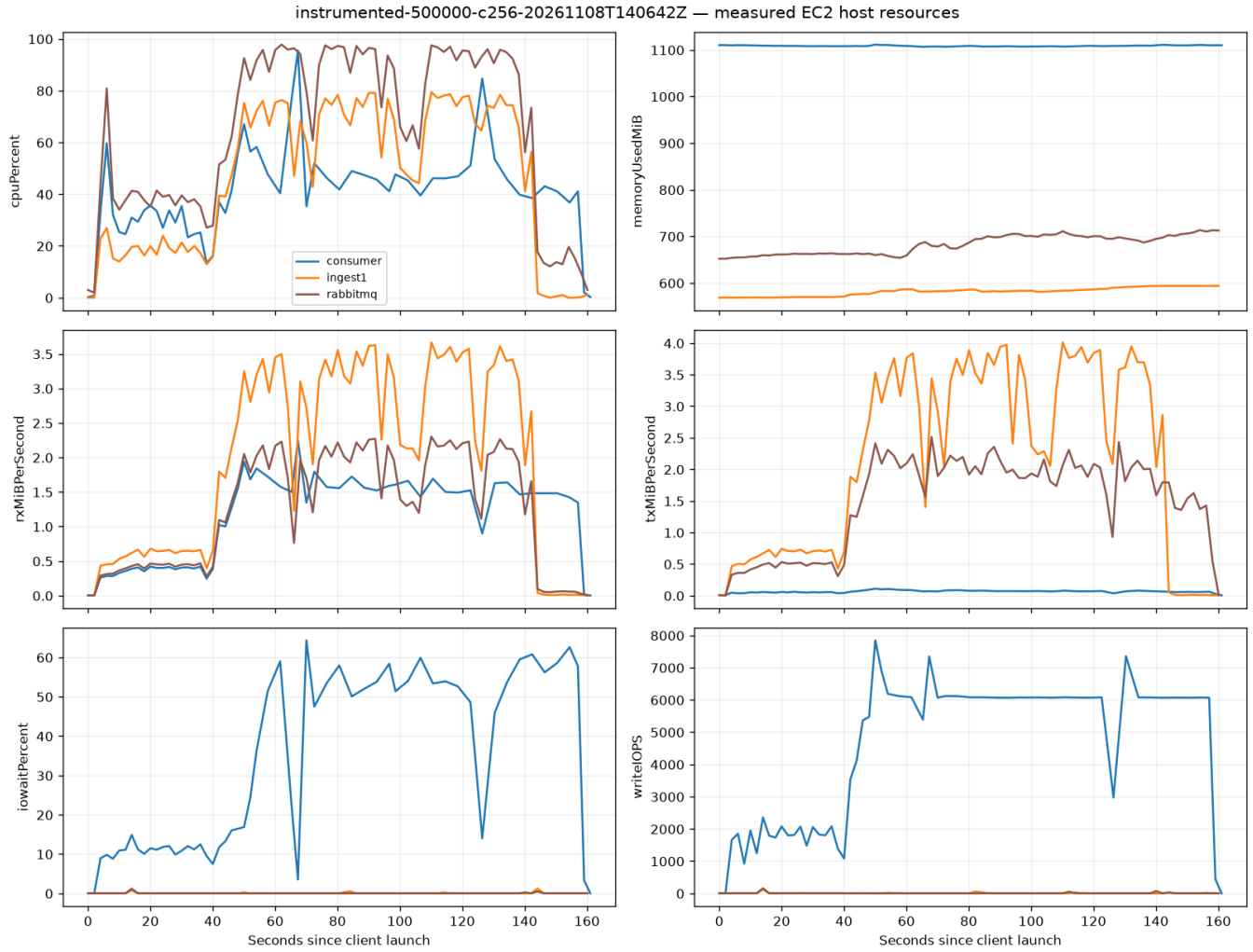
UTC 2026-11-08T14:06:57.686350+00:00 to 2026-11-08T14:09:38.884009+00:00

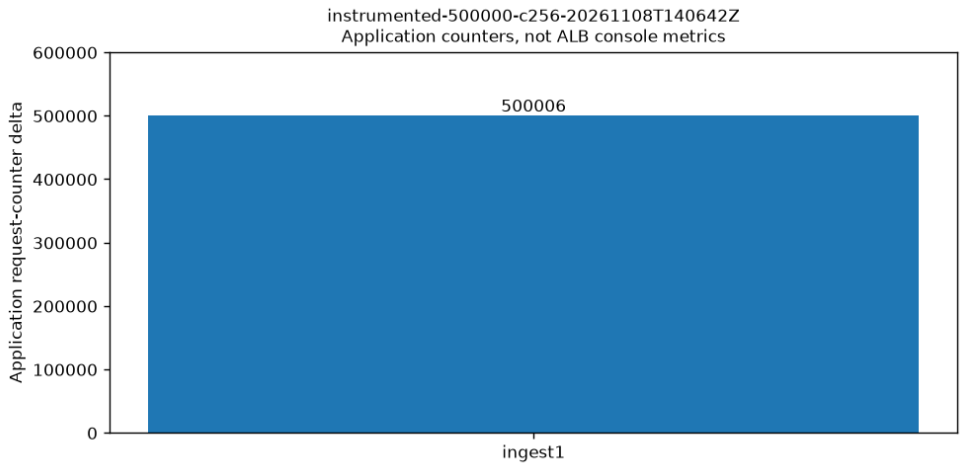
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	4,475.42 / 3,575.16 / 3,192.84
Ingestion / program runtime / drain seconds	139.854 / 160.128 / 16.696
Successful / final failed / application retries	500,000 / 0 / 6
HTTP mean / p95 / p99 ms	55.23 / 100.20 / 275.29
Ready sampled mean / maximum	15,173.65 / 55,463
Unacked / Ready+Unacked maximum	1,280 / 56,743
Broker publish / delivery / ACK sampled mean per second	3,096.52 / 3,047.12 / 3,042.60
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	5,177.43
WebSocket updates / reconnections	914 / 0



Source: instrumented-500000-c256-20261108T140642Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 03 | resource use and distribution

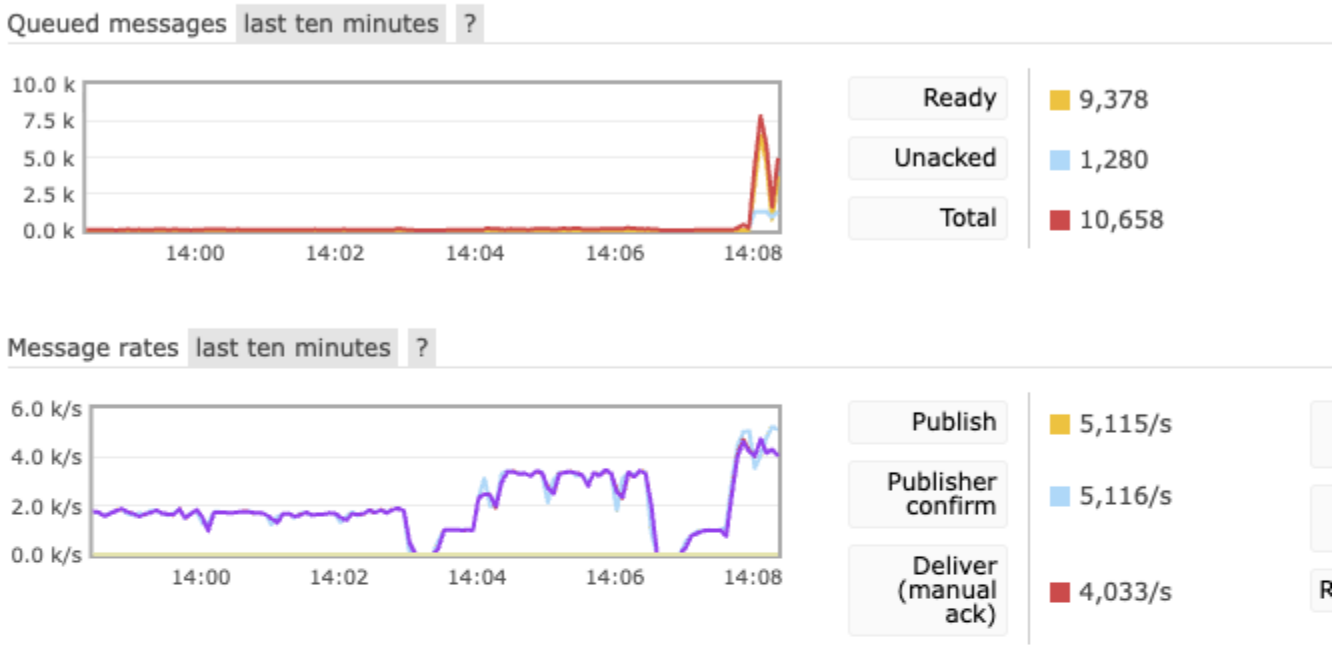




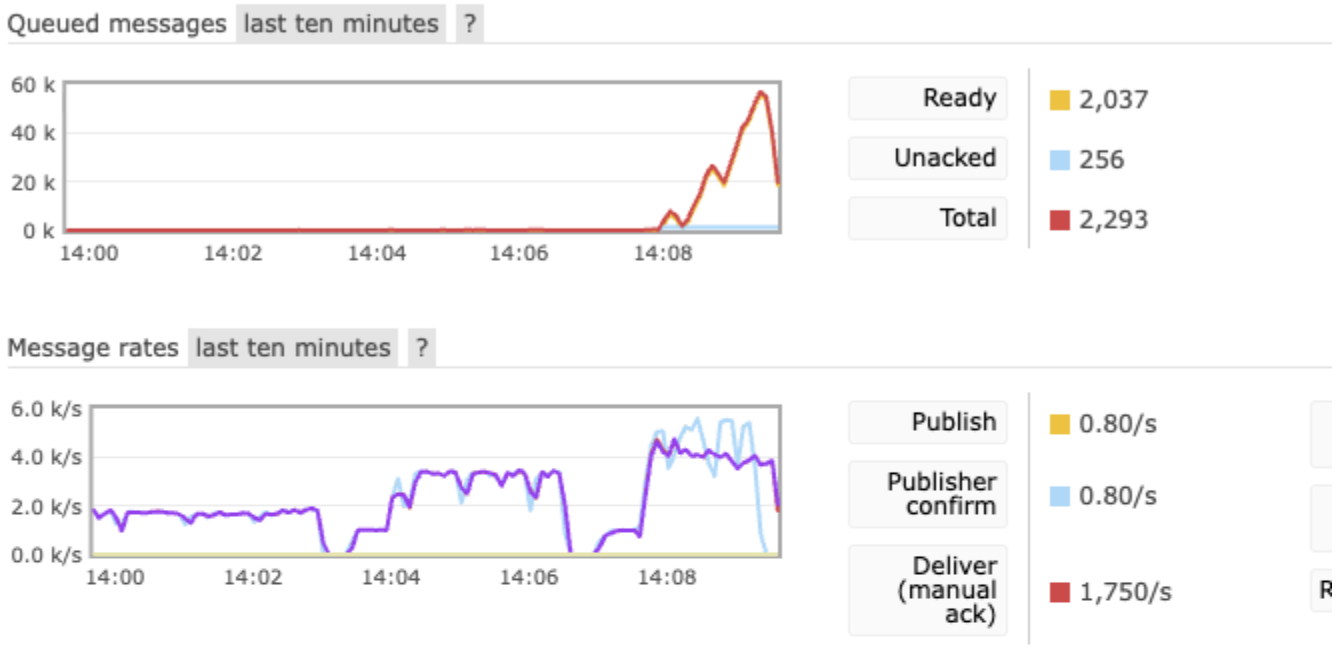
Collection gaps: consumer collection errors: {"timeout": 20}

Run 03 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:06:57.686350+00:00 to 2026-11-08T14:09:38.884009+00:00.

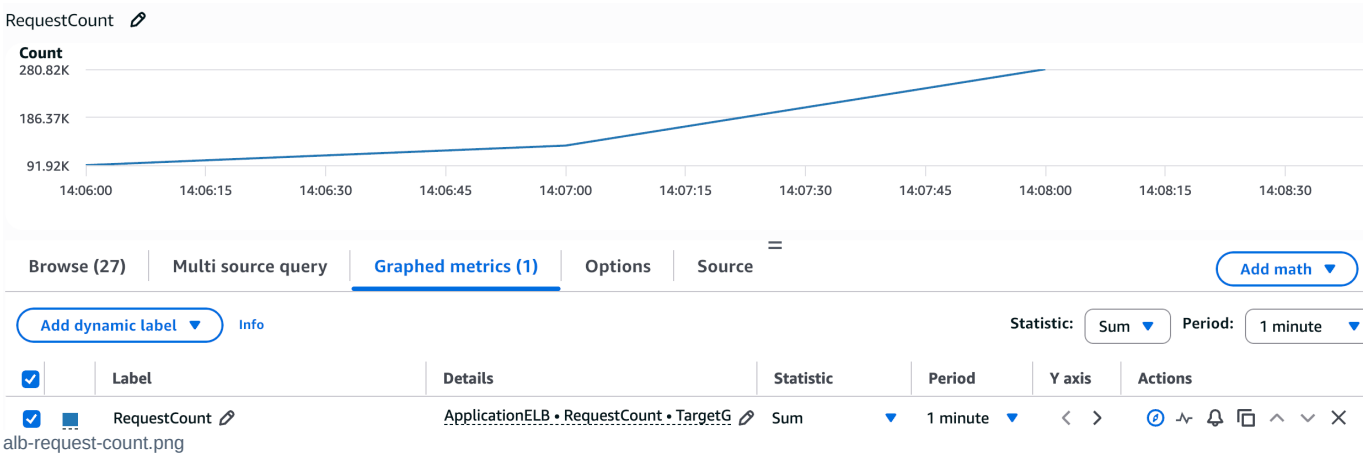


rabbitmq-20261108T140828558224Z.png | Original screenshot retained unchanged.

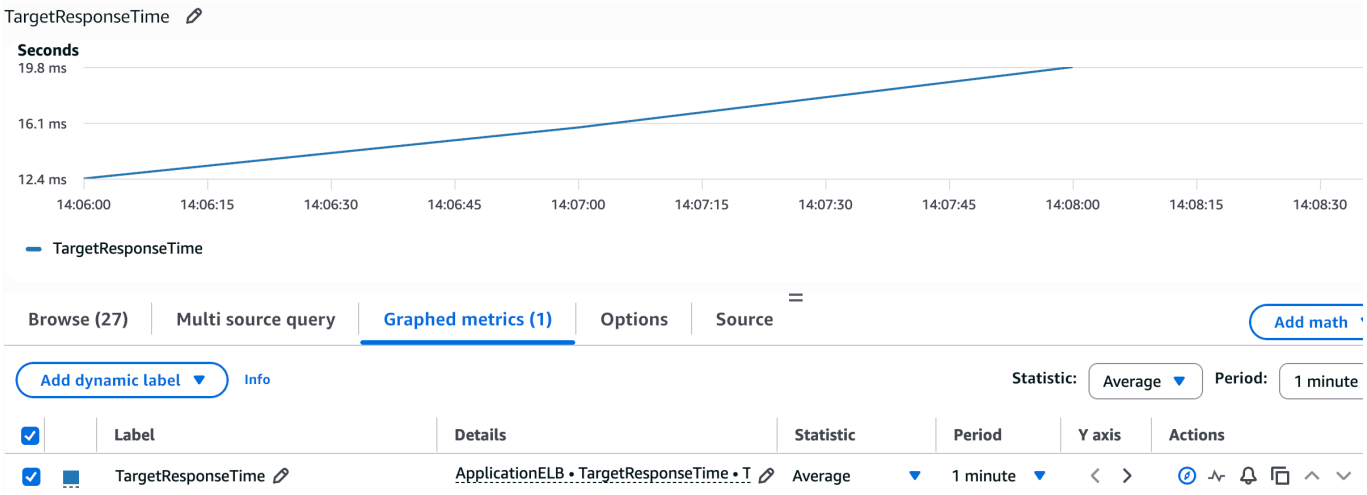


rabbitmq-20261108T140945061337Z-final.png | Original screenshot retained unchanged.

Run 03 | supplied AWS CloudWatch captures



alb-request-count.png



alb-target-response-time.png

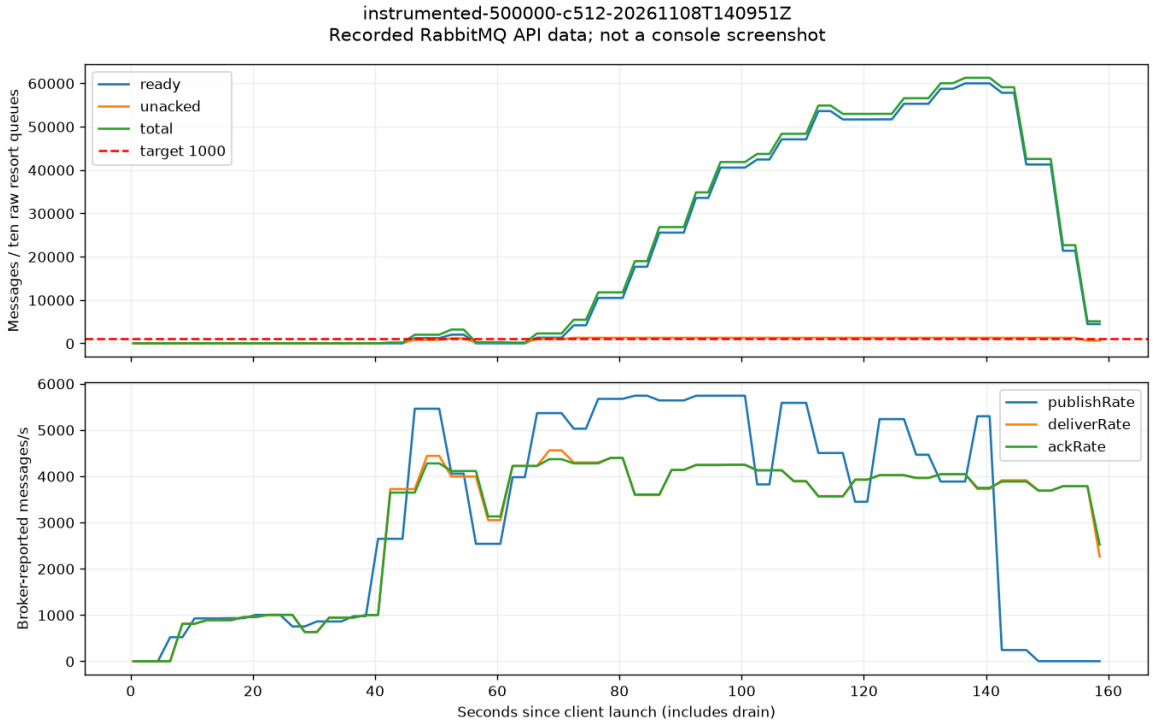
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 04 | 1 ingest(s), 500,000 events

Client threads=512; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

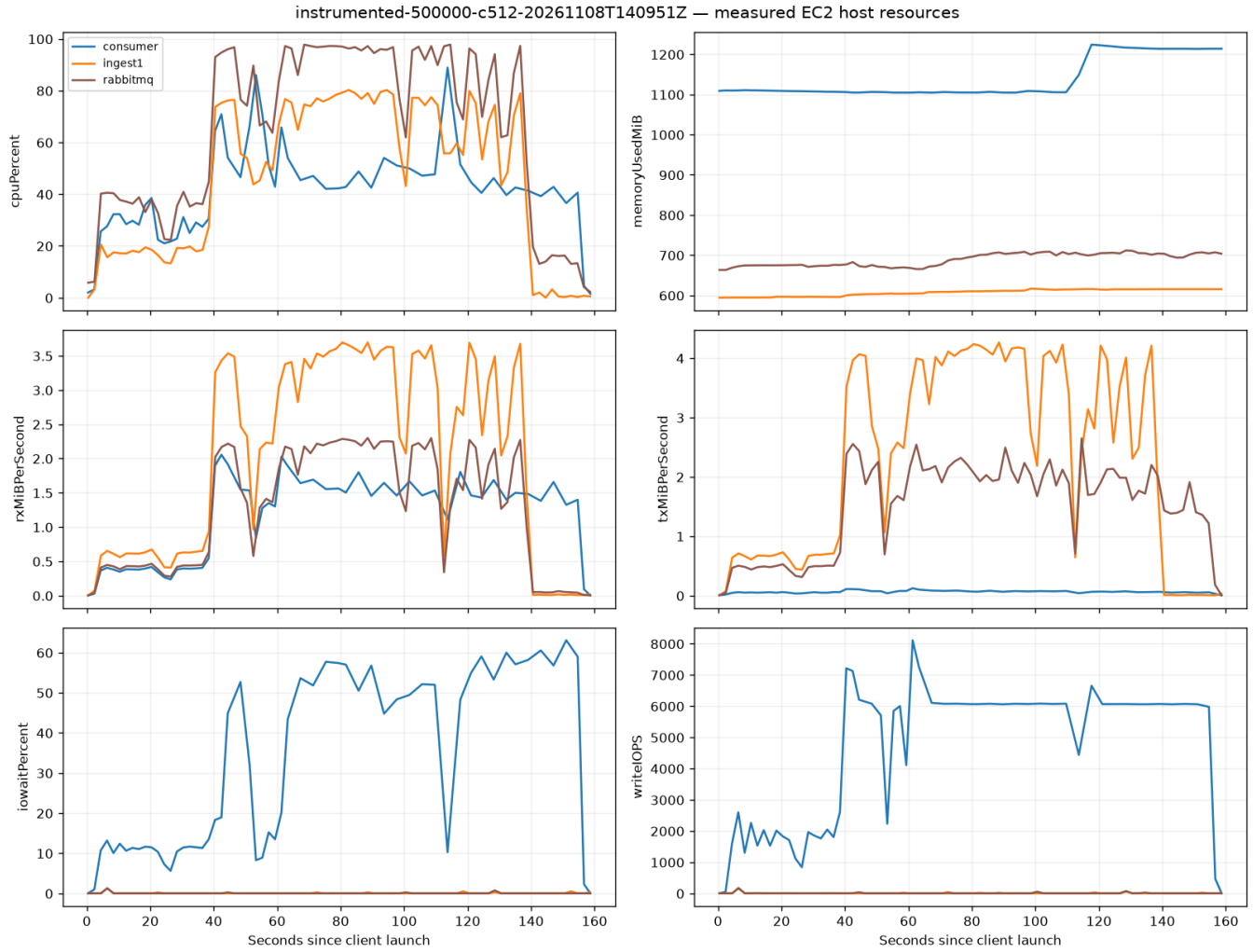
UTC 2026-11-08T14:10:10.758007+00:00 to 2026-11-08T14:12:49.682730+00:00

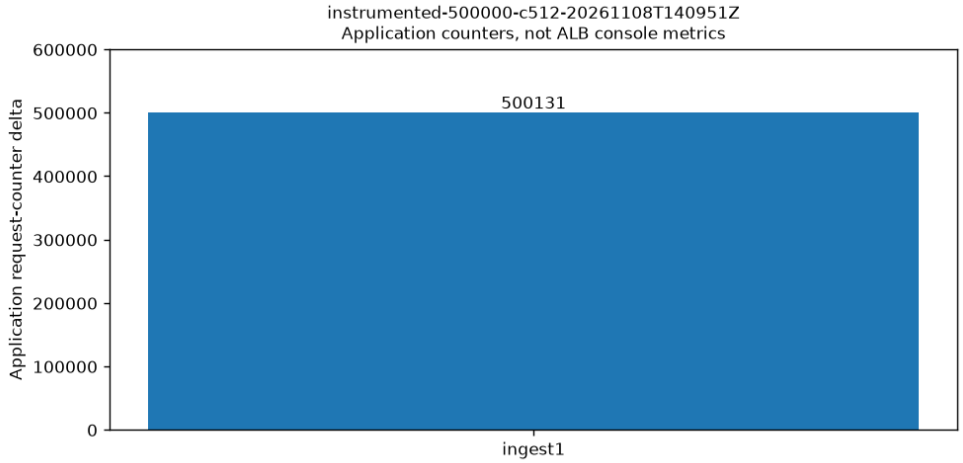
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	4,689.51 / 3,692.81 / 3,235.22
Ingestion / program runtime / drain seconds	135.398 / 157.901 / 19.110
Successful / final failed / application retries	500,000 / 0 / 131
HTTP mean / p95 / p99 ms	104.23 / 285.71 / 469.87
Ready sampled mean / maximum	21,216.08 / 59,967
Unacked / Ready+Unacked maximum	1,280 / 61,247
Broker publish / delivery / ACK sampled mean per second	3,190.88 / 3,101.10 / 3,097.74
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	7,135.63
WebSocket updates / reconnections	913 / 0



Source: instrumented-500000-c512-20261108T140951Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 04 | resource use and distribution

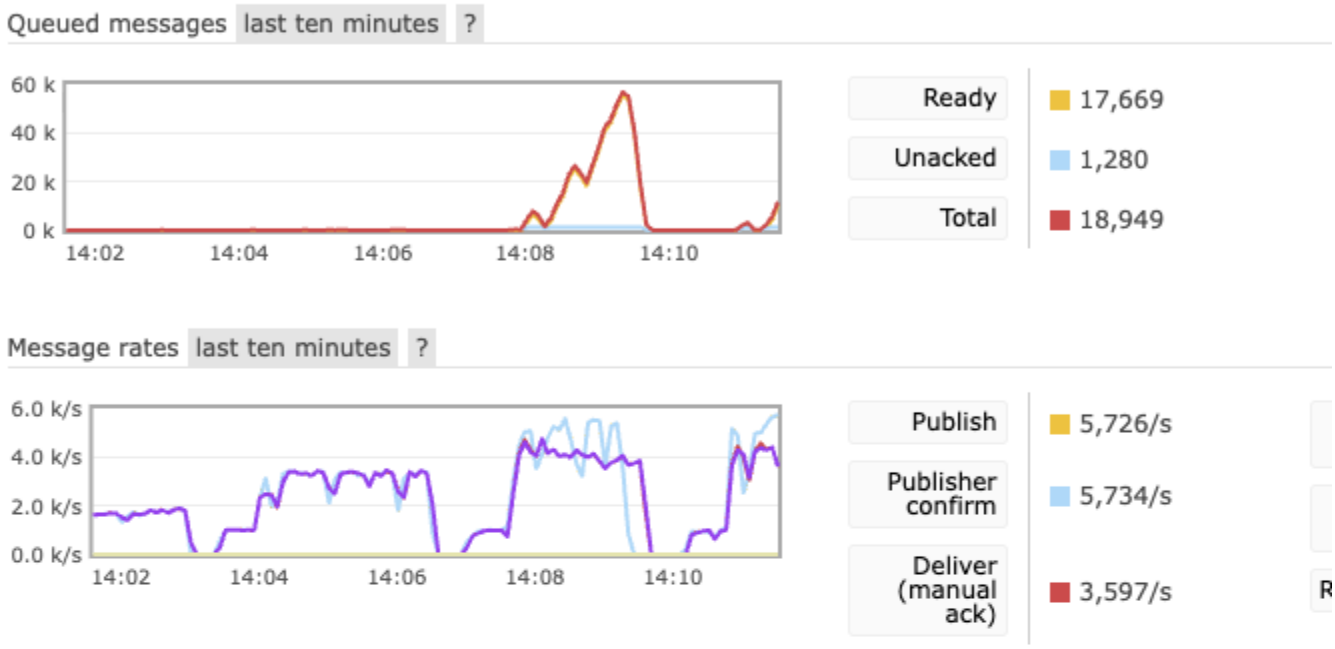




Collection gaps: consumer collection errors: {"timeout": 19}

Run 04 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:10:10.758007+00:00 to 2026-11-08T14:12:49.682730+00:00.

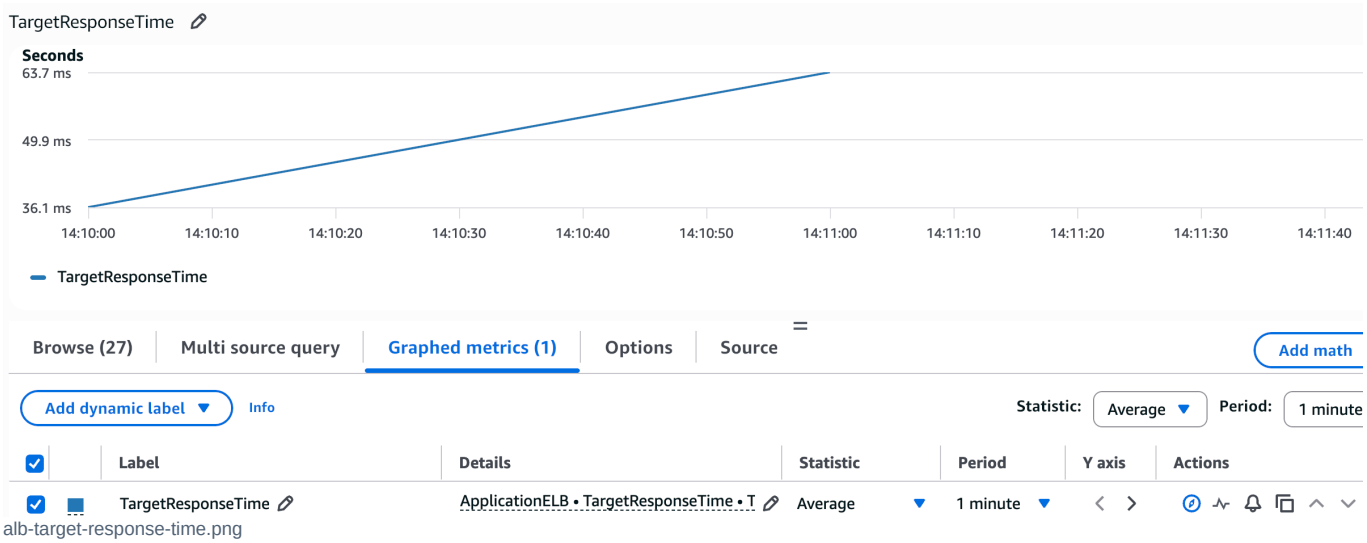


rabbitmq-20261108T141141324212Z.png | Original screenshot retained unchanged.

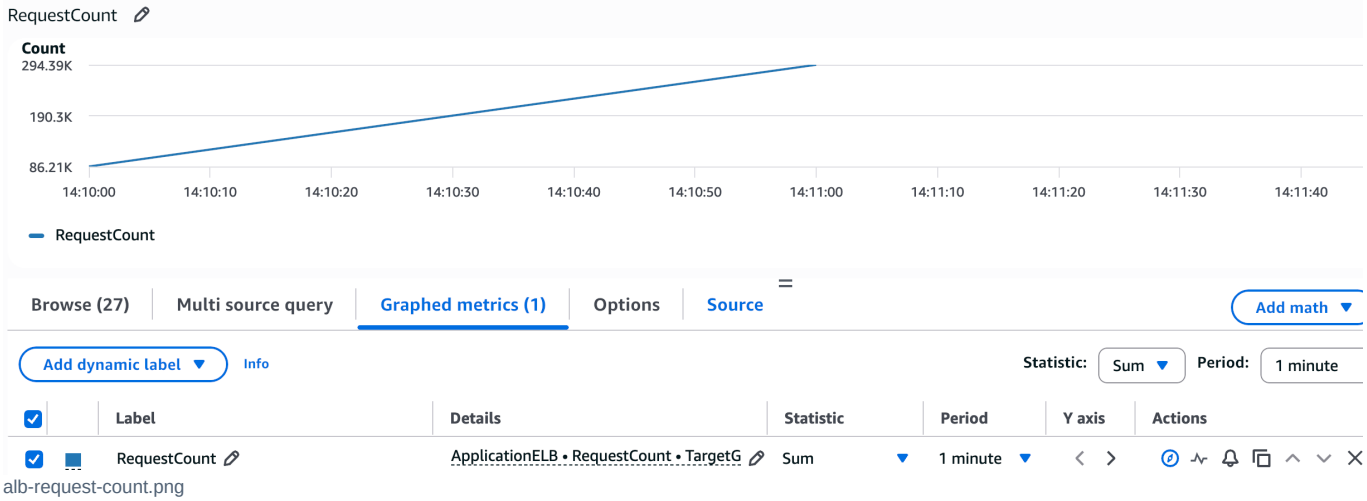


rabbitmq-20261108T141255950142Z-final.png | Original screenshot retained unchanged.

Run 04 | supplied AWS CloudWatch captures



alb-target-response-time.png



alb-request-count.png

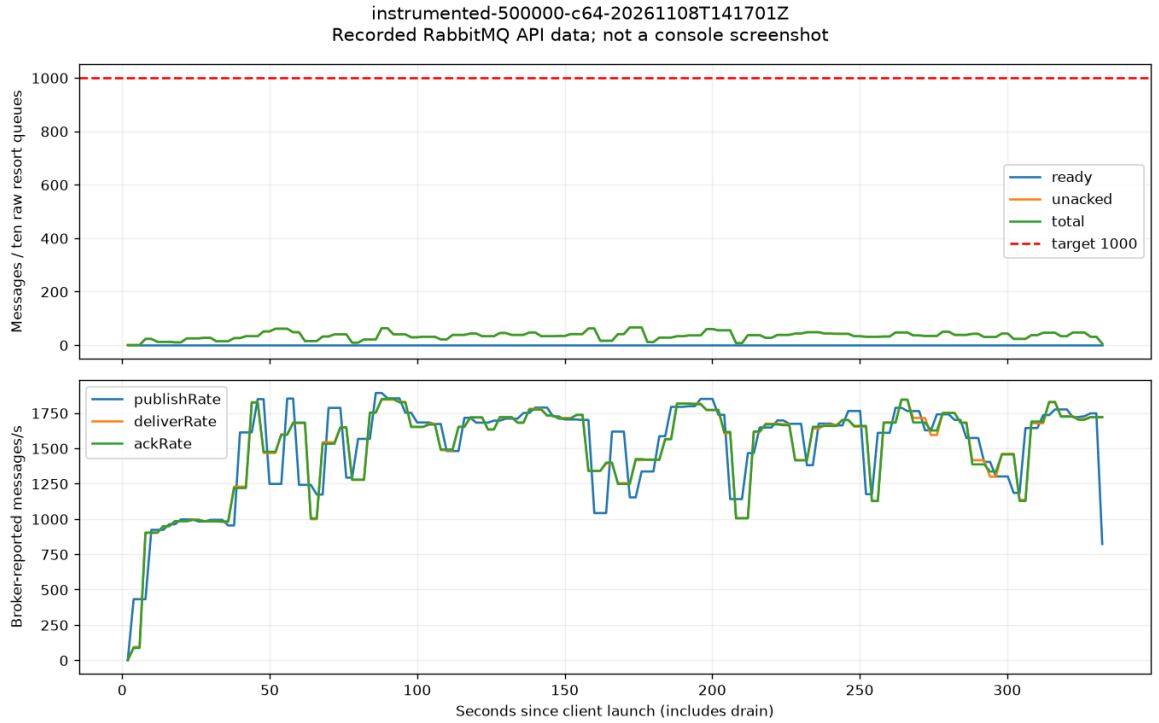
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 05 | 2 ingest(s), 500,000 events

Client threads=64; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

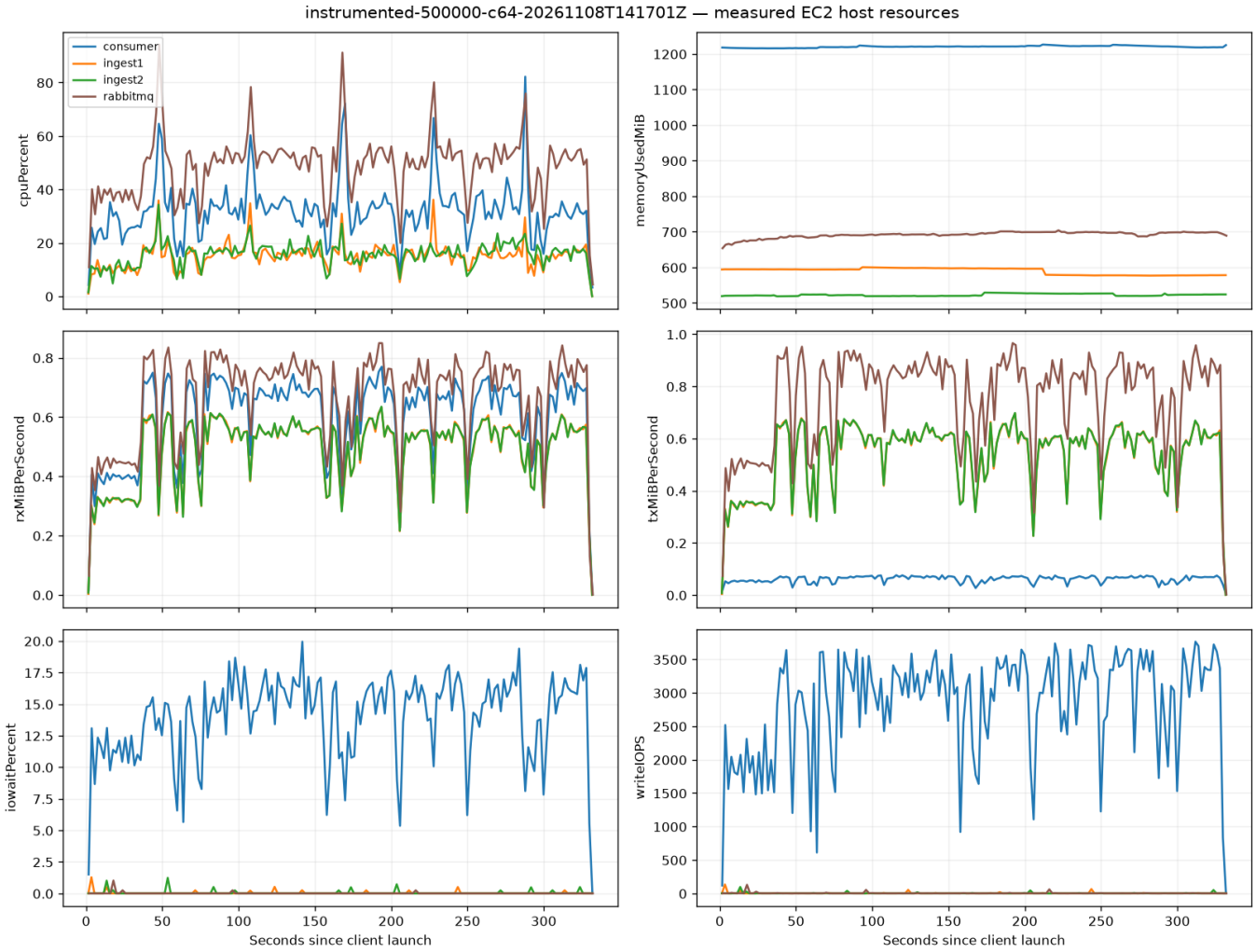
UTC 2026-11-08T14:17:15.532676+00:00 to 2026-11-08T14:22:48.038979+00:00

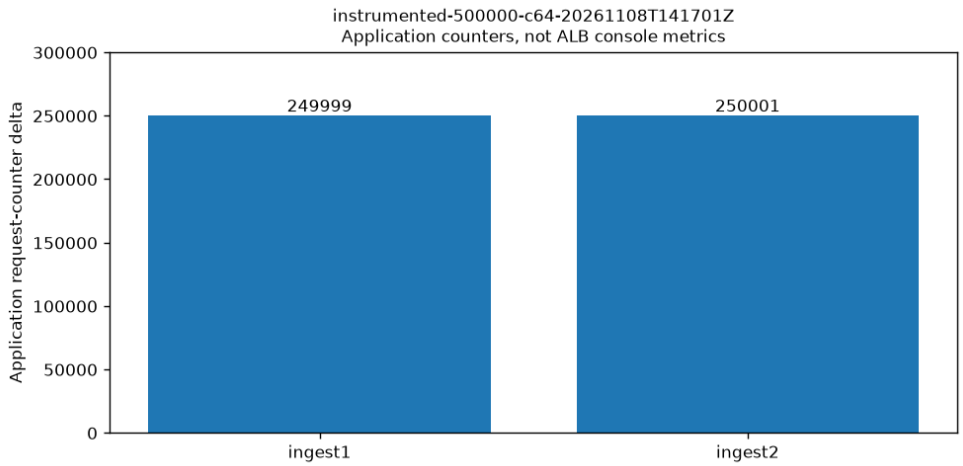
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	1,594.86 / 1,528.59 / 1,524.64
Ingestion / program runtime / drain seconds	327.098 / 331.398 / 0.804
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	39.67 / 54.30 / 153.19
Ready sampled mean / maximum	0.00 / 0
Unacked / Ready+Unacked maximum	66 / 66
Broker publish / delivery / ACK sampled mean per second	1,509.35 / 1,501.61 / 1,500.93
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	18.22
WebSocket updates / reconnections	911 / 0



Source: instrumented-500000-c64-20261108T141701Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 05 | resource use and distribution

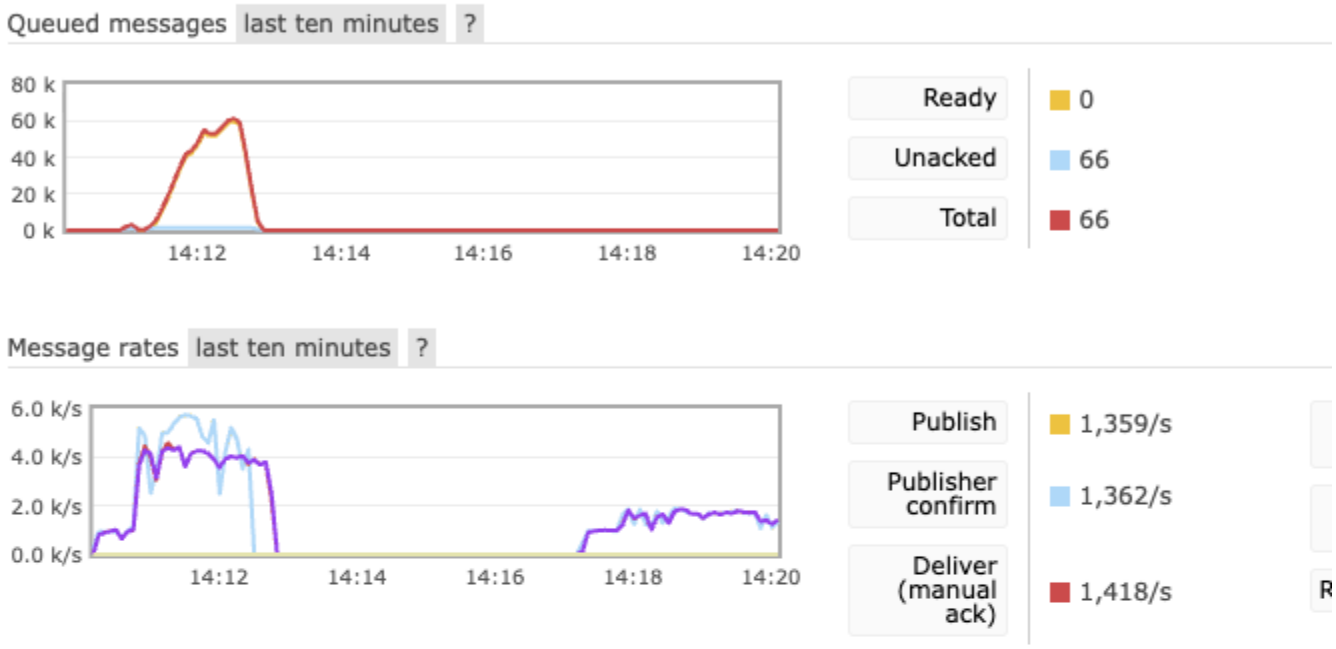




Collection gaps: None detected.

Run 05 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:17:15.532676+00:00 to 2026-11-08T14:22:48.038979+00:00.



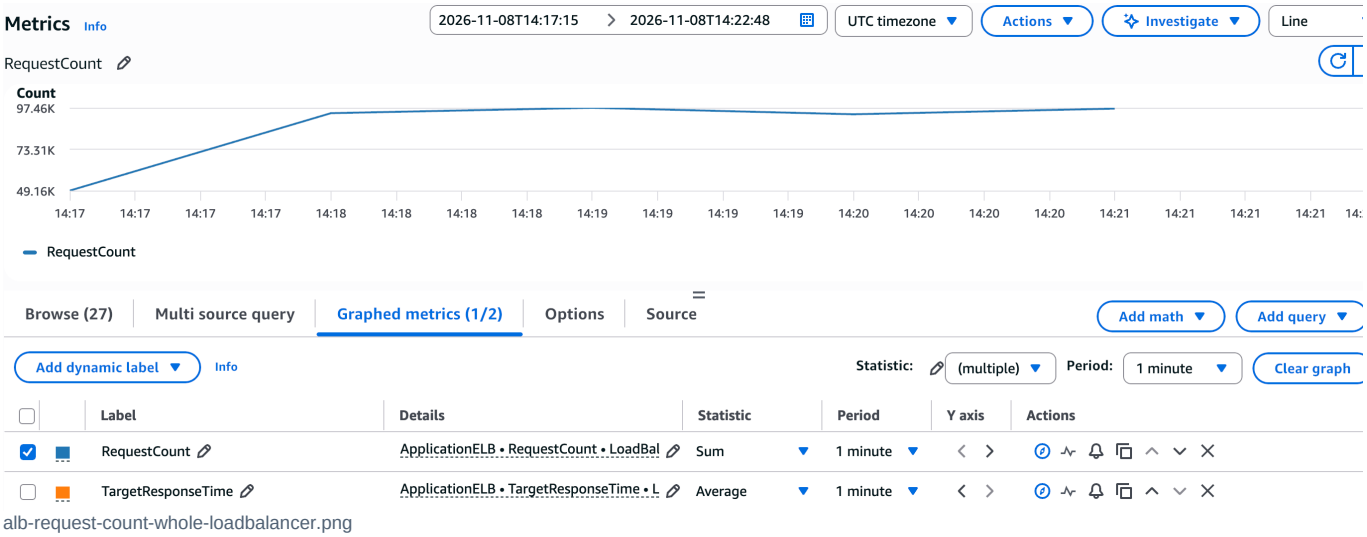
rabbitmq-20261108T142017454100Z.png | Original screenshot retained unchanged.



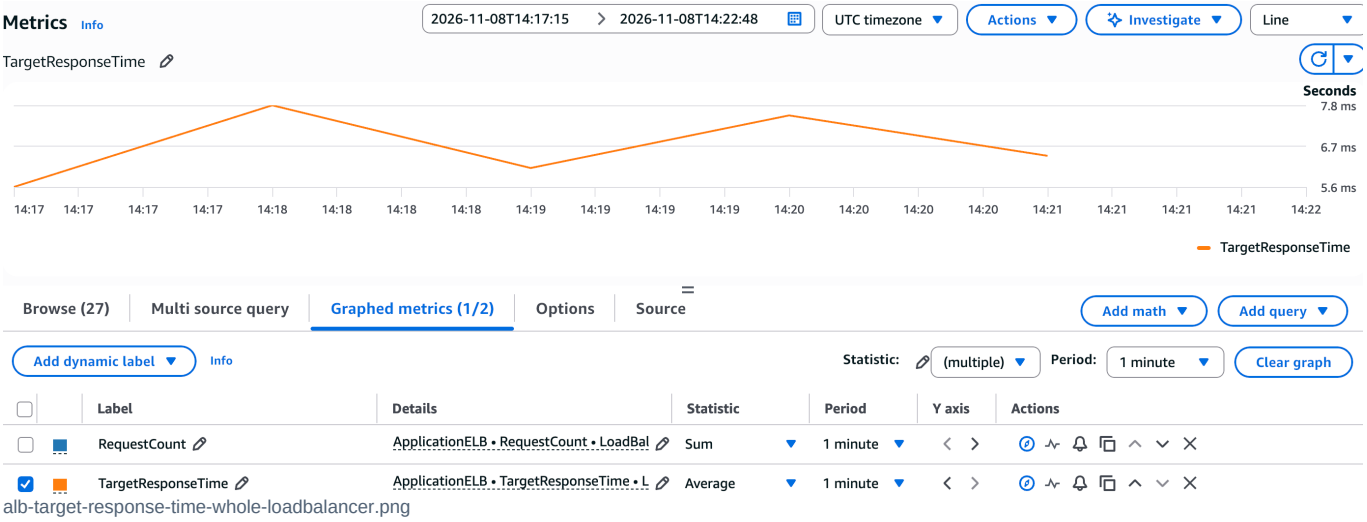
rabbitmq-20261108T142254621796Z-final.png | Original screenshot retained unchanged.

Run 05 | supplied AWS CloudWatch captures

Metric scope: Whole LoadBalancer; no AvailabilityZone or TargetGroup filter.



alb-request-count-whole-loadbalancer.png



alb-target-response-time-whole-loadbalancer.png

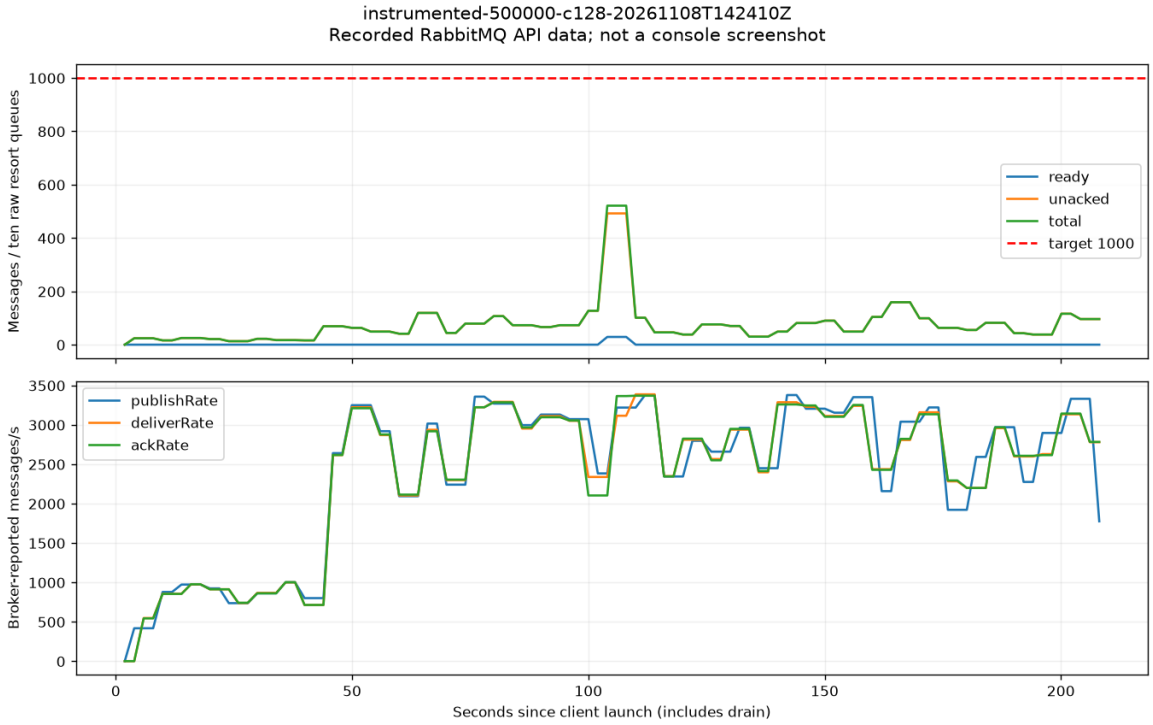
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 06 | 2 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

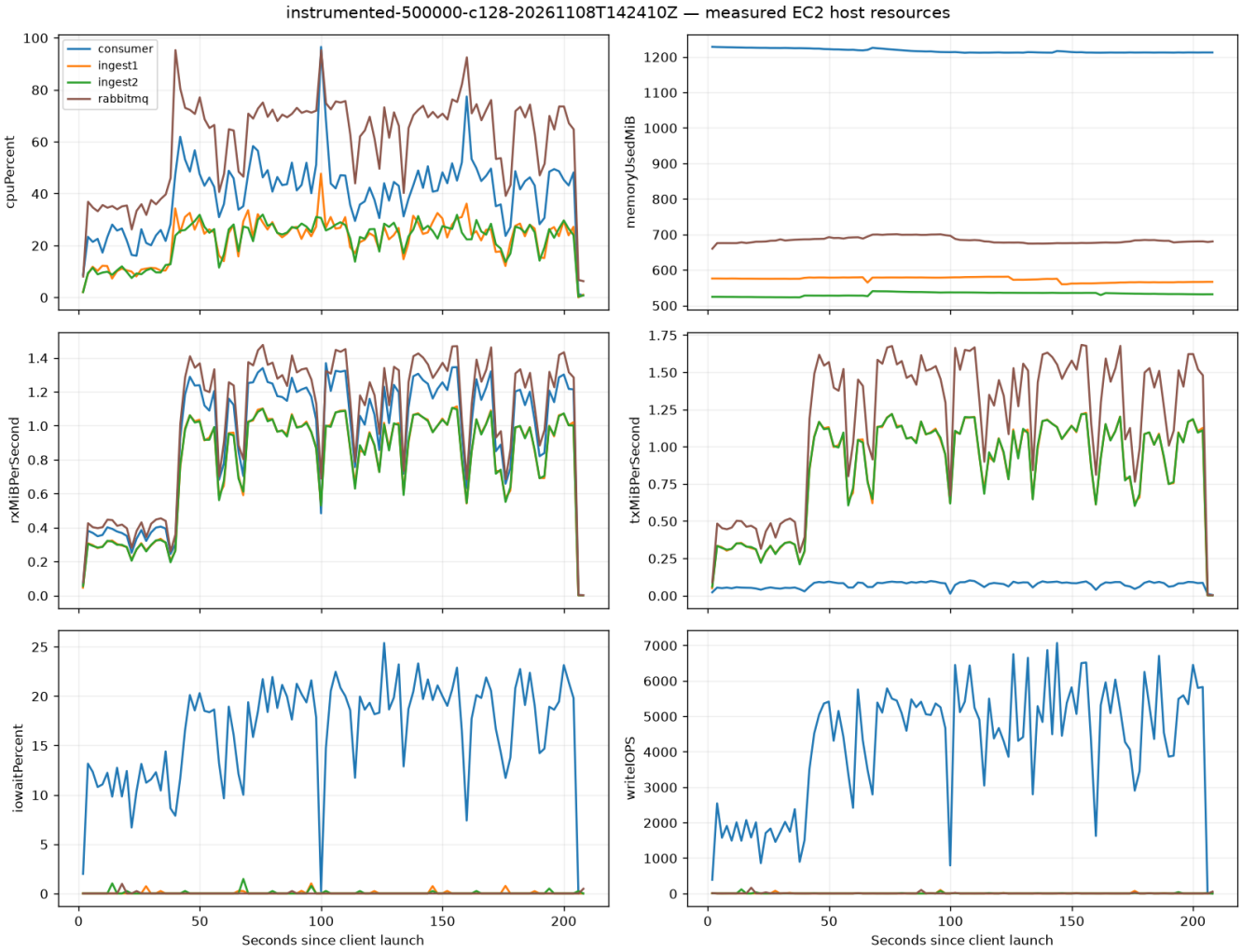
UTC 2026-11-08T14:24:23.473898+00:00 to 2026-11-08T14:27:52.242822+00:00

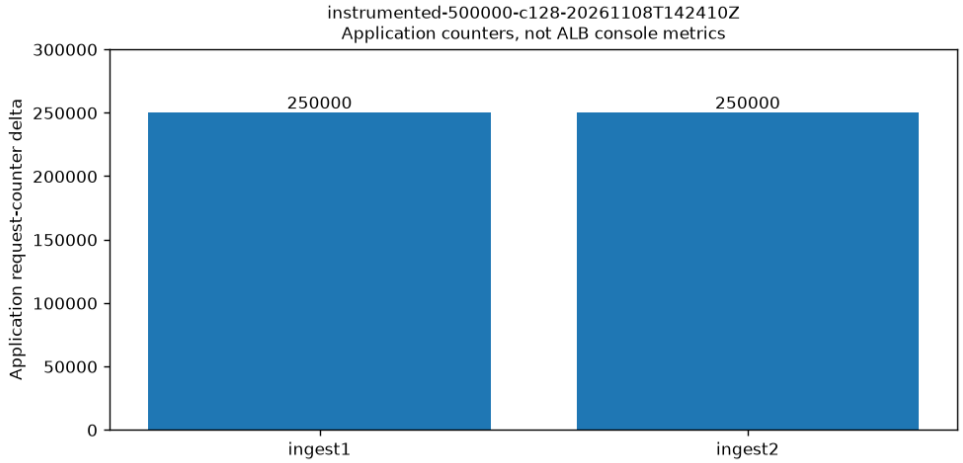
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	2,821.24 / 2,469.74 / 2,460.44
Ingestion / program runtime / drain seconds	202.451 / 207.123 / 0.727
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	44.75 / 70.57 / 191.59
Ready sampled mean / maximum	0.84 / 29
Unacked / Ready+Unacked maximum	492 / 521
Broker publish / delivery / ACK sampled mean per second	2,408.02 / 2,391.88 / 2,388.94
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	29.39
WebSocket updates / reconnections	910 / 0



Source: instrumented-500000-c128-20261108T142410Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 06 | resource use and distribution

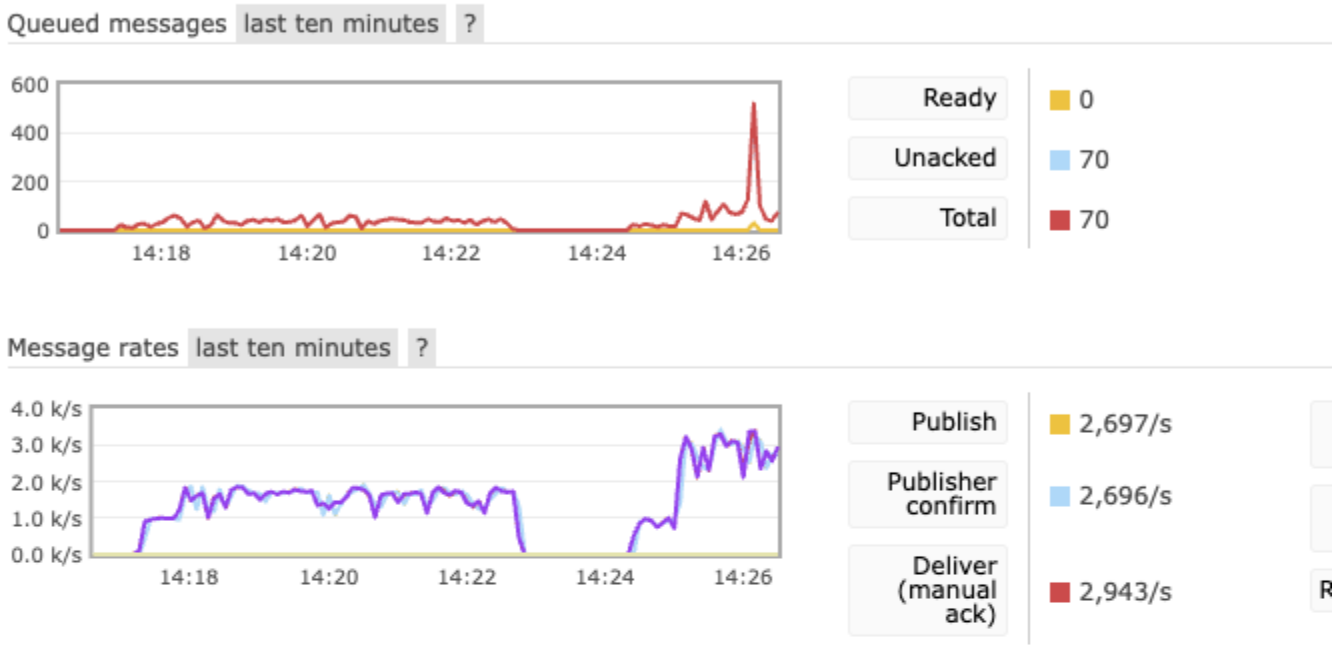




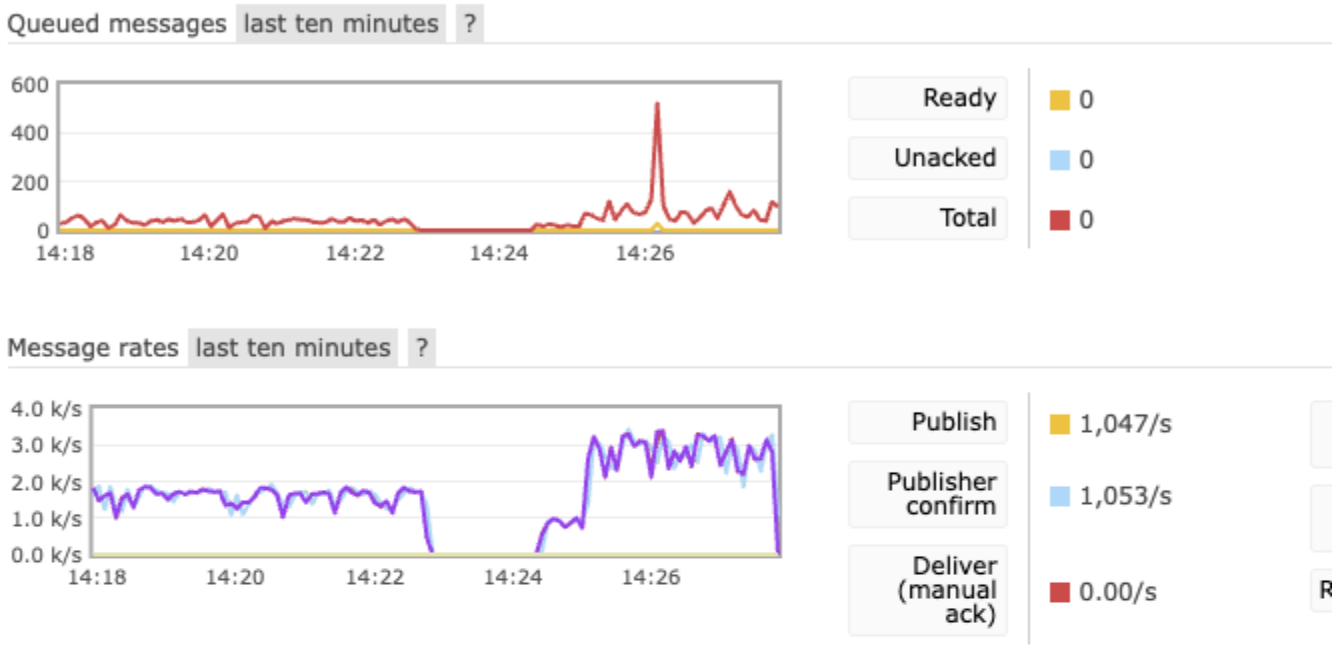
Collection gaps: None detected.

Run 06 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:24:23.473898+00:00 to 2026-11-08T14:27:52.242822+00:00.



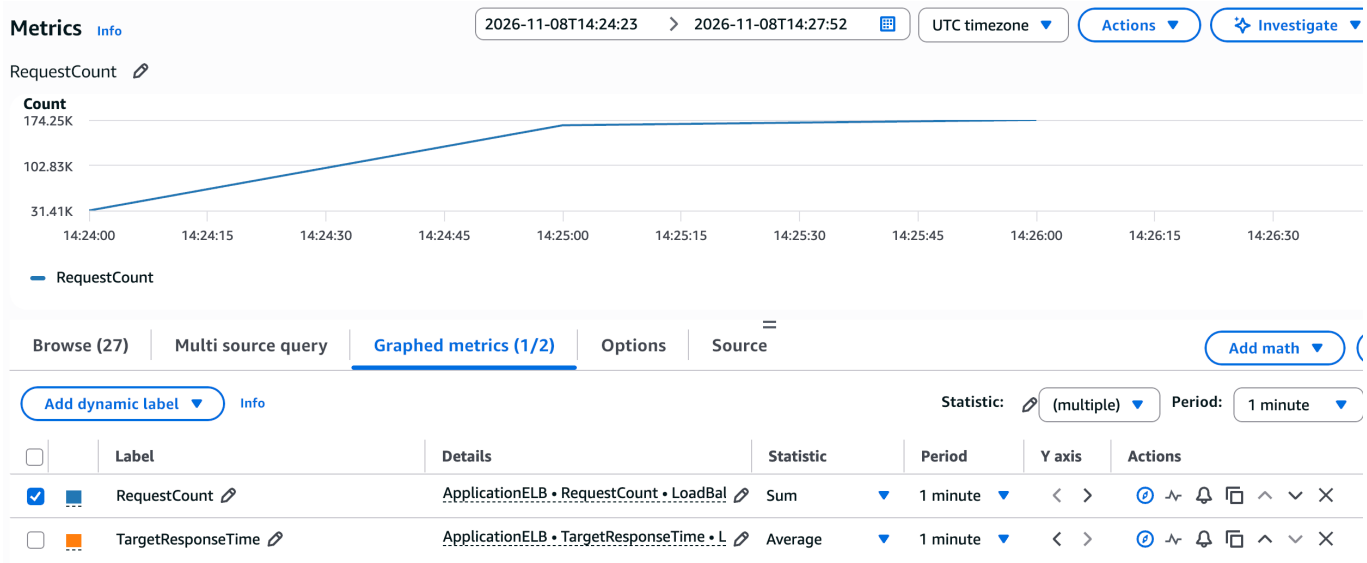
rabbitmq-20261108T142639326312Z.png | Original screenshot retained unchanged.



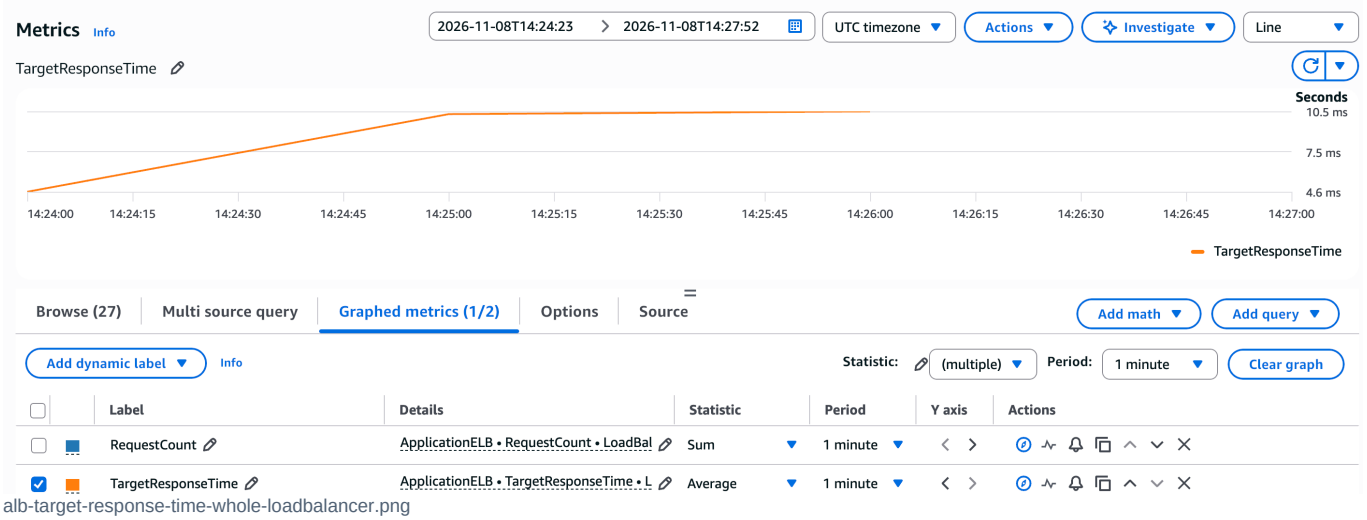
rabbitmq-20261108T142758760836Z-final.png | Original screenshot retained unchanged.

Run 06 | supplied AWS CloudWatch captures

Metric scope: Whole LoadBalancer; no AvailabilityZone or TargetGroup filter.



alb-request-count-whole-loadbalancer.png



alb-target-response-time-whole-loadbalancer.png

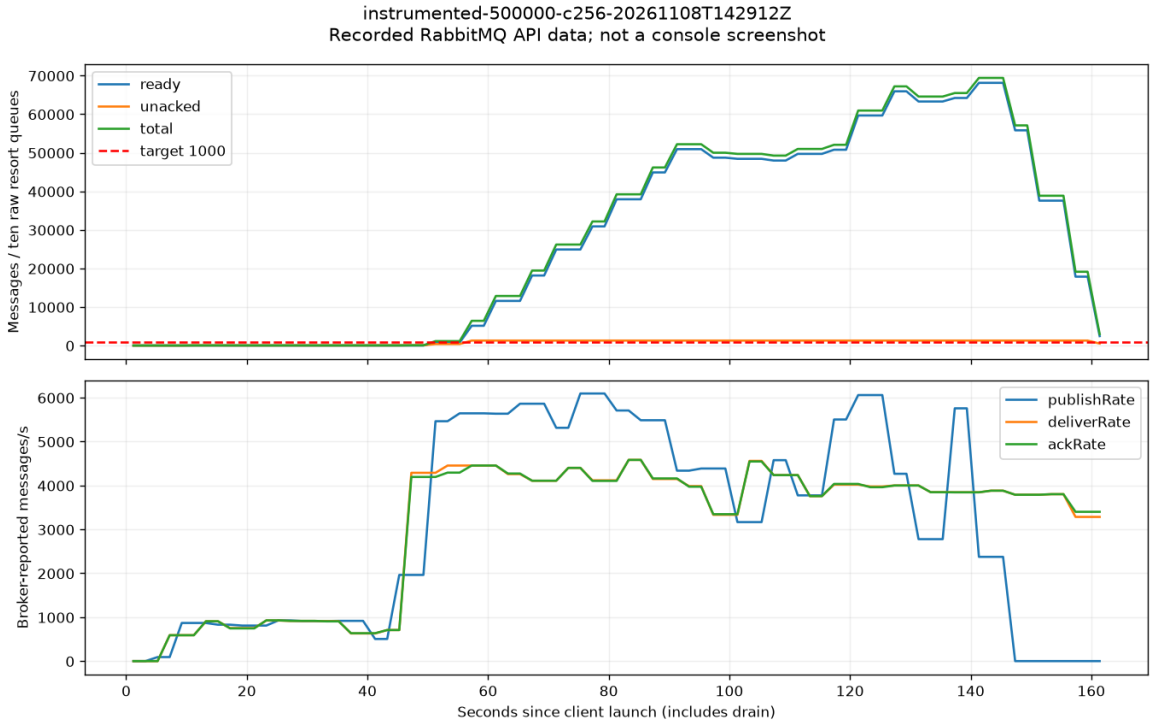
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 07 | 2 ingest(s), 500,000 events

Client threads=256; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

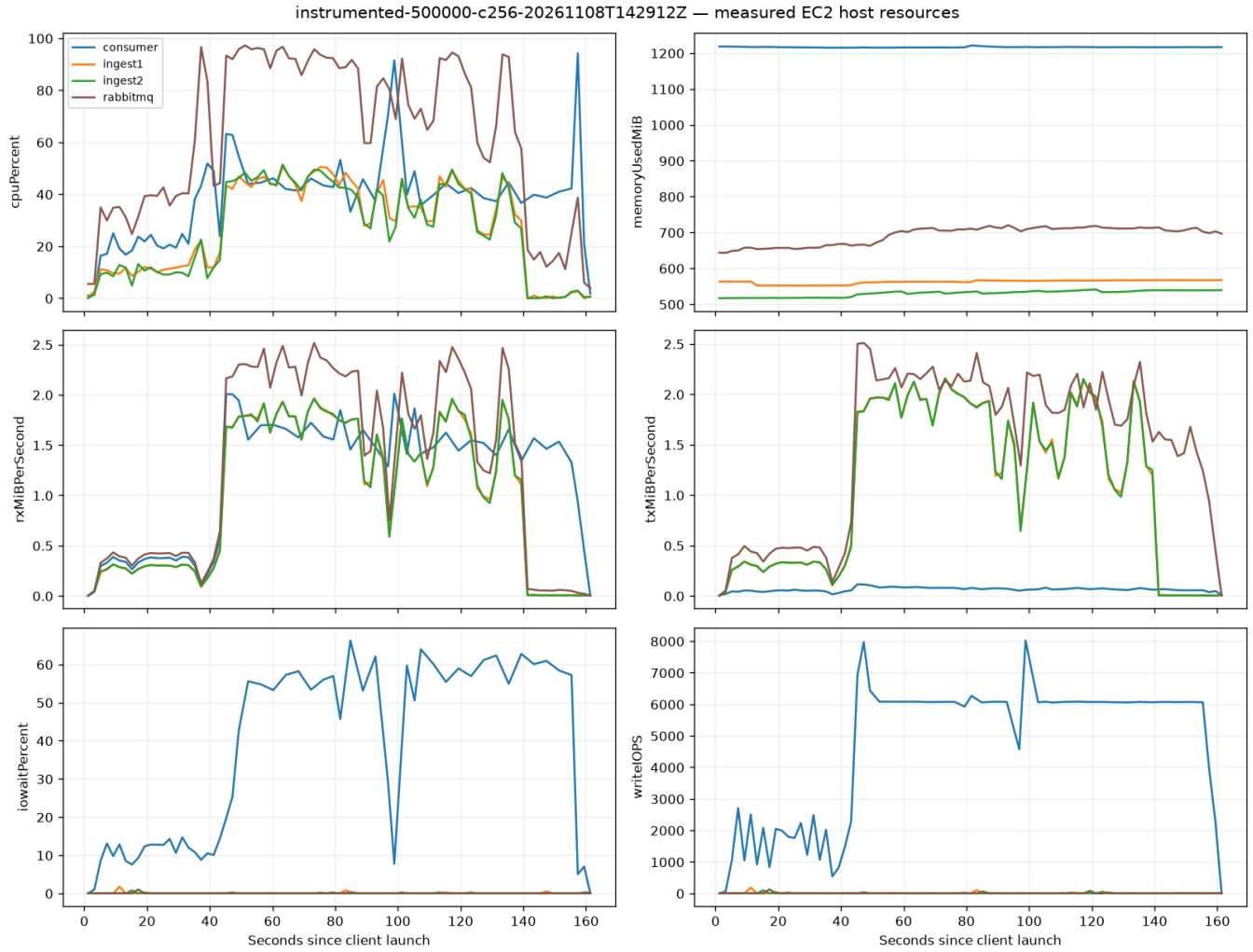
UTC 2026-11-08T14:29:26.256846+00:00 to 2026-11-08T14:32:07.986202+00:00

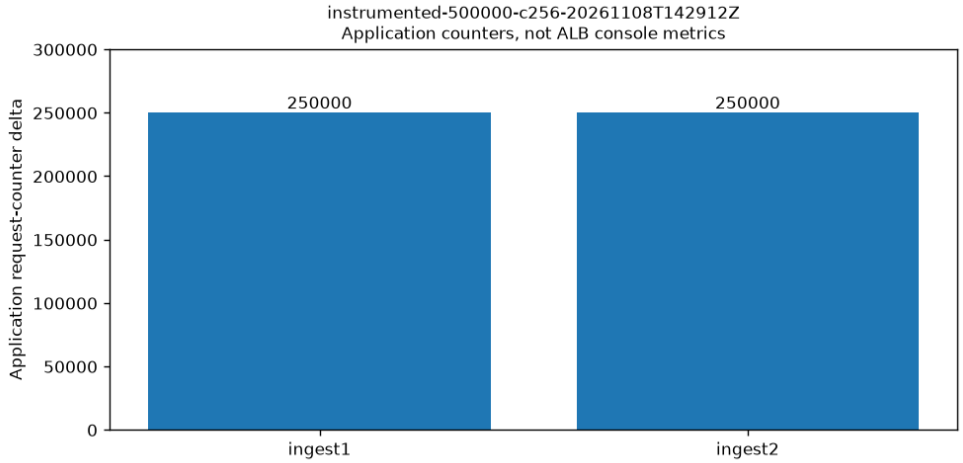
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	4,839.15 / 3,658.98 / 3,183.52
Ingestion / program runtime / drain seconds	136.650 / 160.362 / 20.386
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	52.01 / 96.41 / 272.08
Ready sampled mean / maximum	27,919.26 / 68,120
Unacked / Ready+Unacked maximum	1,280 / 69,400
Broker publish / delivery / ACK sampled mean per second	3,099.88 / 3,079.26 / 3,076.42
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	9,306.91
WebSocket updates / reconnections	904 / 0



Source: instrumented-500000-c256-20261108T142912Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 07 | resource use and distribution

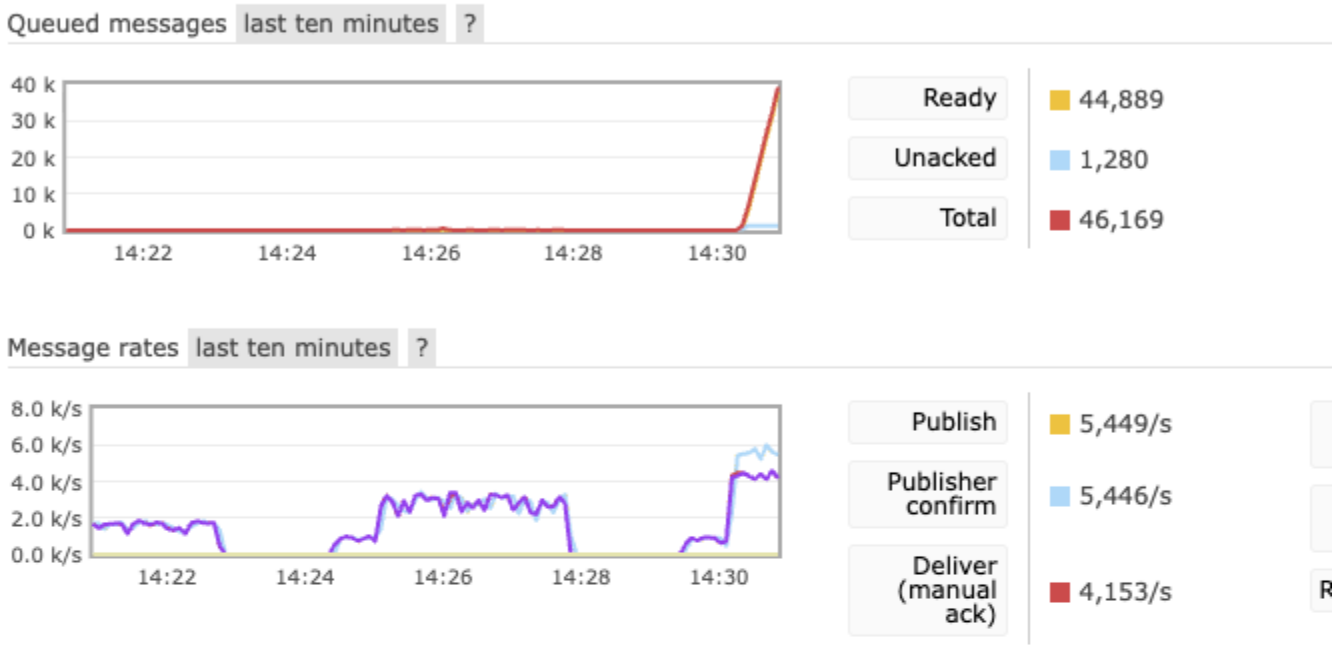




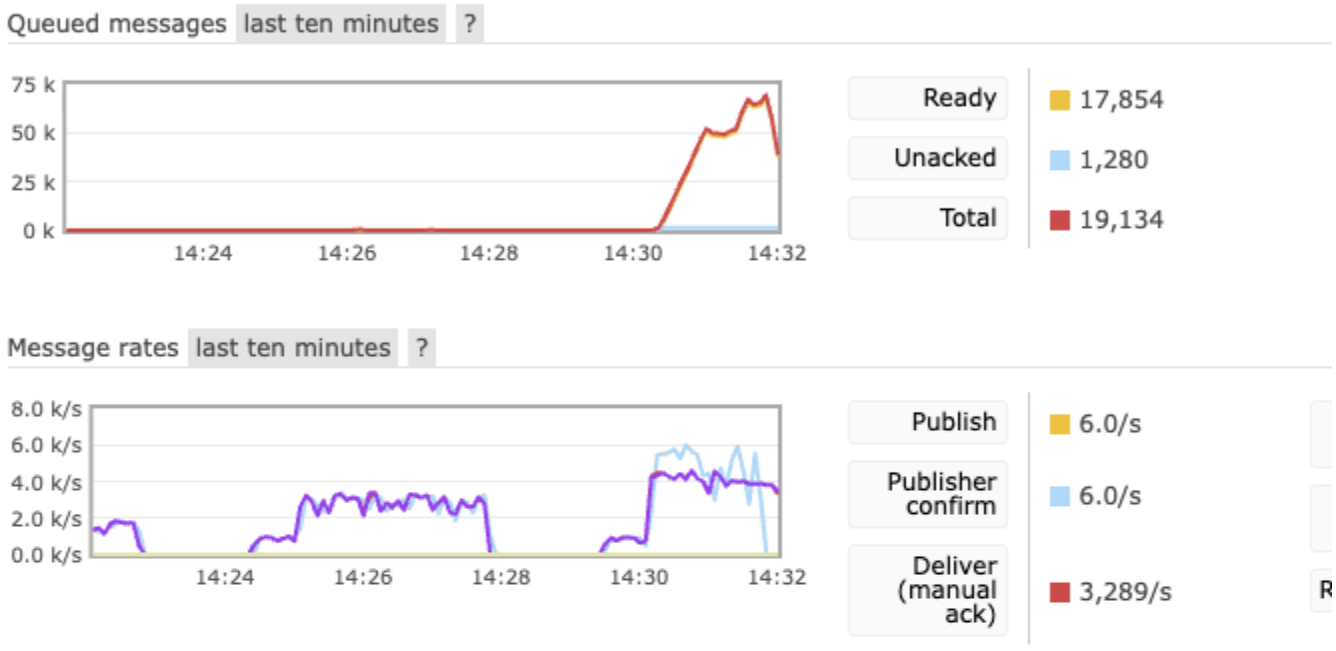
Collection gaps: consumer collection errors: {"timeout": 22}

Run 07 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:29:26.256846+00:00 to 2026-11-08T14:32:07.986202+00:00.



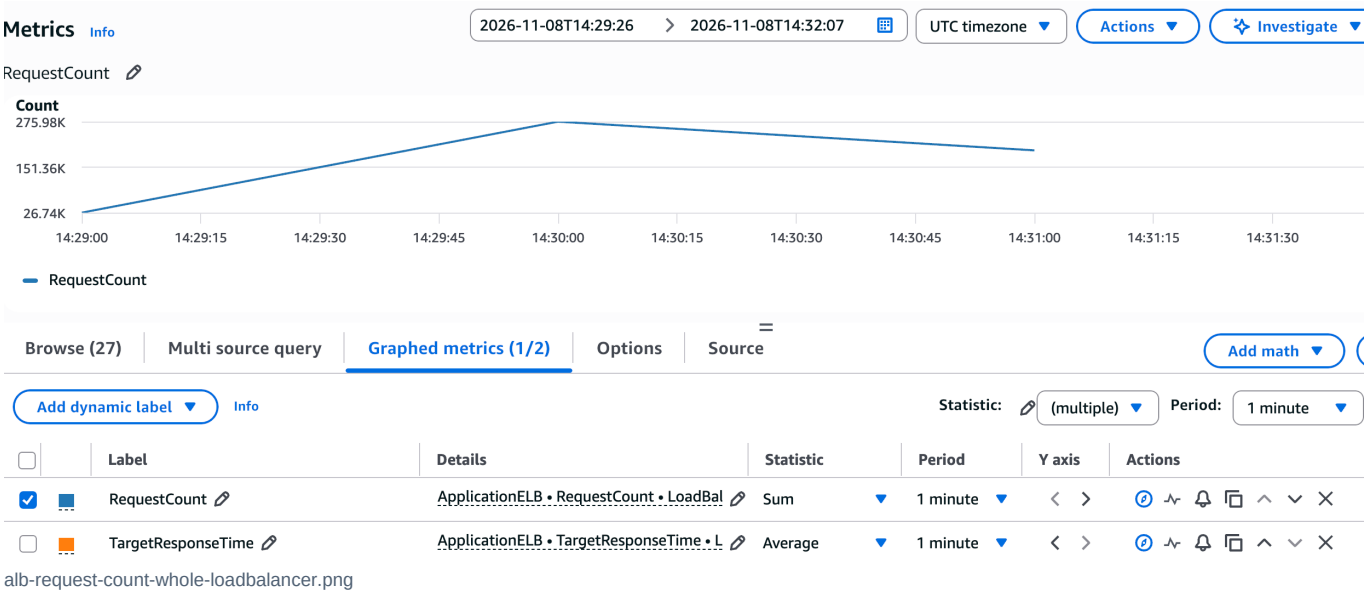
rabbitmq-20261108T143057065814Z.png | Original screenshot retained unchanged.



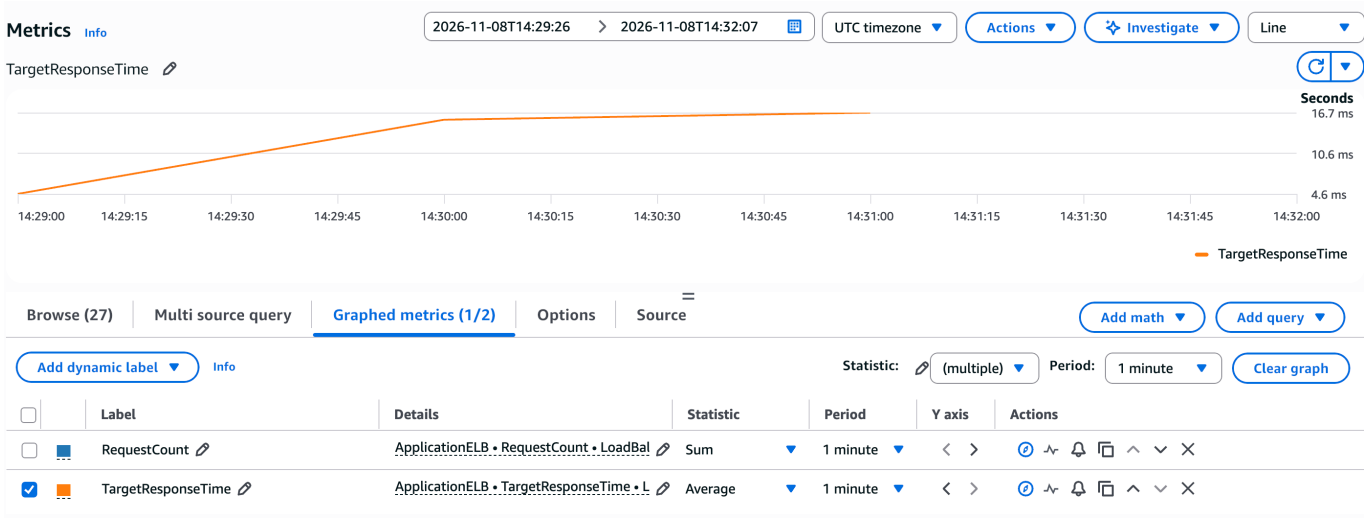
rabbitmq-20261108T143214353302Z-final.png | Original screenshot retained unchanged.

Run 07 | supplied AWS CloudWatch captures

Metric scope: Whole LoadBalancer; no AvailabilityZone or TargetGroup filter.



alb-request-count-whole-loadbalancer.png



alb-target-response-time-whole-loadbalancer.png

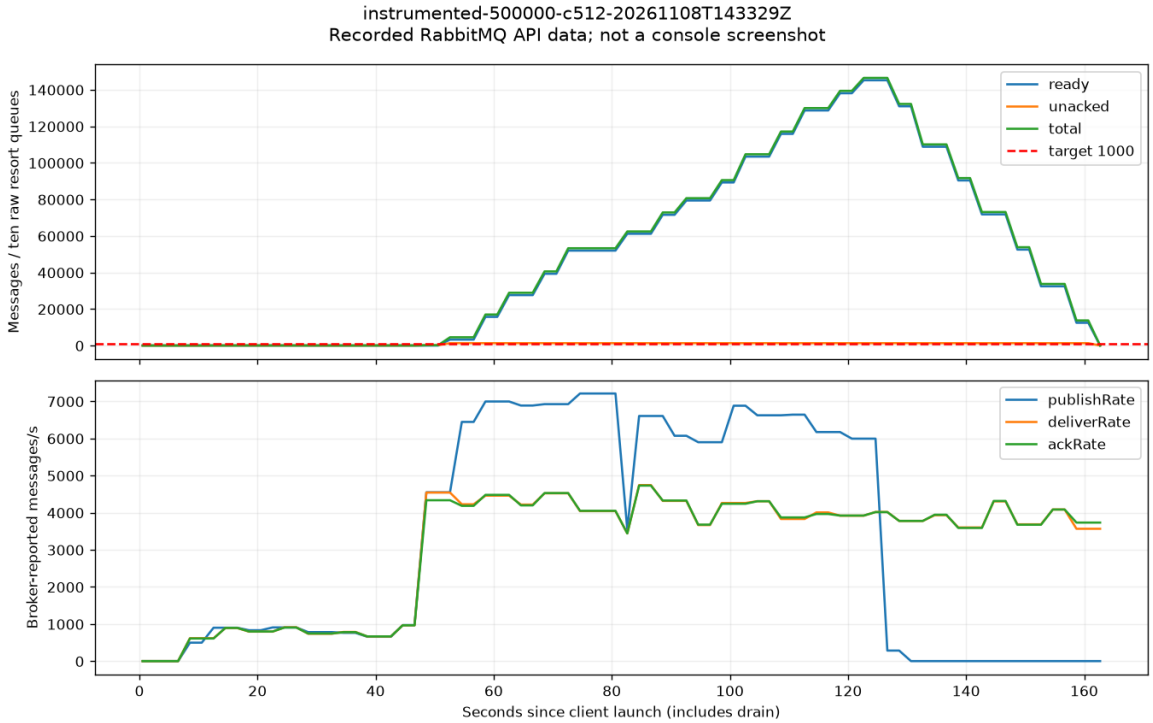
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 08 | 2 ingest(s), 500,000 events

Client threads=512; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

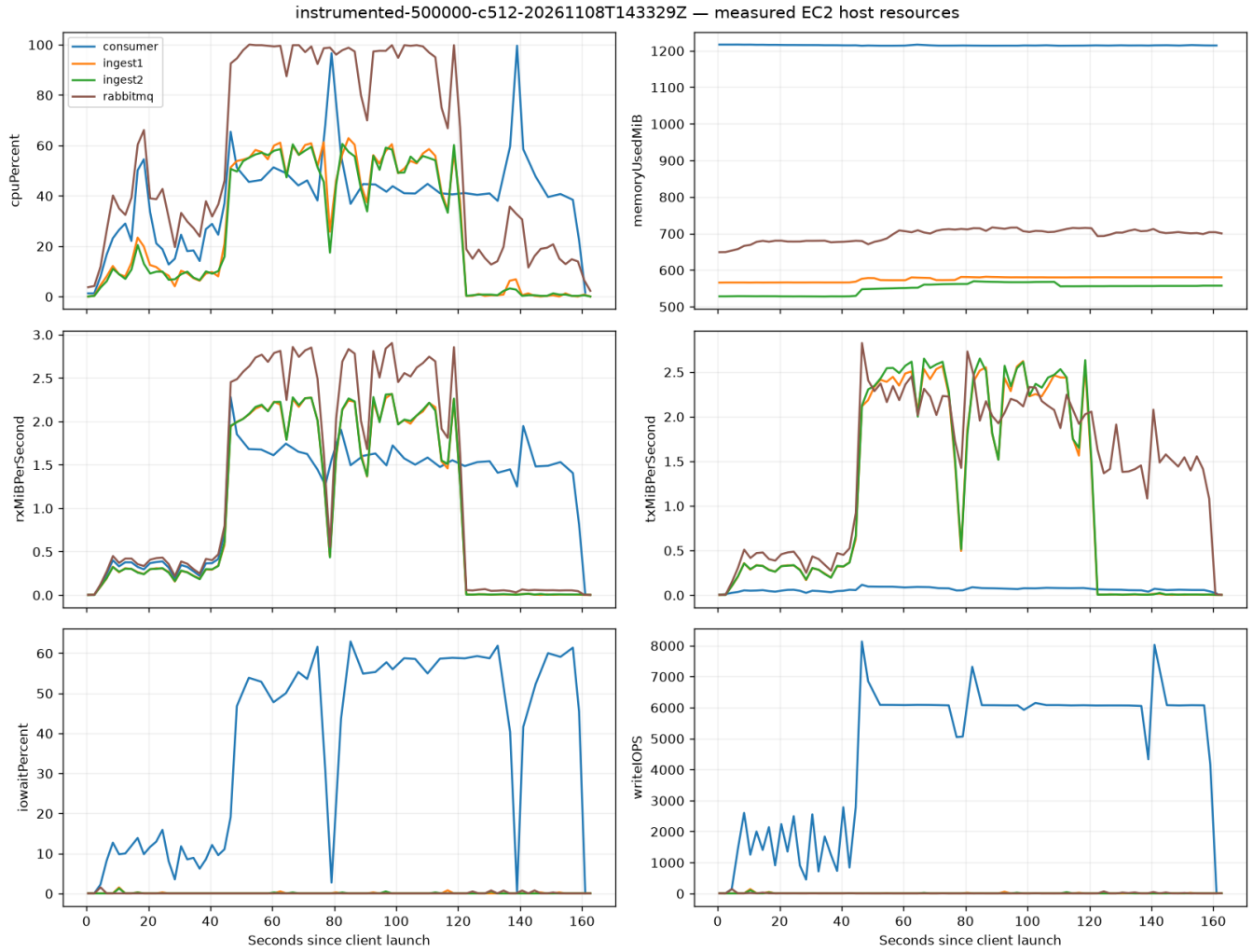
UTC 2026-11-08T14:33:45.429523+00:00 to 2026-11-08T14:36:28.147351+00:00

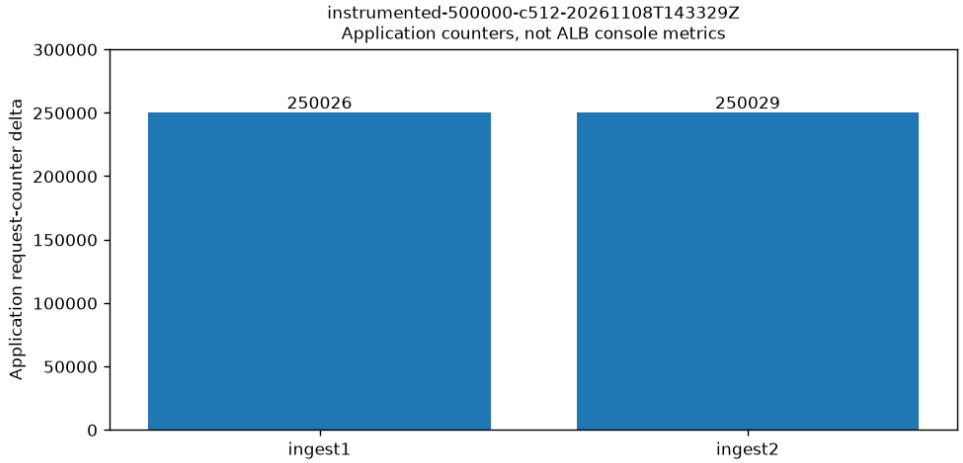
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	6,135.42 / 4,267.82 / 3,168.37
Ingestion / program runtime / drain seconds	117.156 / 161.400 / 40.631
Successful / final failed / application retries	500,000 / 0 / 55
HTTP mean / p95 / p99 ms	80.52 / 172.69 / 314.39
Ready sampled mean / maximum	49,459.70 / 145,164
Unacked / Ready+Unacked maximum	1,280 / 146,444
Broker publish / delivery / ACK sampled mean per second	3,217.80 / 3,067.16 / 3,065.21
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	16,665.74
WebSocket updates / reconnections	903 / 0



Source: instrumented-500000-c512-20261108T143329Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 08 | resource use and distribution

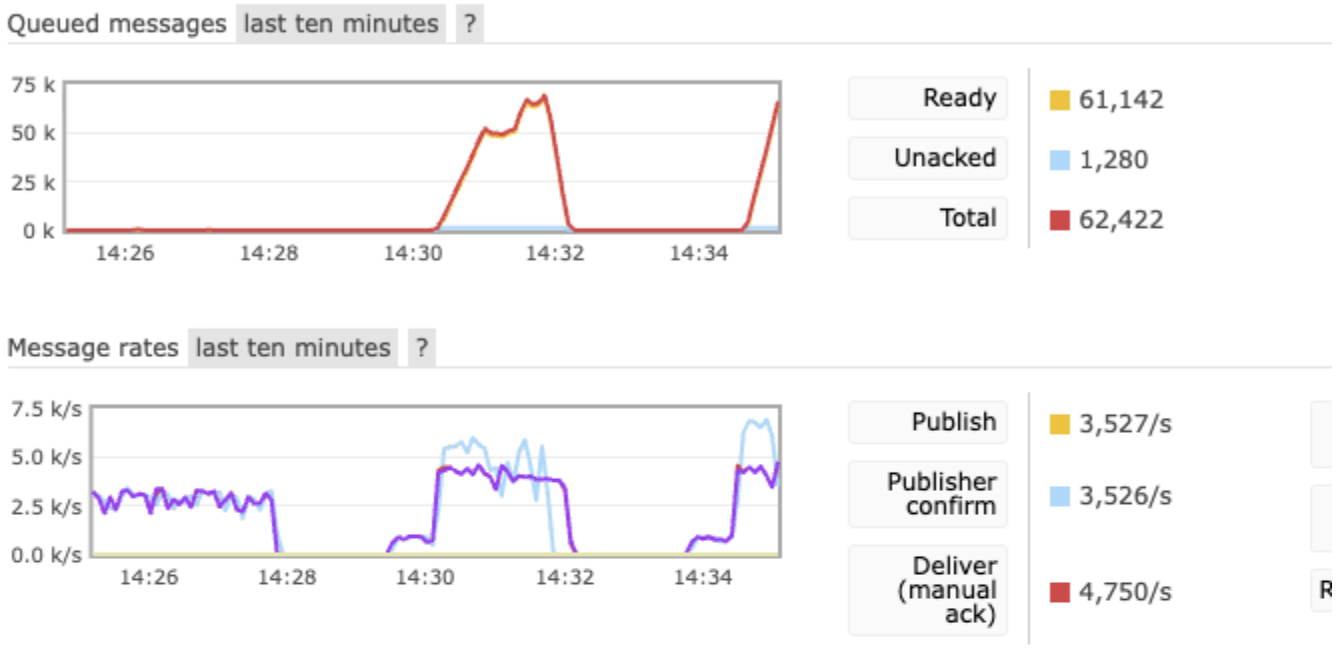




Collection gaps: consumer collection errors: {"timeout": 16}

Run 08 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:33:45.429523+00:00 to 2026-11-08T14:36:28.147351+00:00.



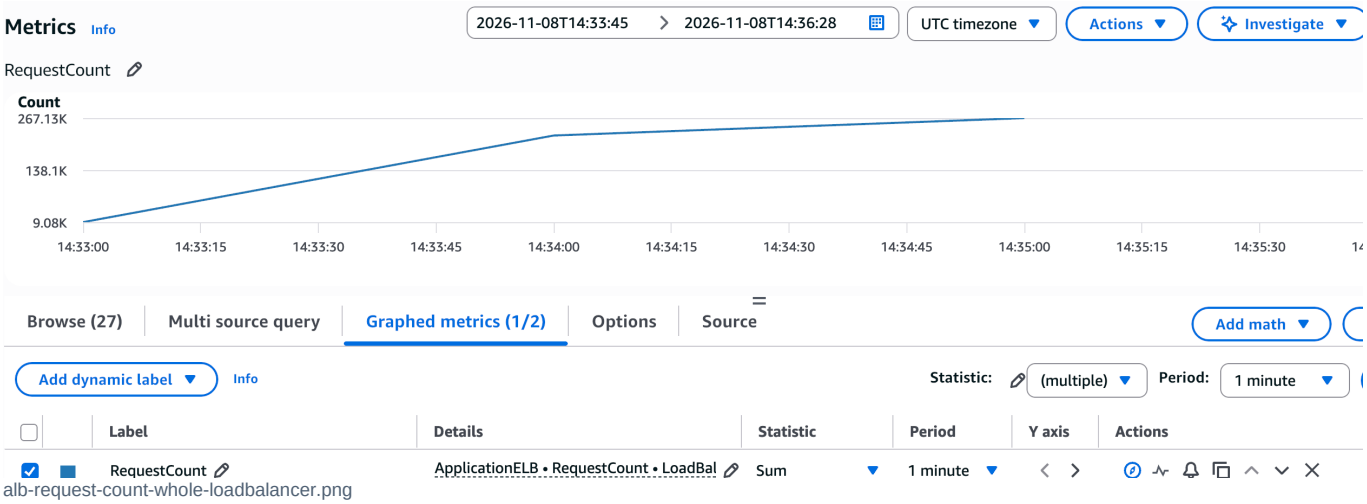
rabbitmq-20261108T143515830322Z.png | Original screenshot retained unchanged.



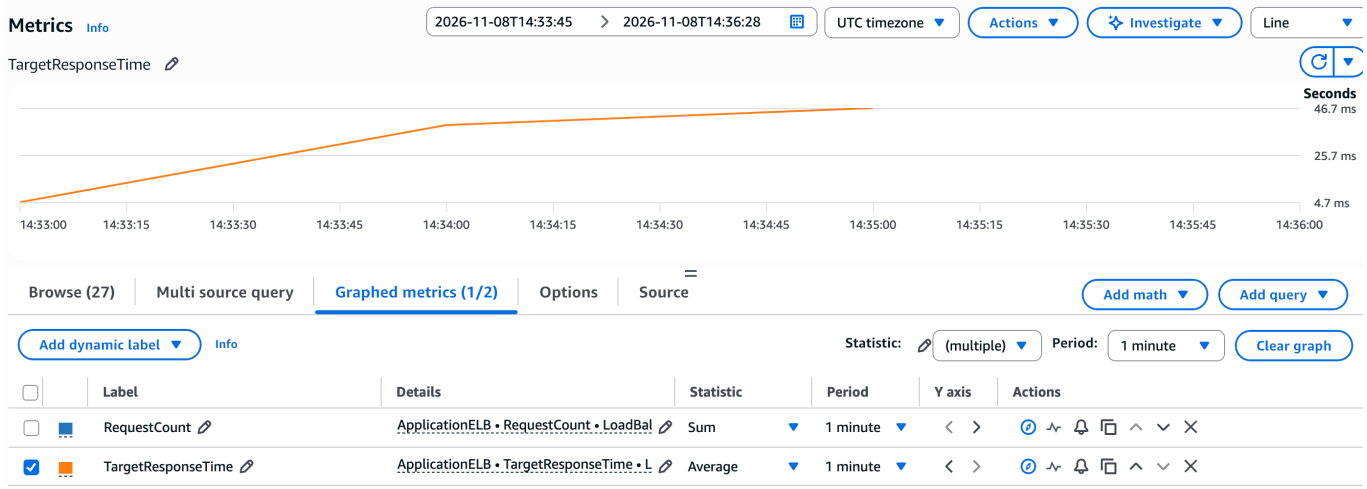
rabbitmq-20261108T143634584570Z-final.png | Original screenshot retained unchanged.

Run 08 | supplied AWS CloudWatch captures

Metric scope: Whole LoadBalancer; no AvailabilityZone or TargetGroup filter.



alb-request-count-whole-loadbalancer.png



alb-target-response-time-whole-loadbalancer.png

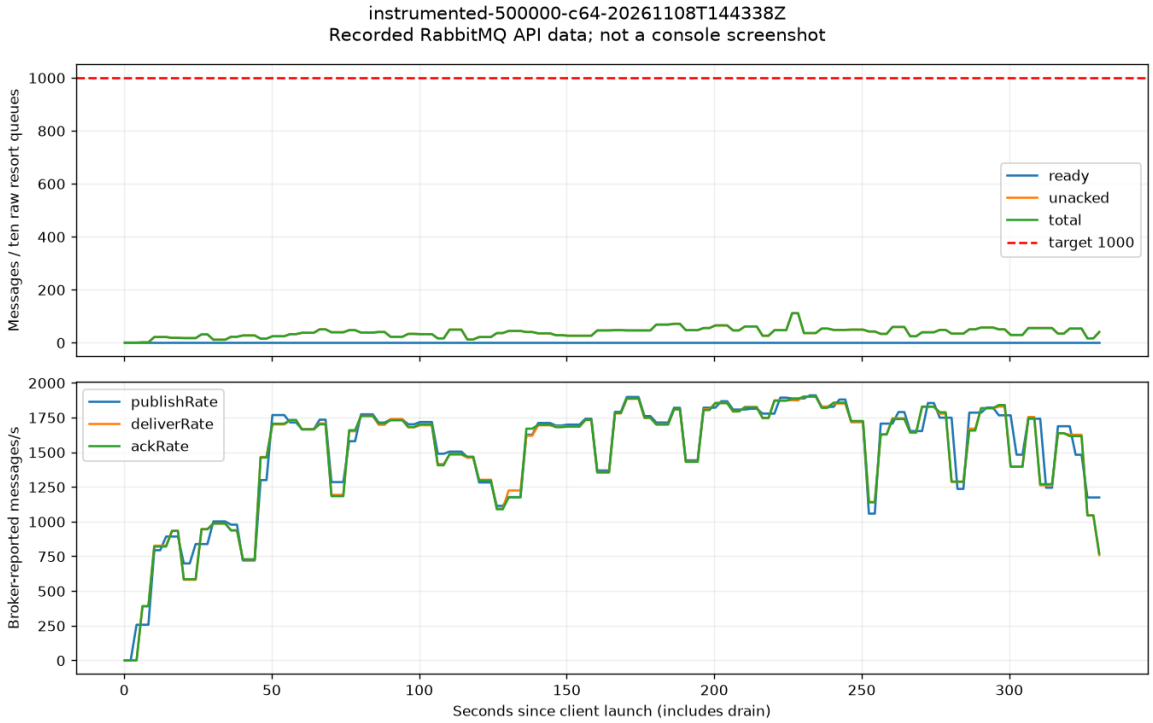
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 09 | 4 ingest(s), 500,000 events

Client threads=64; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

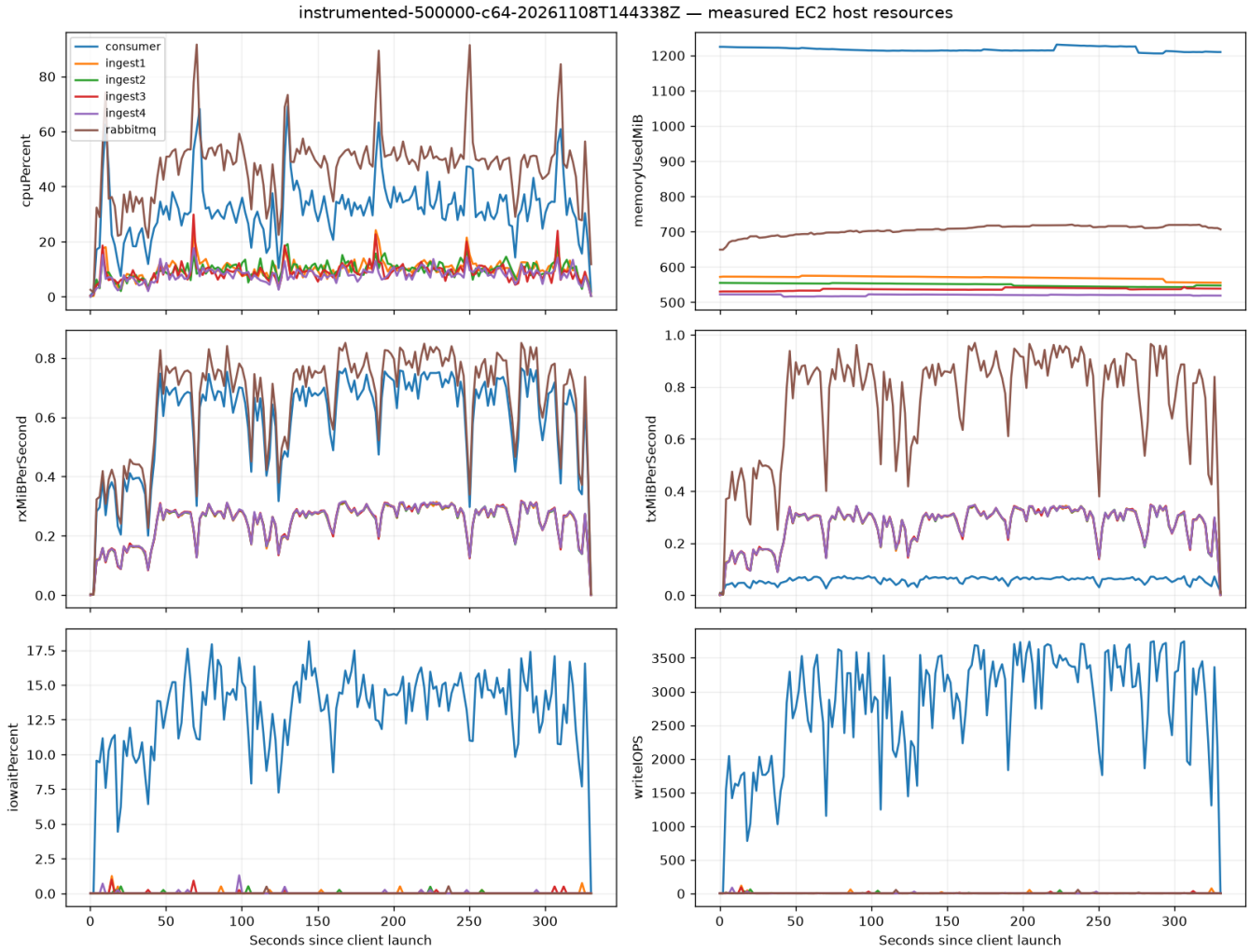
UTC 2026-11-08T14:43:53.547961+00:00 to 2026-11-08T14:49:25.707862+00:00

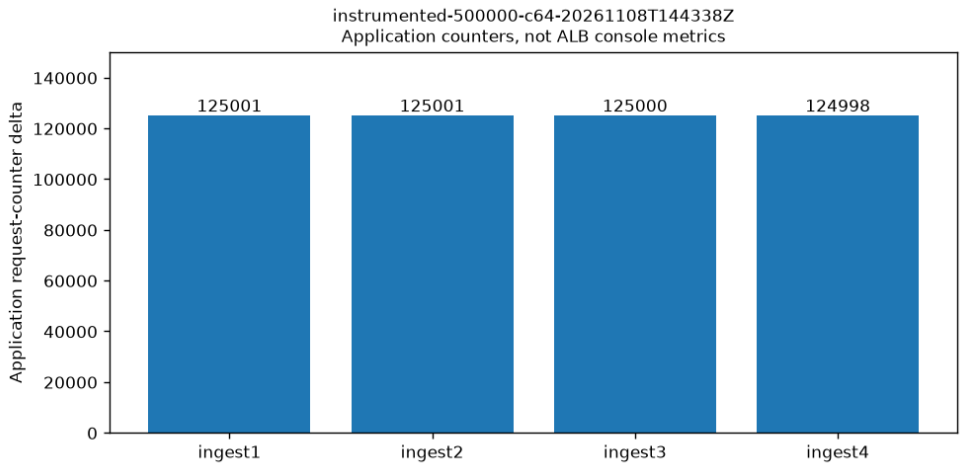
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	1,631.83 / 1,536.61 / 1,527.74
Ingestion / program runtime / drain seconds	325.391 / 330.596 / 0.844
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	39.13 / 56.60 / 155.48
Ready sampled mean / maximum	0.00 / 0
Unacked / Ready+Unacked maximum	111 / 111
Broker publish / delivery / ACK sampled mean per second	1,509.88 / 1,493.06 / 1,492.61
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	18.87
WebSocket updates / reconnections	905 / 0



Source: instrumented-500000-c64-20261108T144338Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 09 | resource use and distribution





Collection gaps: None detected.

Run 09 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:43:53.547961+00:00 to 2026-11-08T14:49:25.707862+00:00.



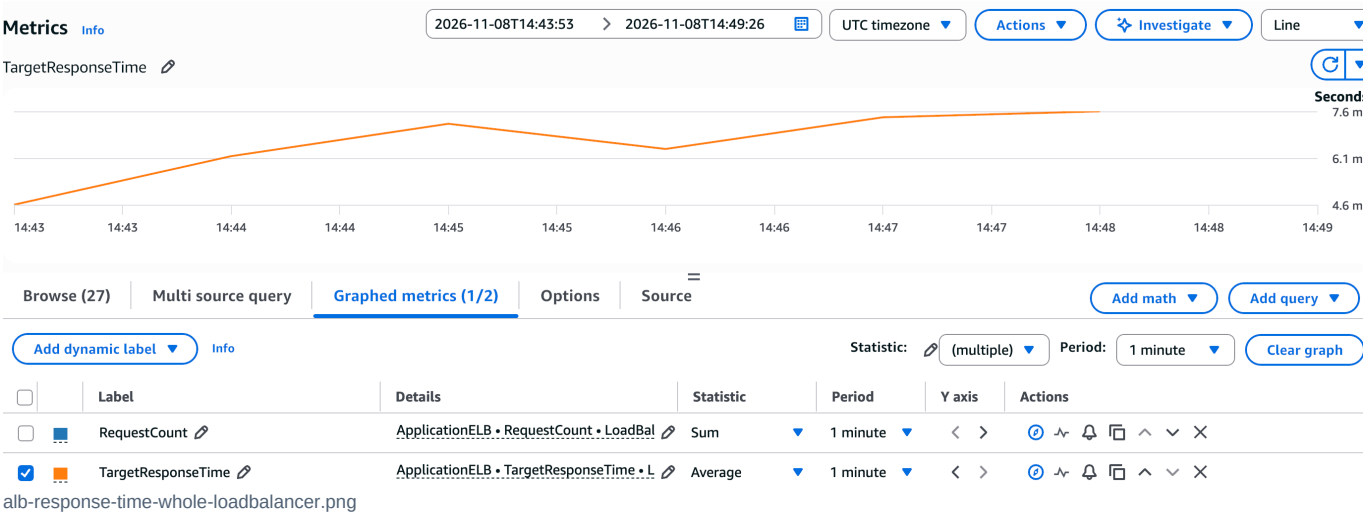
rabbitmq-20261108T144654750139Z.png | Original screenshot retained unchanged.



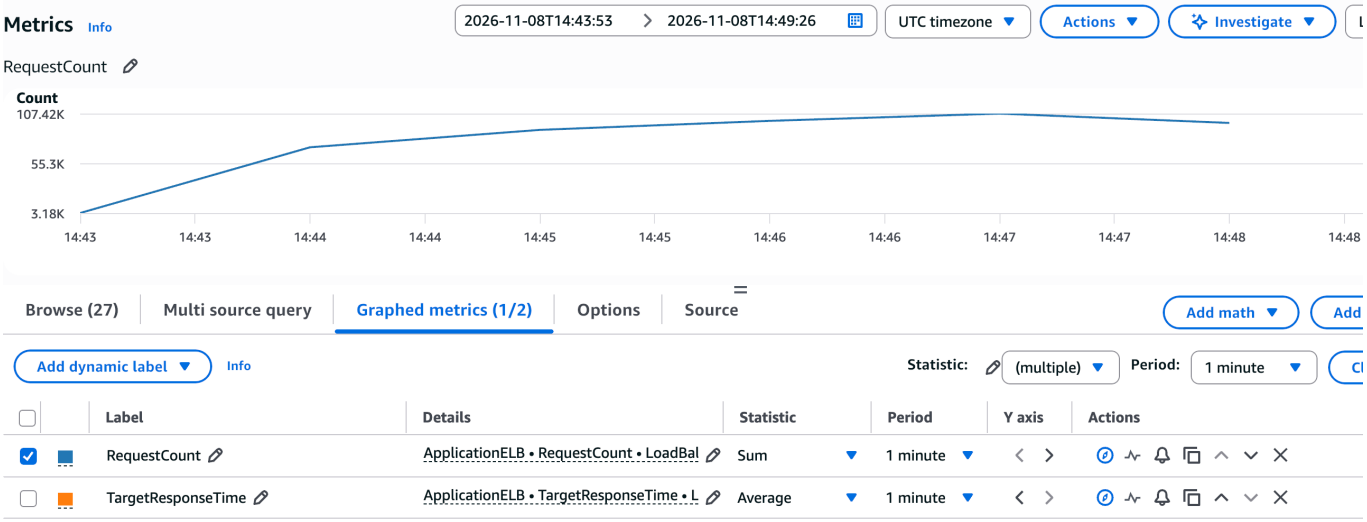
rabbitmq-20261108T144931869234Z-final.png | Original screenshot retained unchanged.

Run 09 | supplied AWS CloudWatch captures

Metric scope: LoadBalancer aggregate.



alb-response-time-whole-loadbalancer.png



alb-request-count-whole-loadbalancer.png

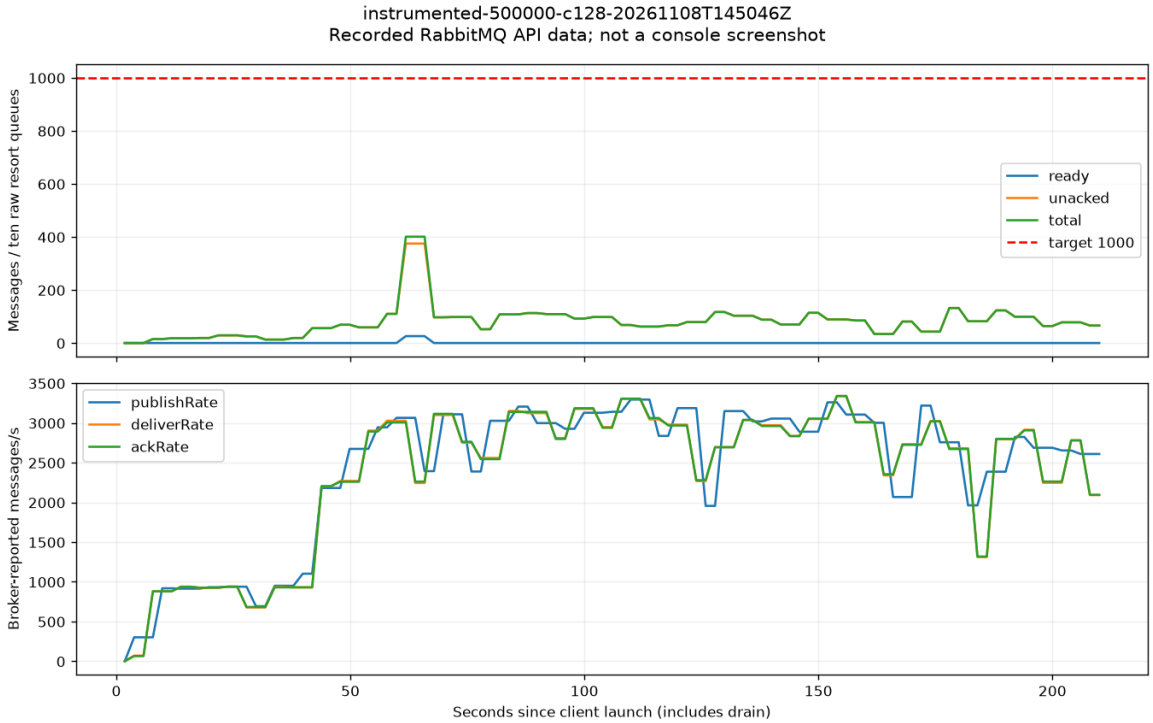
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 10 | 4 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

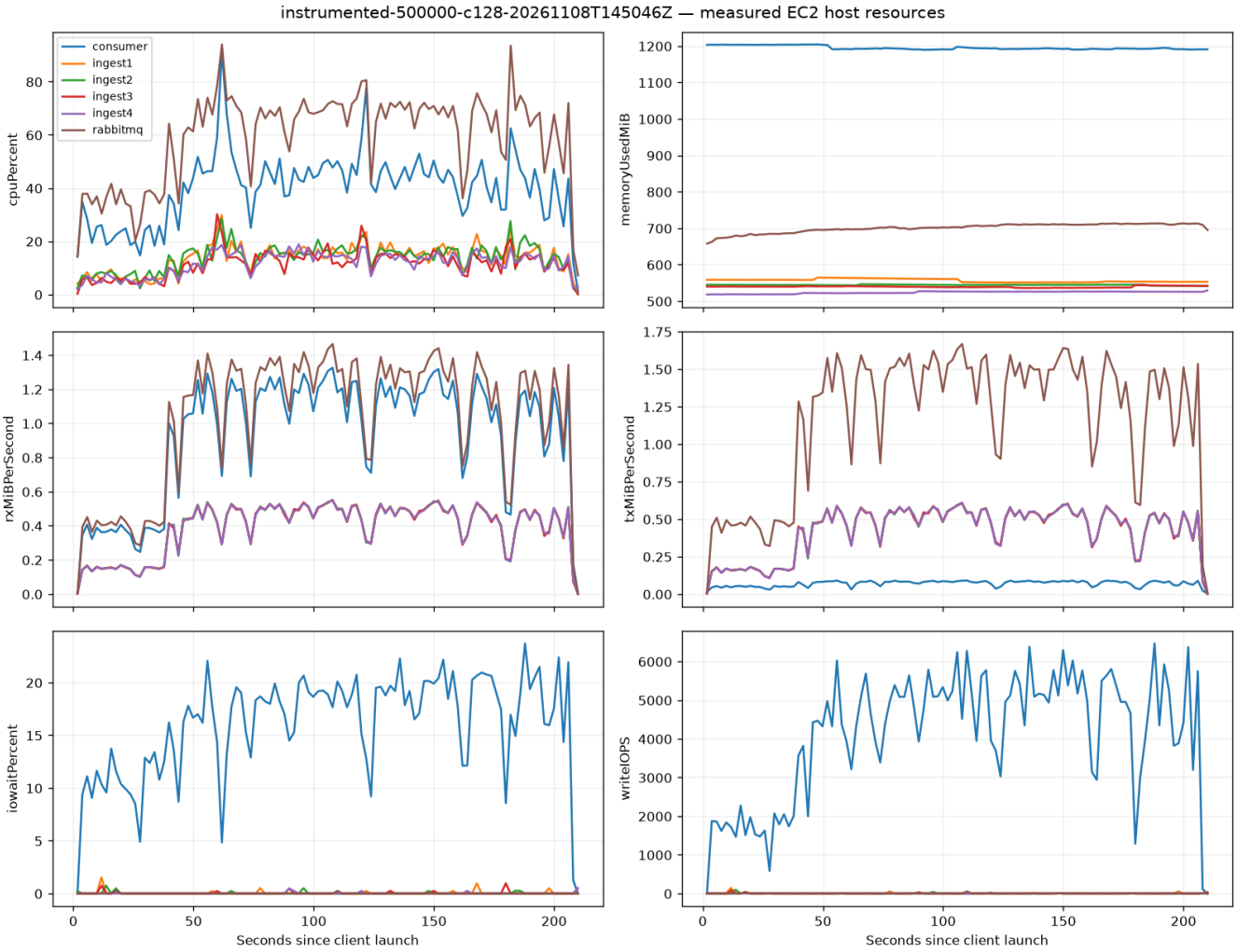
UTC 2026-11-08T14:51:01.754633+00:00 to 2026-11-08T14:54:31.871816+00:00

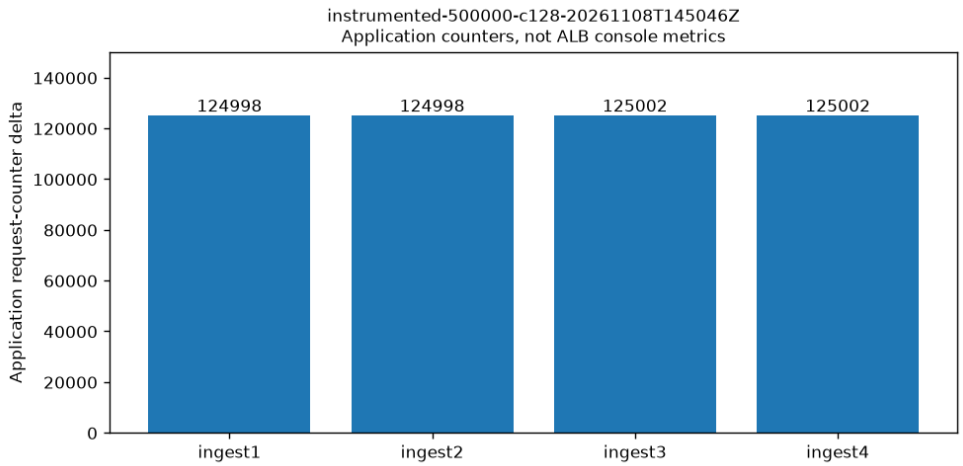
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	2,781.22 / 2,443.44 / 2,433.20
Ingestion / program runtime / drain seconds	204.630 / 208.626 / 0.798
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	45.35 / 71.79 / 202.31
Ready sampled mean / maximum	0.74 / 26
Unacked / Ready+Unacked maximum	375 / 401
Broker publish / delivery / ACK sampled mean per second	2,420.33 / 2,372.64 / 2,372.49
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	24.43
WebSocket updates / reconnections	910 / 0



Source: instrumented-500000-c128-20261108T145046Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 10 | resource use and distribution

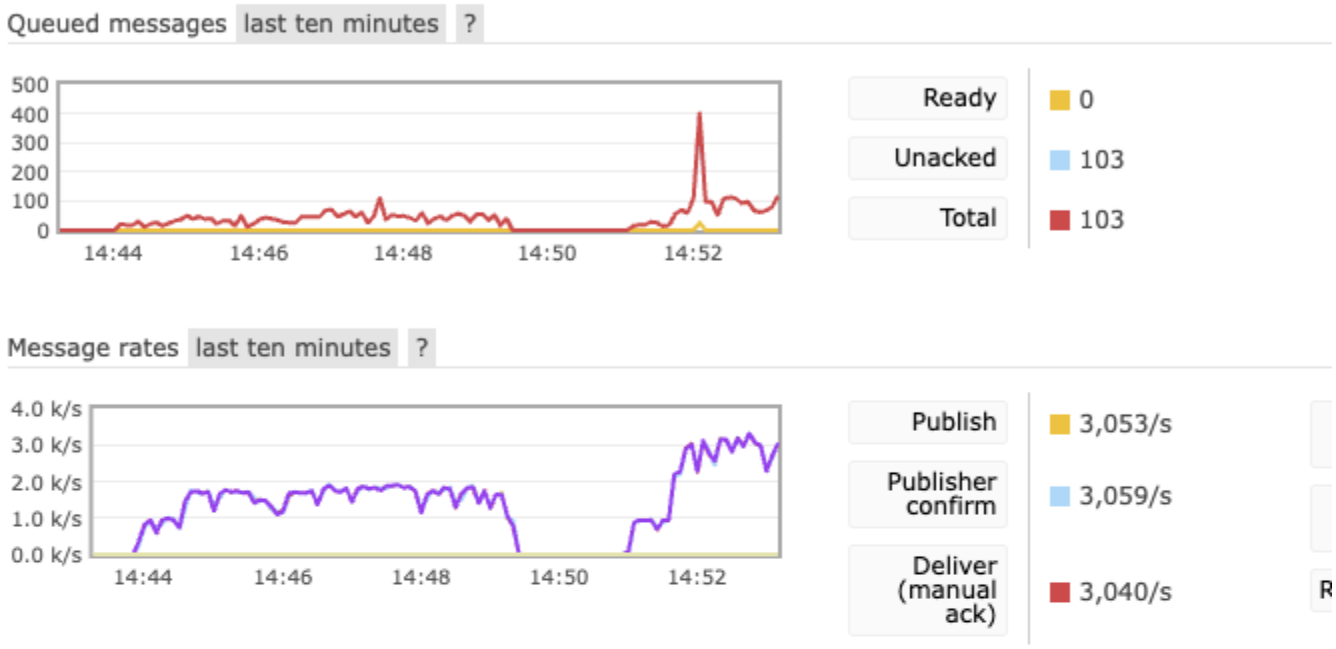




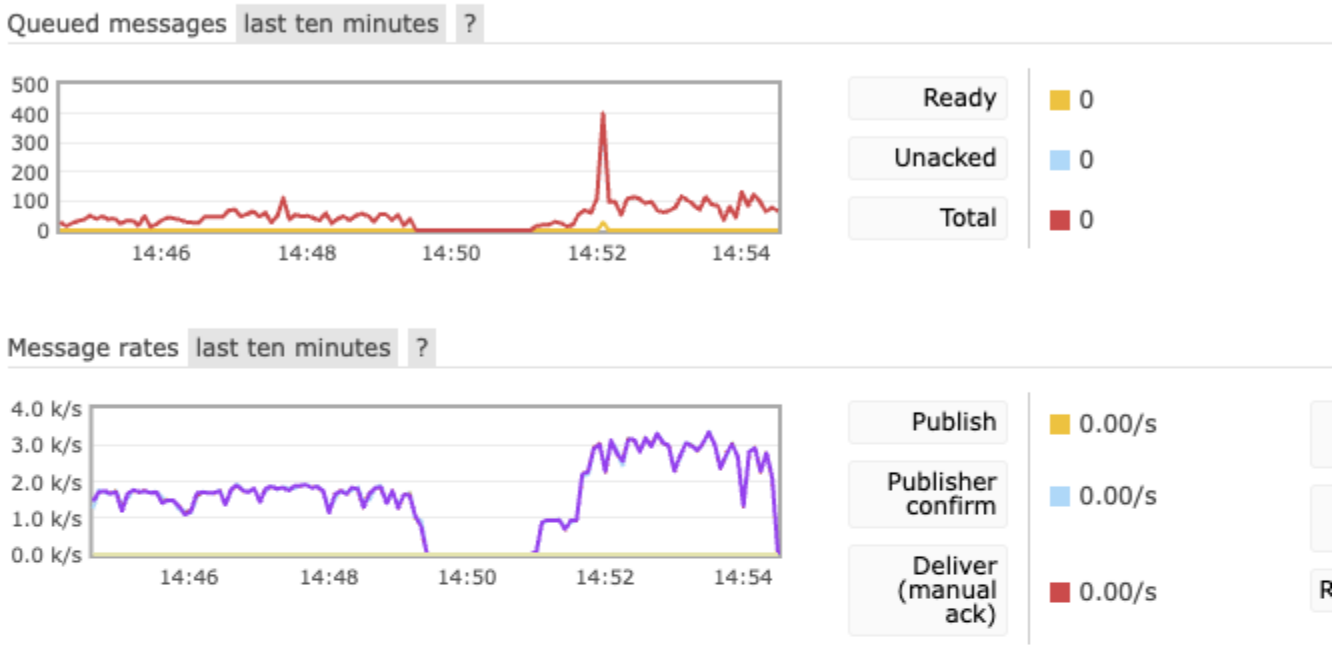
Collection gaps: None detected.

Run 10 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:51:01.754633+00:00 to 2026-11-08T14:54:31.871816+00:00.



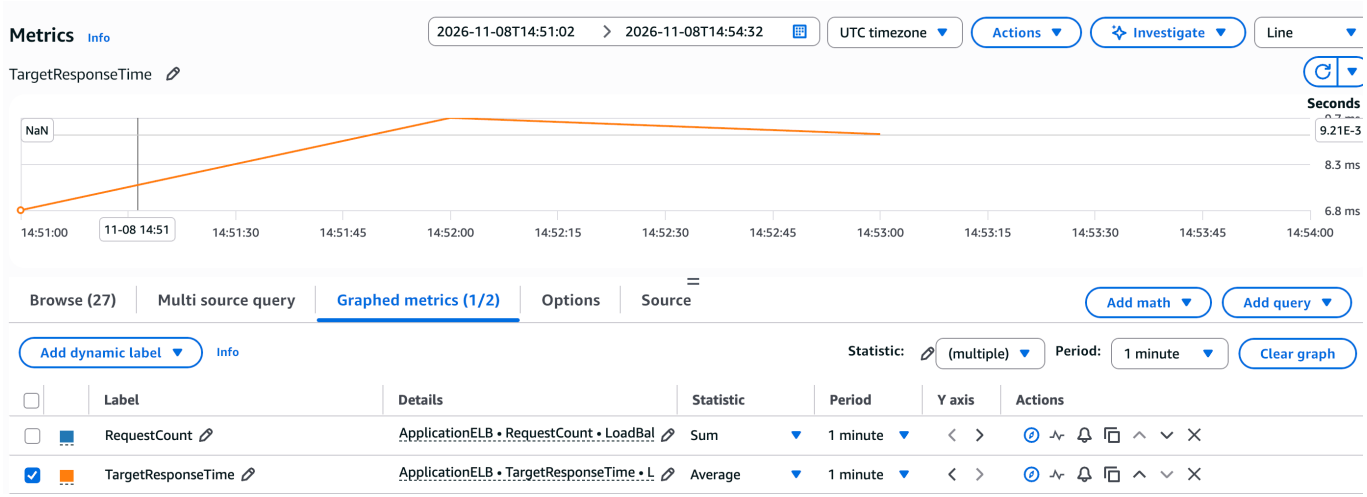
rabbitmq-20261108T145317285102Z.png | Original screenshot retained unchanged.



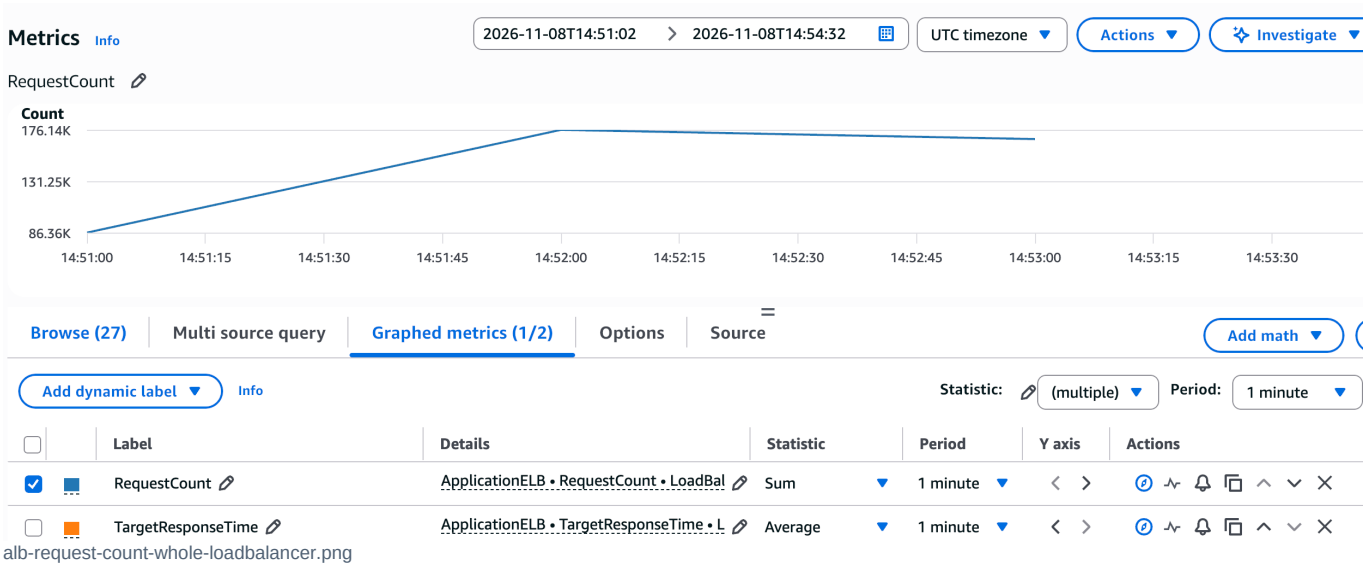
rabbitmq-20261108T145438041284Z-final.png | Original screenshot retained unchanged.

Run 10 | supplied AWS CloudWatch captures

Metric scope: LoadBalancer aggregate.



alb-response-time-whole-loadbalancer.png



alb-request-count-whole-loadbalancer.png

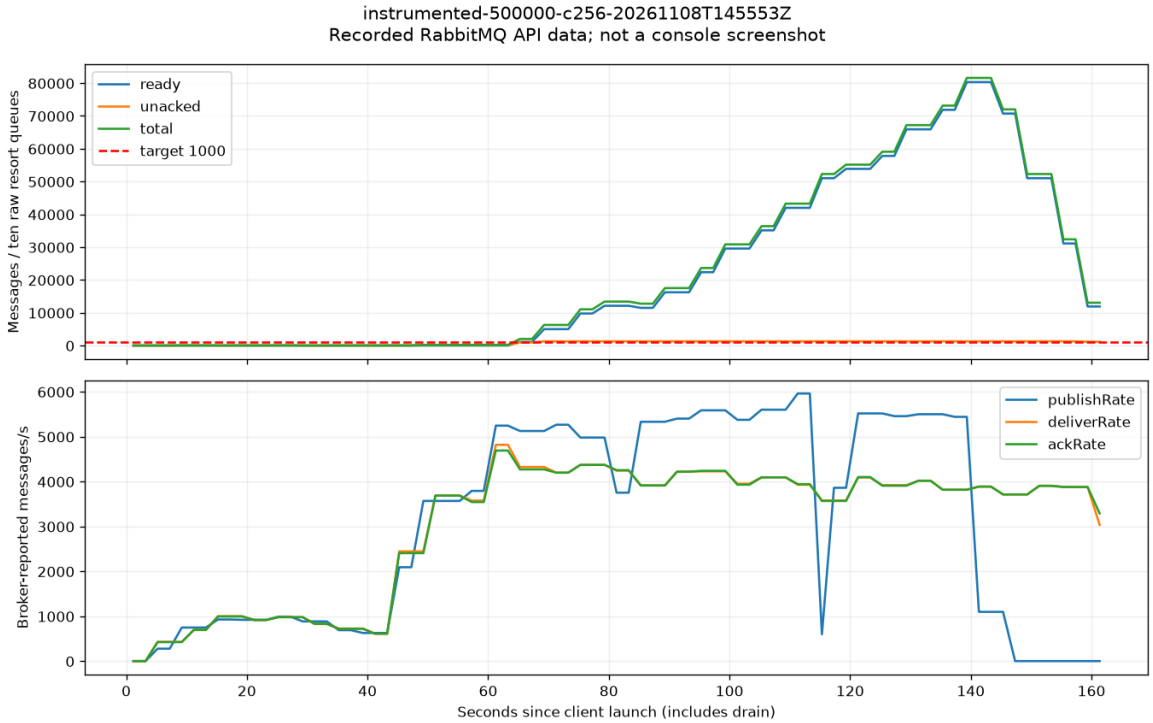
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 11 | 4 ingest(s), 500,000 events

Client threads=256; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

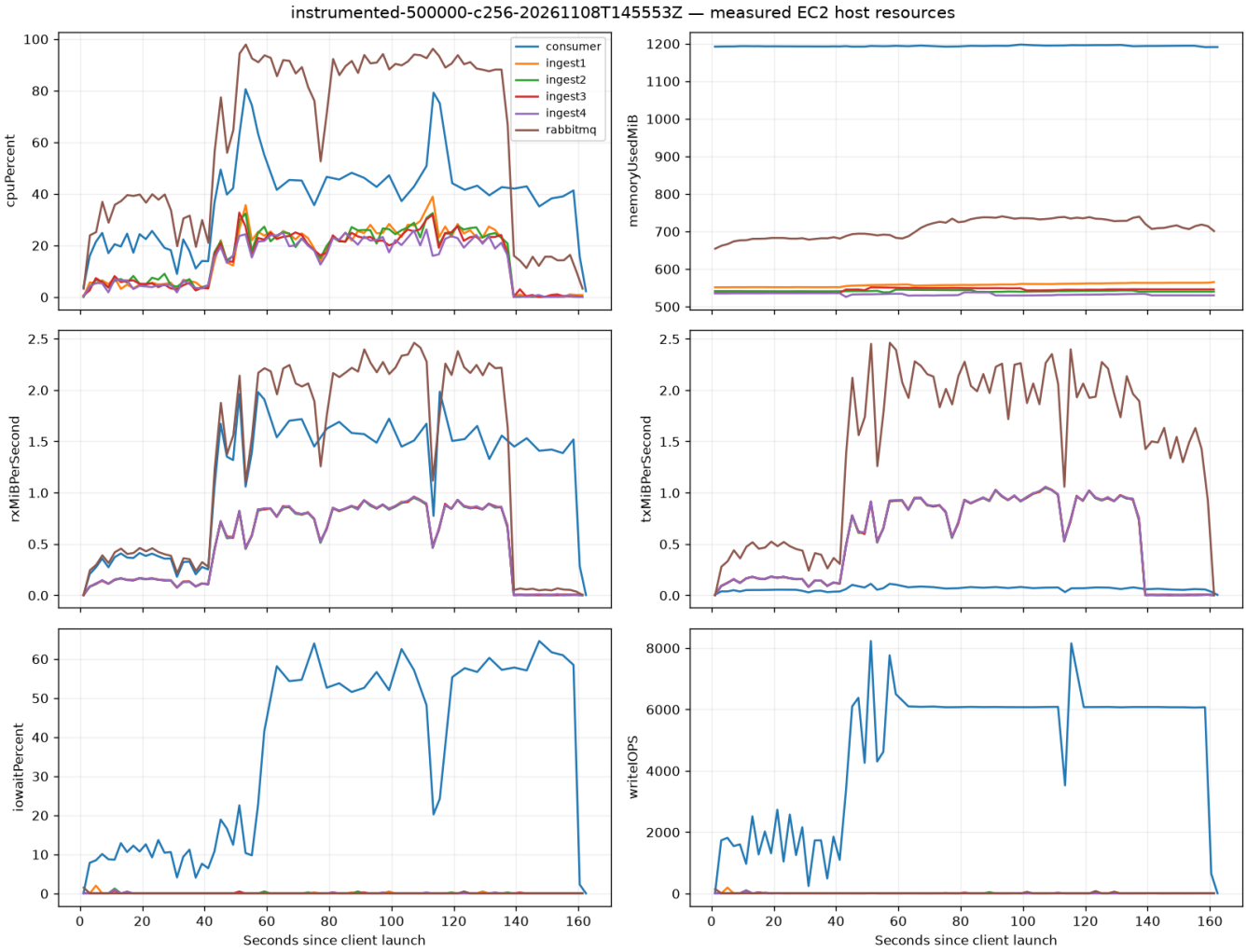
UTC 2026-11-08T14:56:09.658997+00:00 to 2026-11-08T14:58:52.520694+00:00

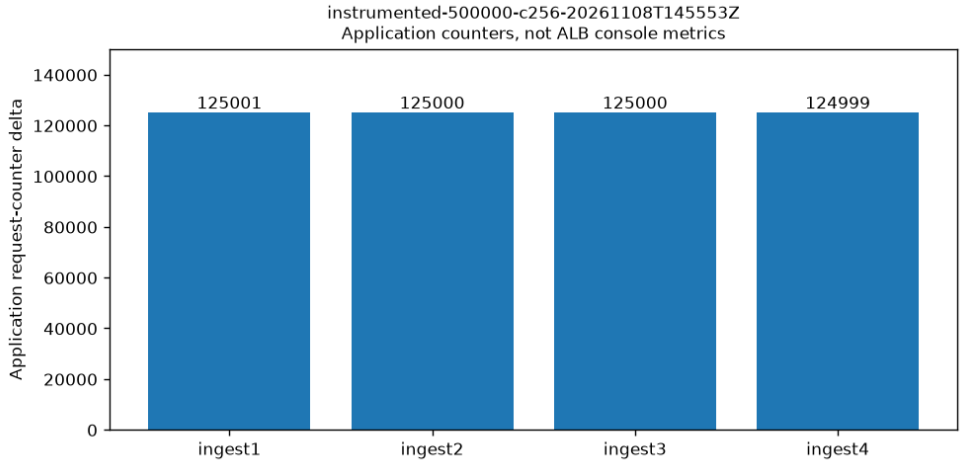
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	4,924.34 / 3,711.86 / 3,154.89
Ingestion / program runtime / drain seconds	134.703 / 161.983 / 23.719
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	51.12 / 82.12 / 194.77
Ready sampled mean / maximum	22,422.90 / 80,309
Unacked / Ready+Unacked maximum	1,280 / 81,589
Broker publish / delivery / ACK sampled mean per second	3,082.02 / 3,037.82 / 3,033.15
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	7,641.57
WebSocket updates / reconnections	913 / 0



Source: instrumented-500000-c256-20261108T145553Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 11 | resource use and distribution

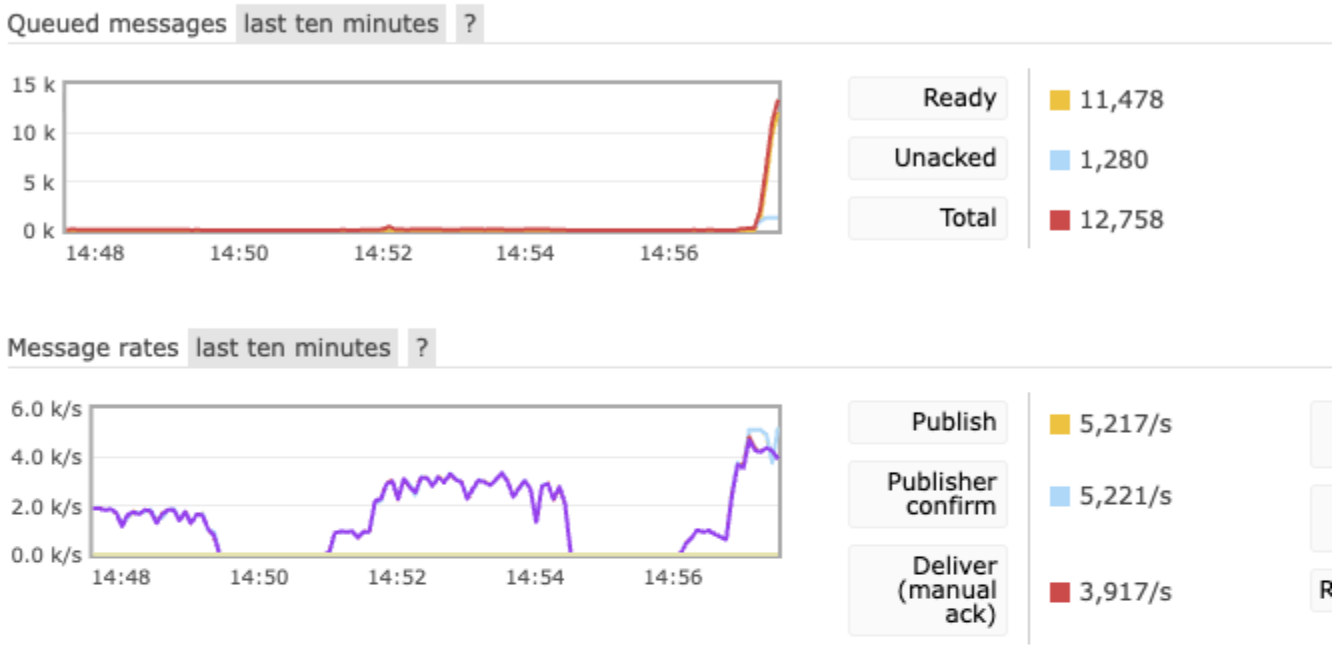




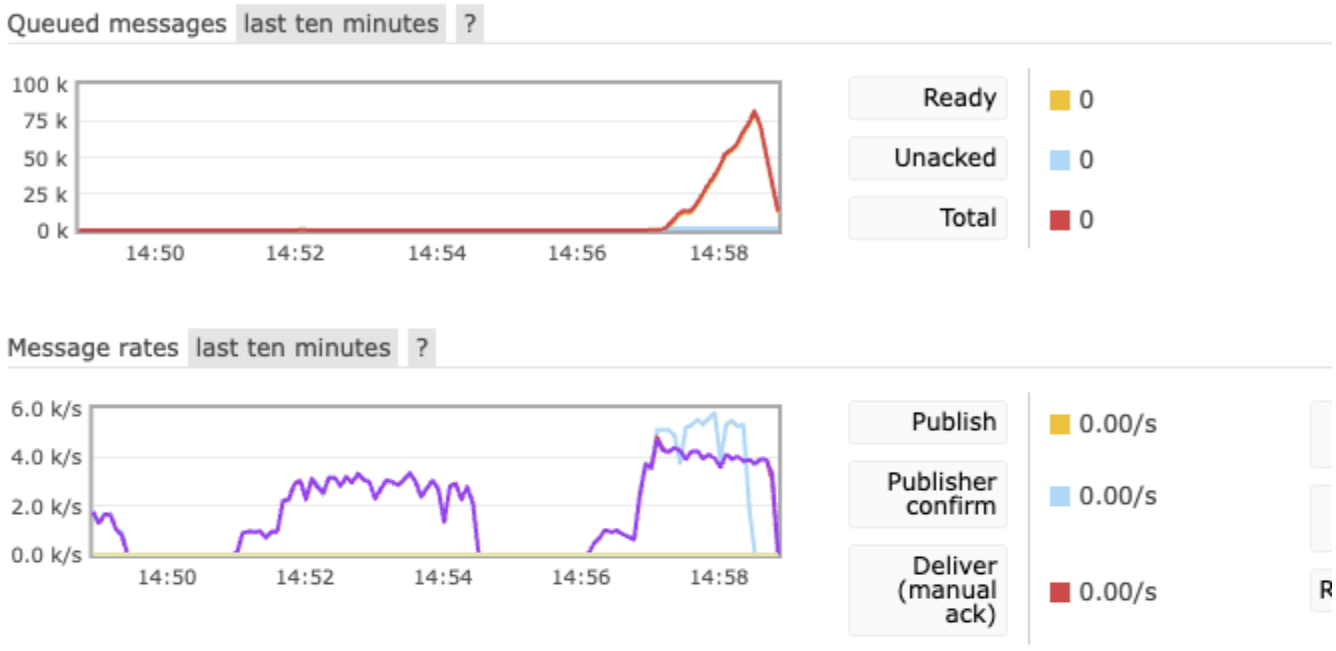
Collection gaps: consumer collection errors: {"timeout": 21}

Run 11 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T14:56:09.658997+00:00 to 2026-11-08T14:58:52.520694+00:00.



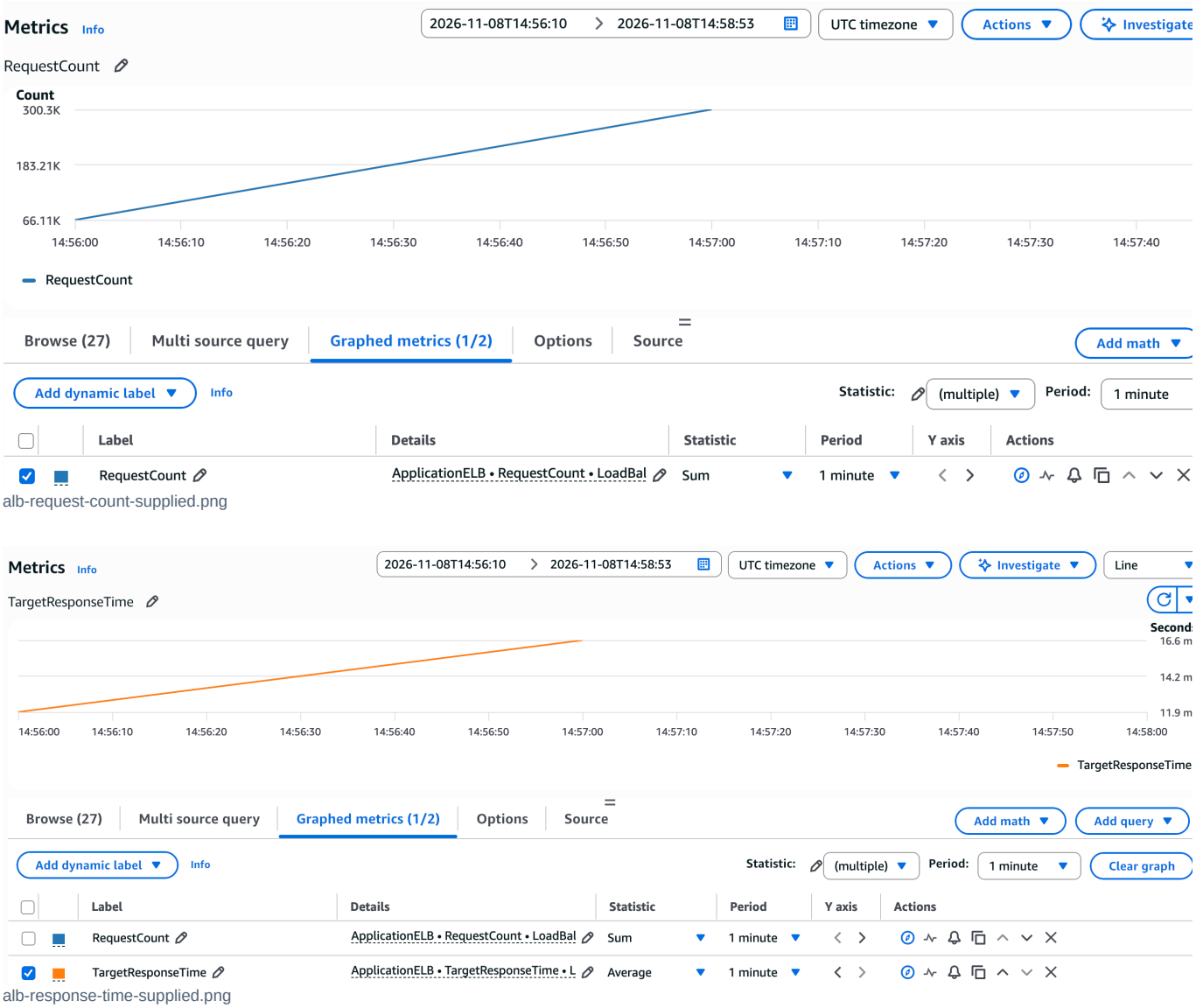
rabbitmq-20261108T145739597113Z.png | Original screenshot retained unchanged.



rabbitmq-20261108T145858916926Z-final.png | Original screenshot retained unchanged.

Run 11 | supplied AWS CloudWatch captures

Metric scope: LoadBalancer aggregate.



alb-request-count-supplied.png

alb-response-time-supplied.png

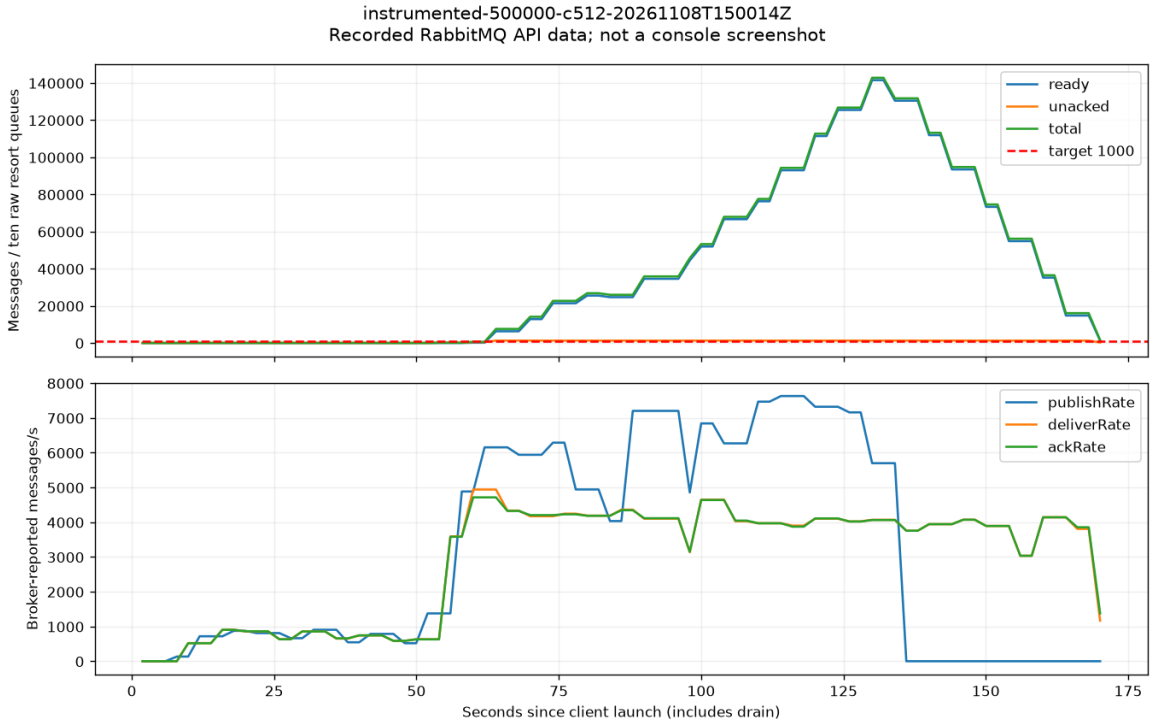
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 12 | 4 ingest(s), 500,000 events

Client threads=512; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

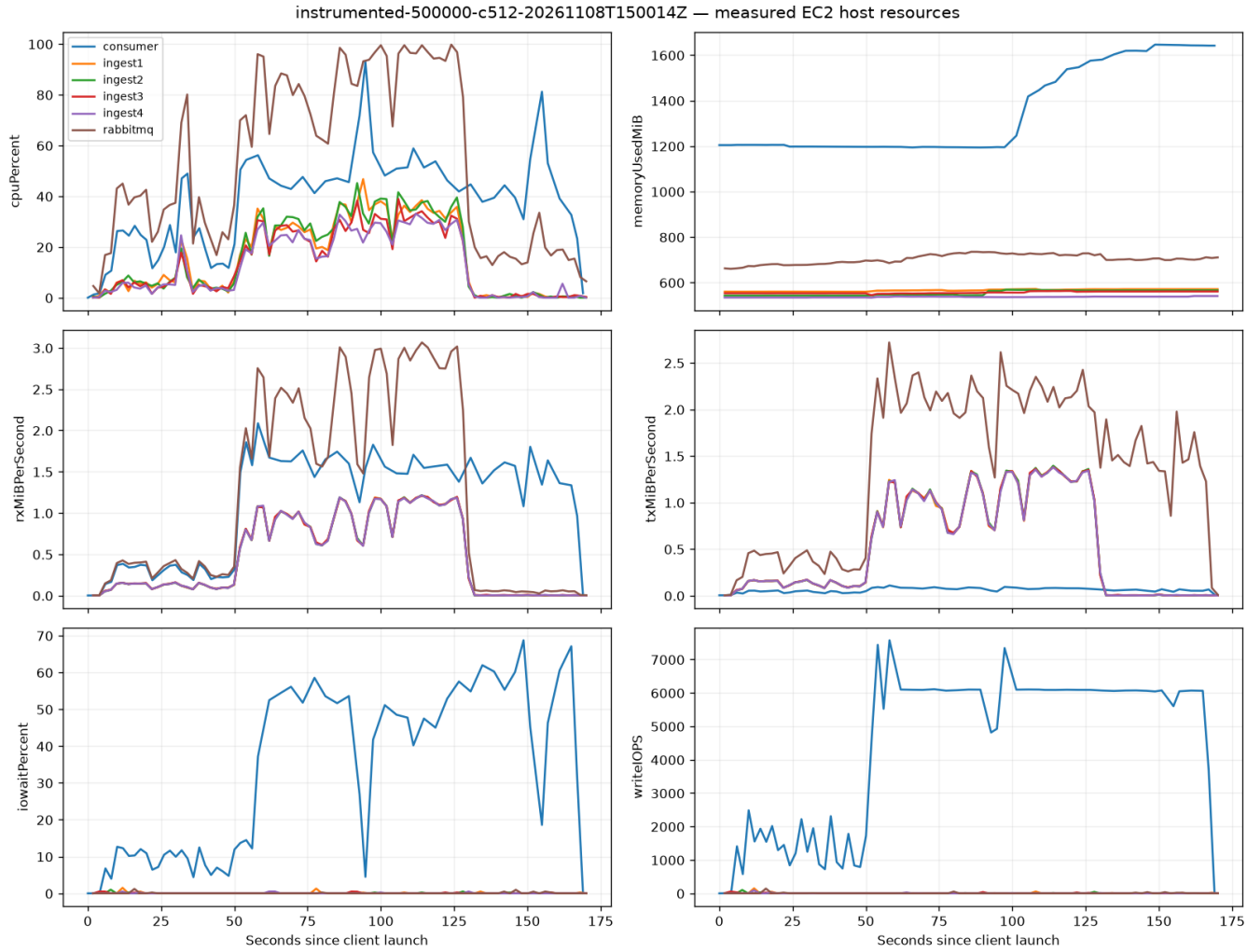
UTC 2026-11-08T15:00:30.011623+00:00 to 2026-11-08T15:03:20.129150+00:00

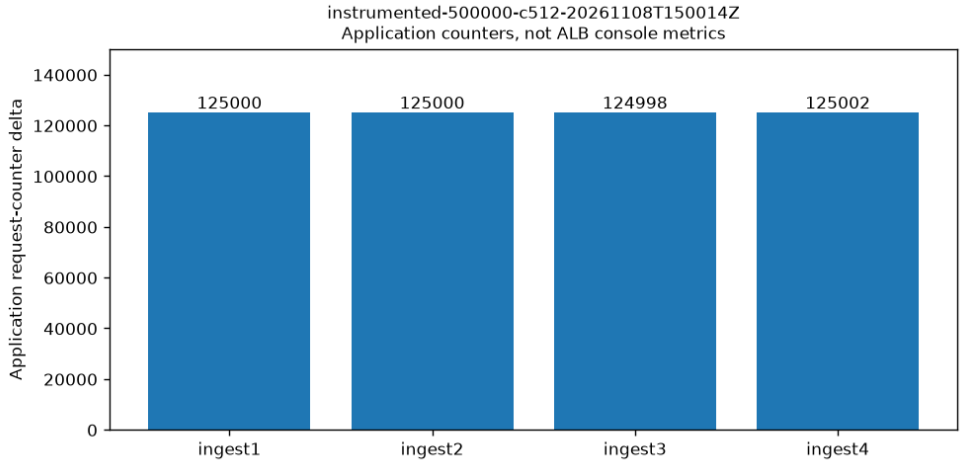
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	5,899.50 / 4,025.14 / 3,068.38
Ingestion / program runtime / drain seconds	124.219 / 167.748 / 38.567
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	83.85 / 186.00 / 331.42
Ready sampled mean / maximum	39,491.69 / 141,495
Unacked / Ready+Unacked maximum	1,280 / 142,775
Broker publish / delivery / ACK sampled mean per second	3,121.07 / 2,936.10 / 2,931.31
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	13,798.40
WebSocket updates / reconnections	909 / 0



Source: instrumented-500000-c512-20261108T150014Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 12 | resource use and distribution

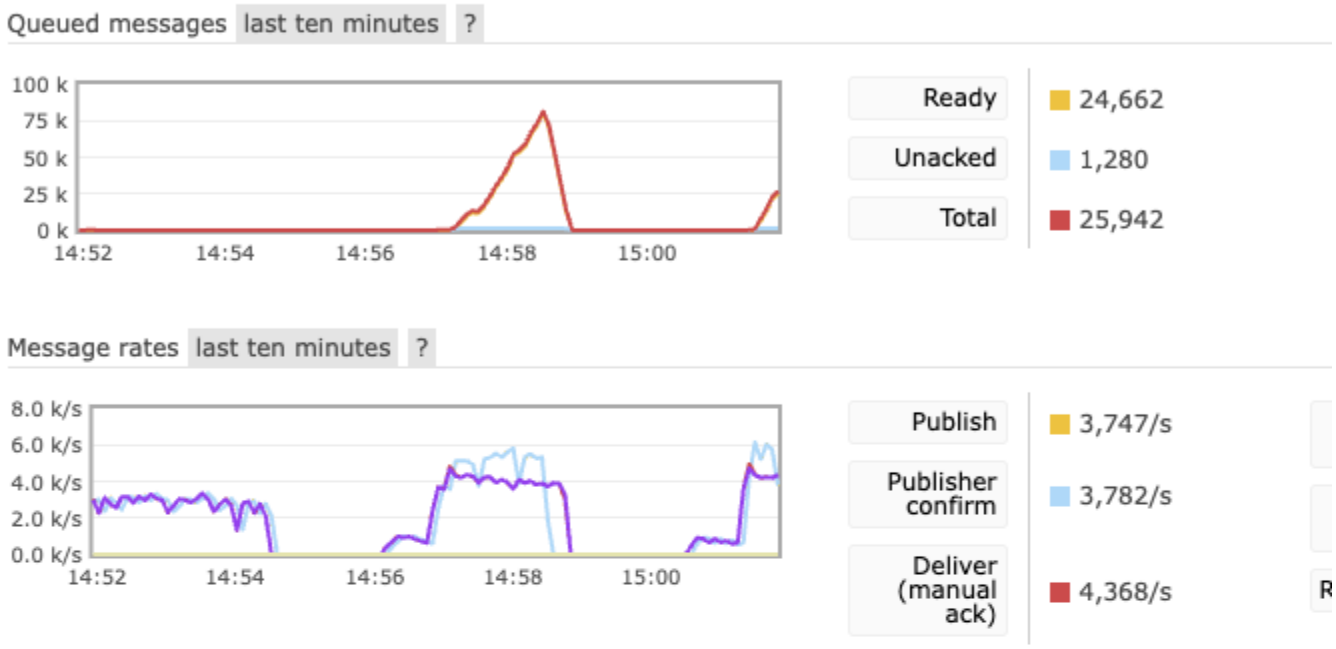




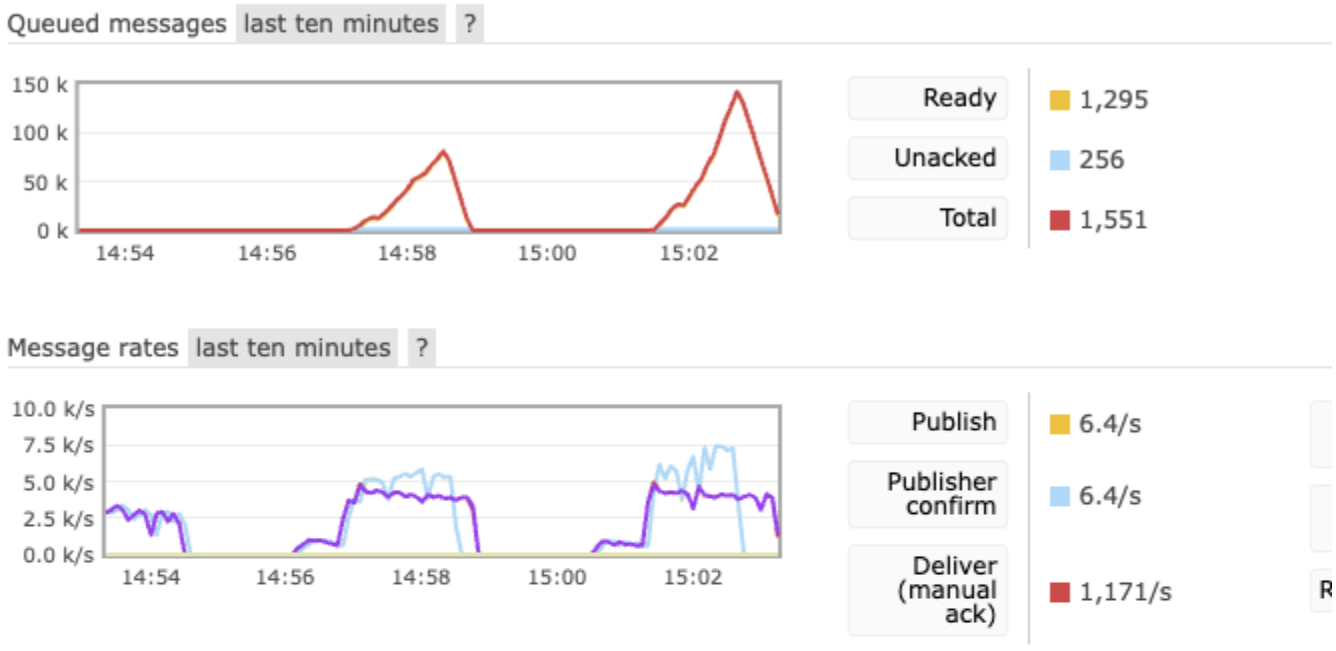
Collection gaps: consumer collection errors: {"timeout": 13}

Run 12 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:00:30.011623+00:00 to 2026-11-08T15:03:20.129150+00:00.



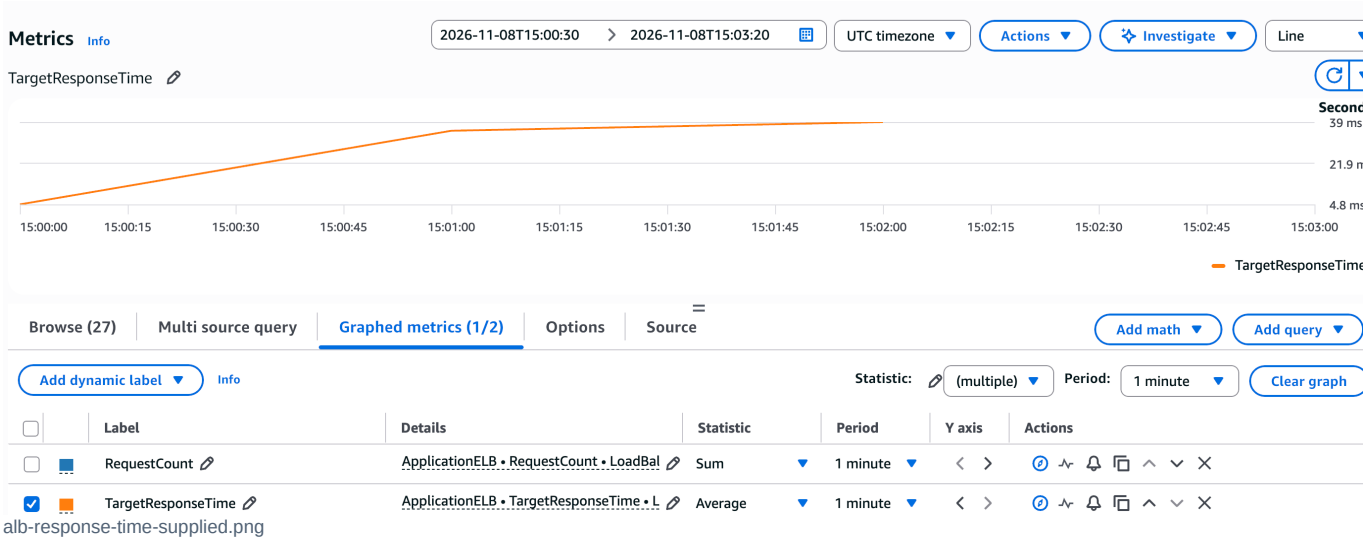
rabbitmq-20261108T150200973877Z.png | Original screenshot retained unchanged.



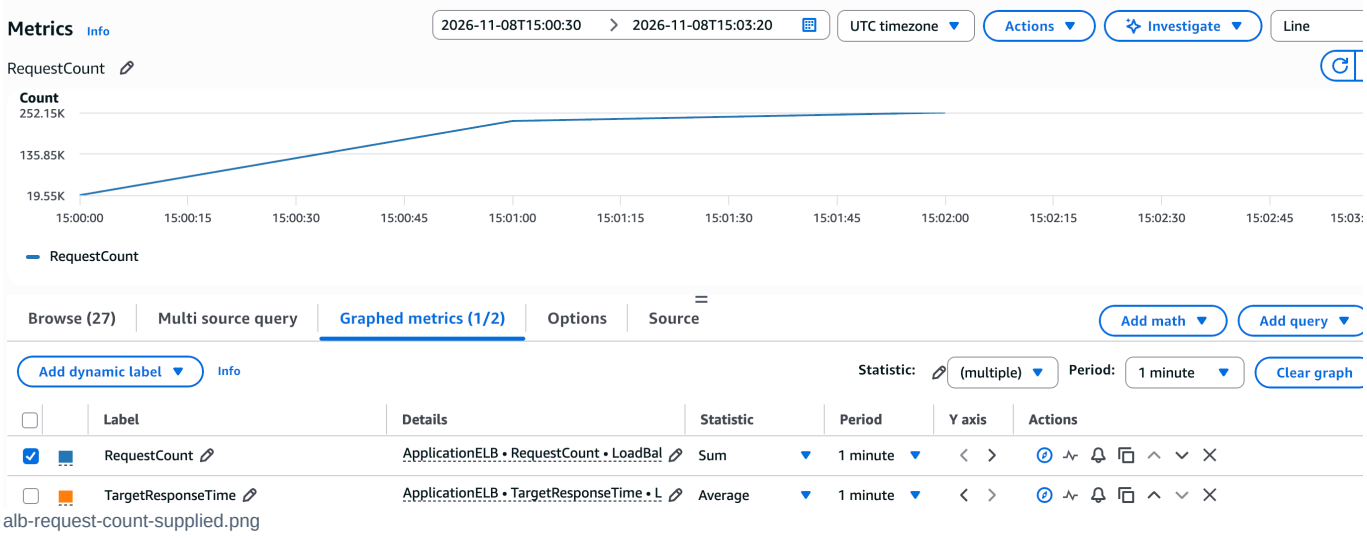
rabbitmq-20261108T150326534269Z-final.png | Original screenshot retained unchanged.

Run 12 | supplied AWS CloudWatch captures

Metric scope: LoadBalancer aggregate.



alb-response-time-supplied.png



alb-request-count-supplied.png

Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

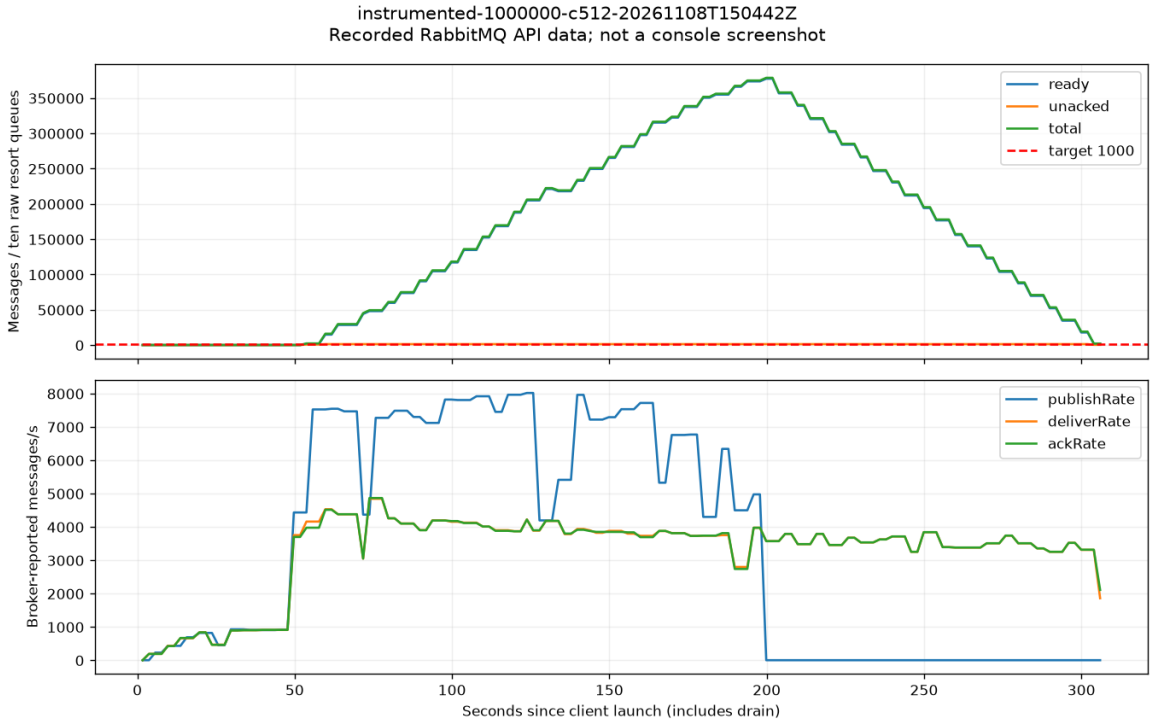
Run 13 | 4 ingest(s), 1,000,000 events

Client threads=512; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

UTC 2026-11-08T15:04:55.794833+00:00 to 2026-11-08T15:10:02.047298+00:00

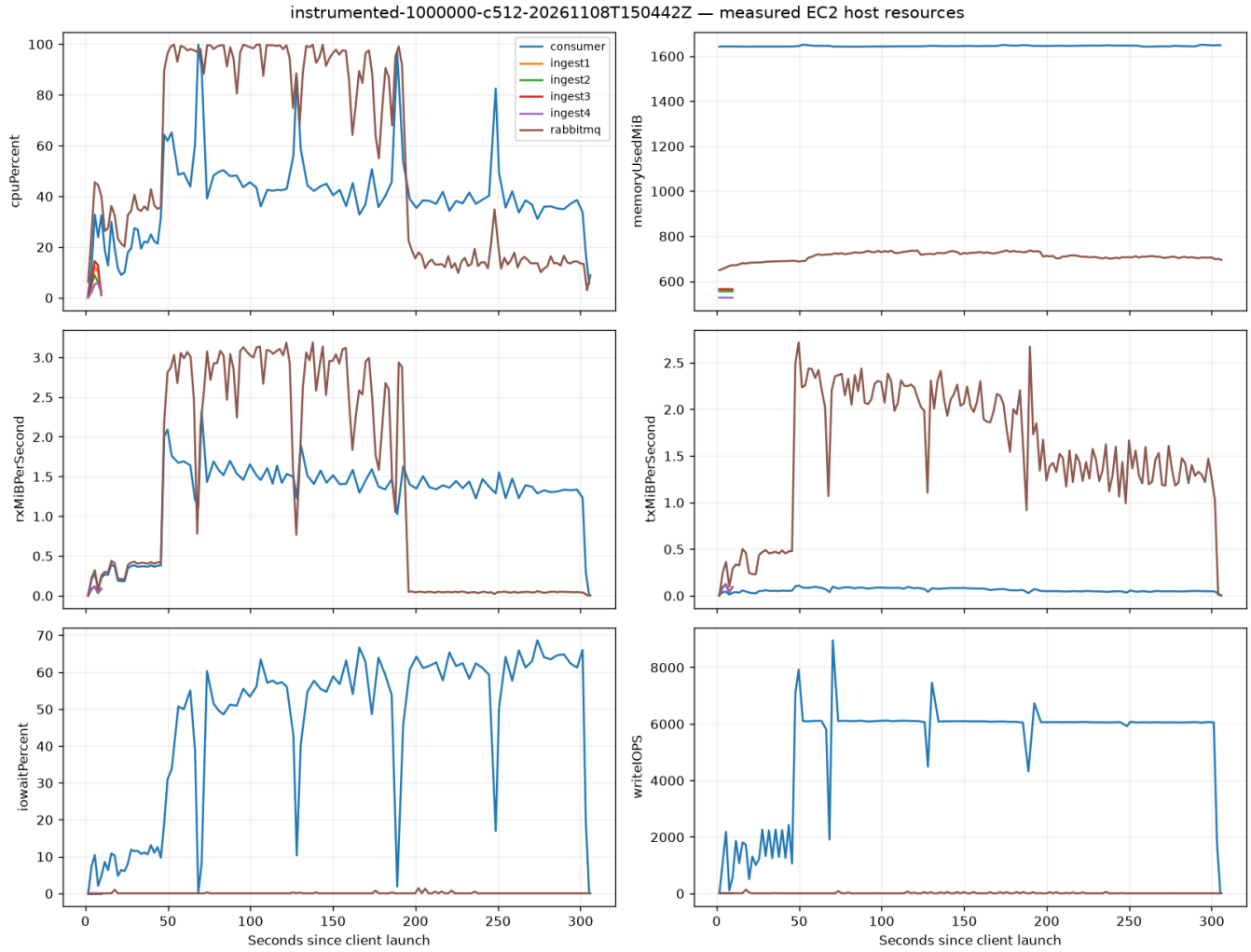
Application distribution coverage is incomplete: recorded counter deltas total 3,716, versus 1,000,077 client attempts. The distribution chart covers only observed counter deltas and must not be read as whole-run balance. The cause of the mismatch was not established.

Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	6,551.14 / 5,247.98 / 3,323.33
Ingestion / program runtime / drain seconds	190.550 / 304.875 / 110.256
Successful / final failed / application retries	1,000,000 / 0 / 77
HTTP mean / p95 / p99 ms	76.88 / 142.22 / 312.63
Ready sampled mean / maximum	158,344.33 / 377,327
Unacked / Ready+Unacked maximum	1,280 / 378,607
Broker publish / delivery / ACK sampled mean per second	3,370.36 / 3,275.11 / 3,271.22
Additional processed / duplicate count	1000000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	49,377.24
WebSocket updates / reconnections	1023 / 0



Source: instrumented-1000000-c512-20261108T150442Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 13 | resource use and distribution

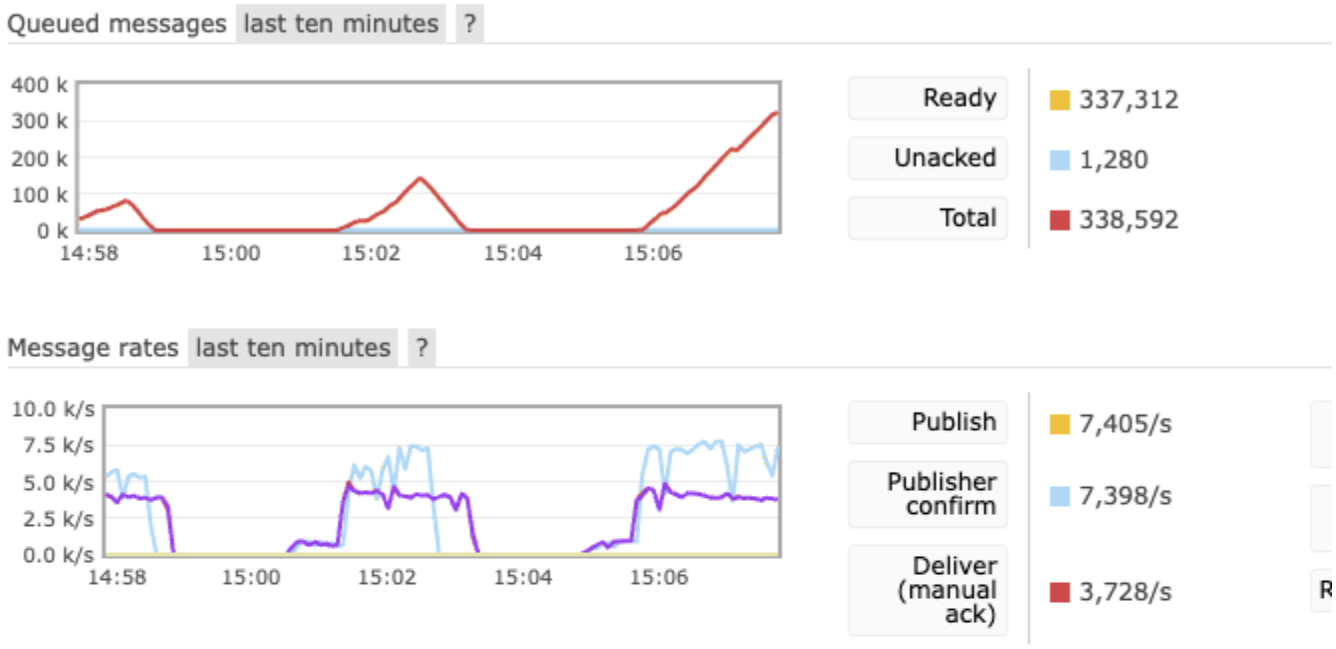


Whole-run per-ingest distribution is not shown: counter coverage is incomplete. The archived chart reflects only the observed deltas.

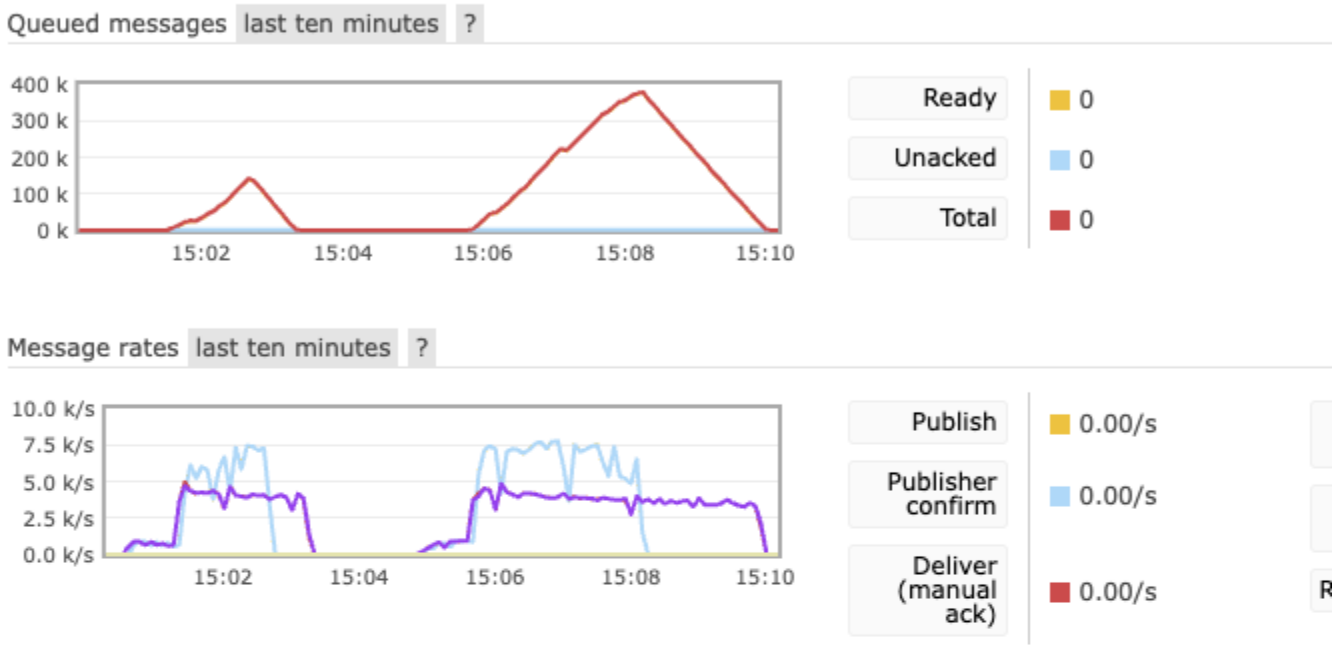
Collection gaps: consumer collection errors: {"timeout": 44}

Run 13 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:04:55.794833+00:00 to 2026-11-08T15:10:02.047298+00:00.



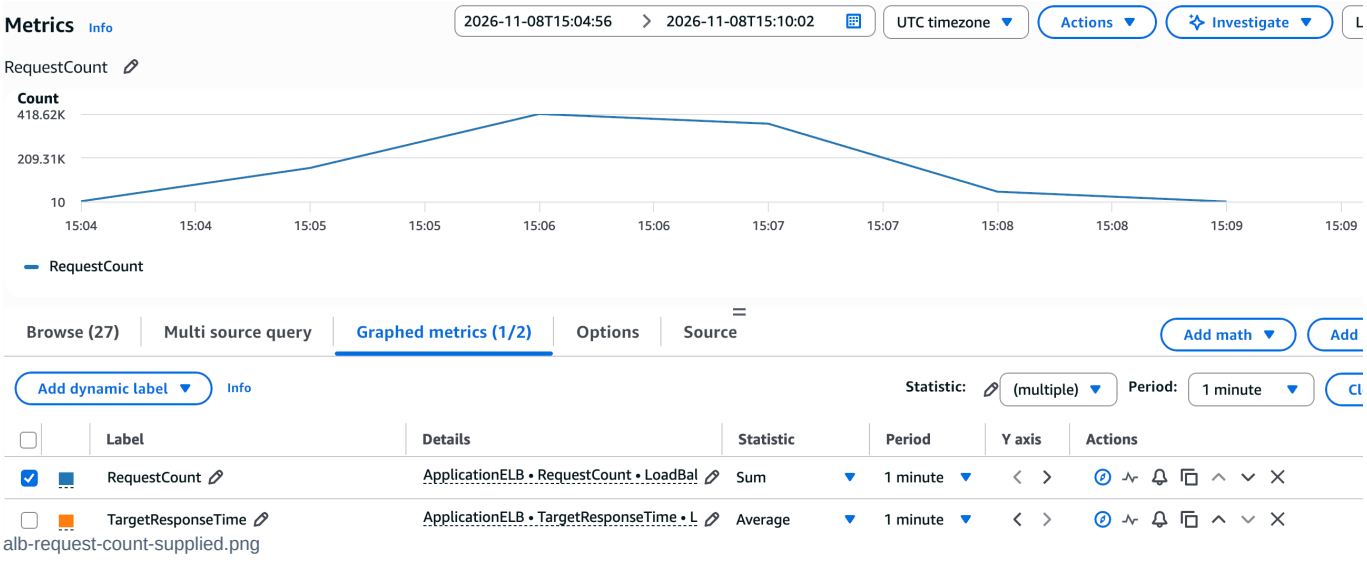
rabbitmq-20261108T1507573001562.png | Original screenshot retained unchanged.



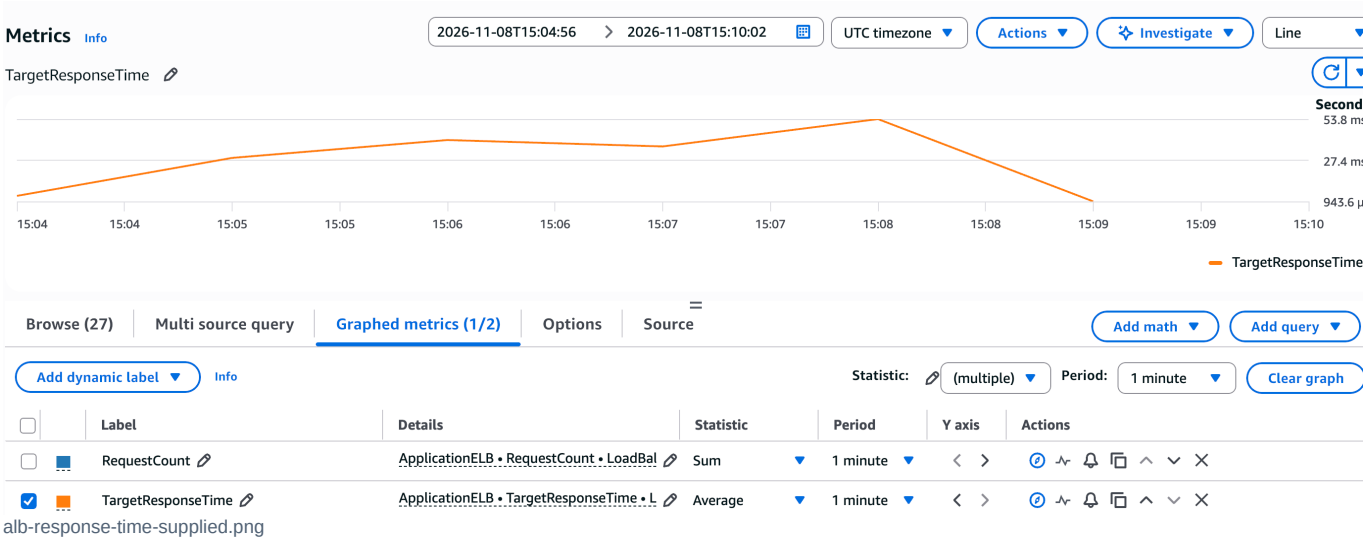
rabbitmq-20261108T1510162800592-final.png | Original screenshot retained unchanged.

Run 13 | supplied AWS CloudWatch captures

Metric scope: LoadBalancer aggregate.



alb-request-count-supplied.png



alb-response-time-supplied.png

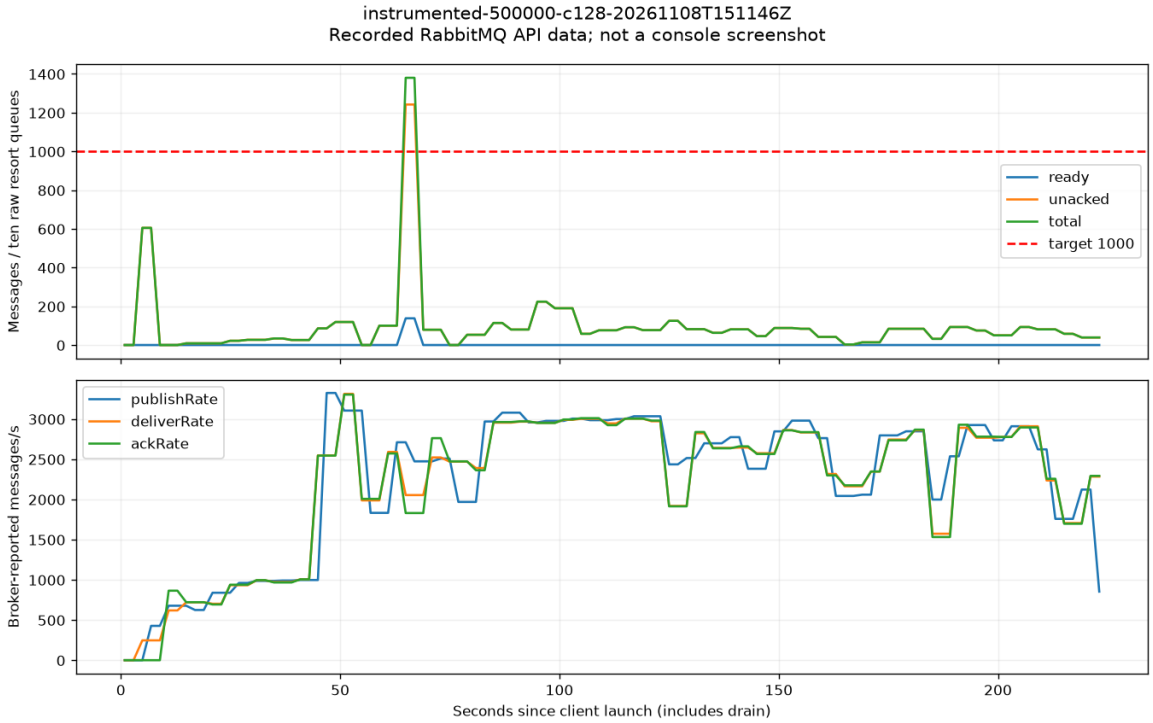
Statistics/time-range verification is recorded in screenshot-context.json. CloudWatch minute buckets can overlap test boundaries; client logs provide exact application event counts.

Run 14 | 4 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 5, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

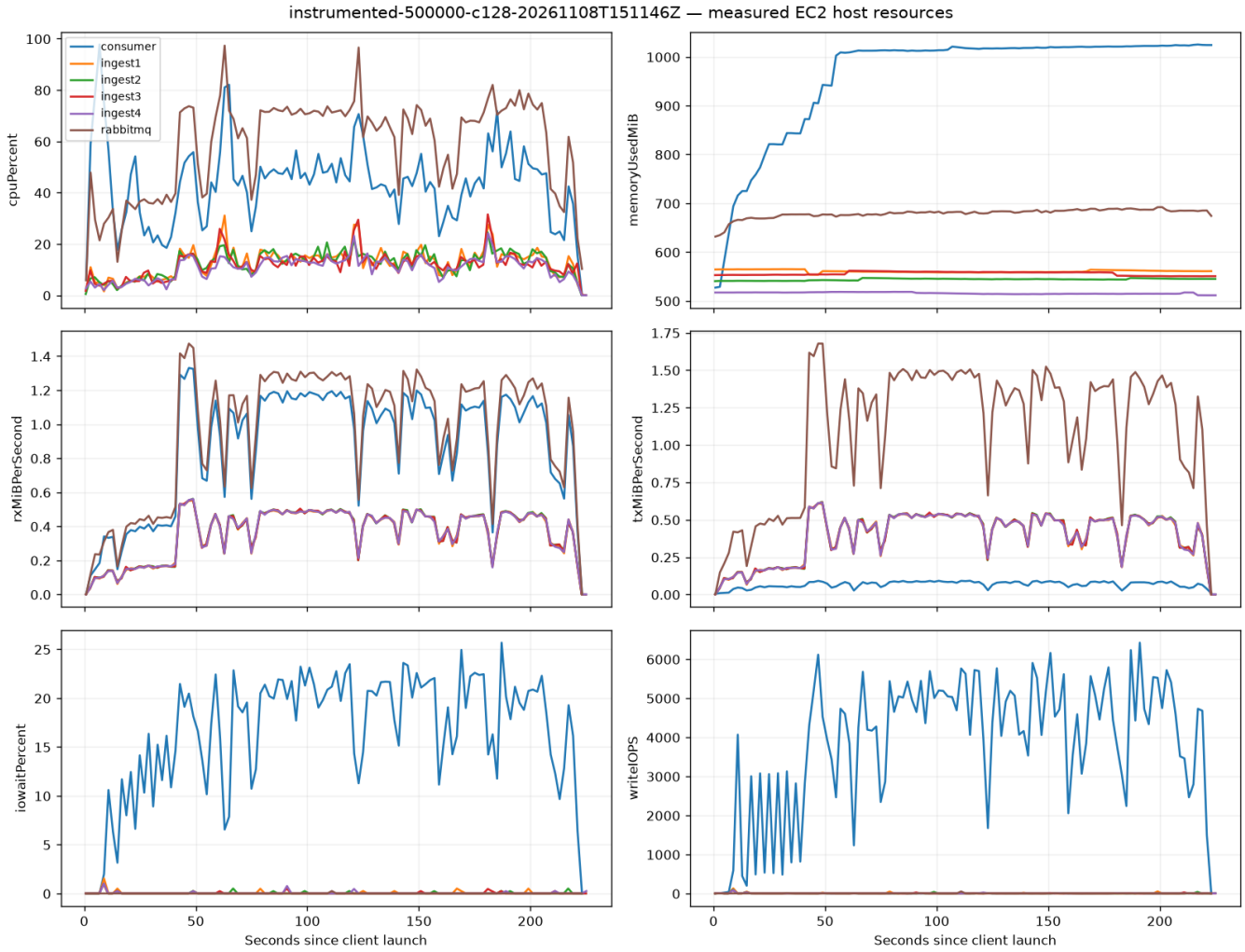
UTC 2026-11-08T15:12:00.633818+00:00 to 2026-11-08T15:15:45.724134+00:00

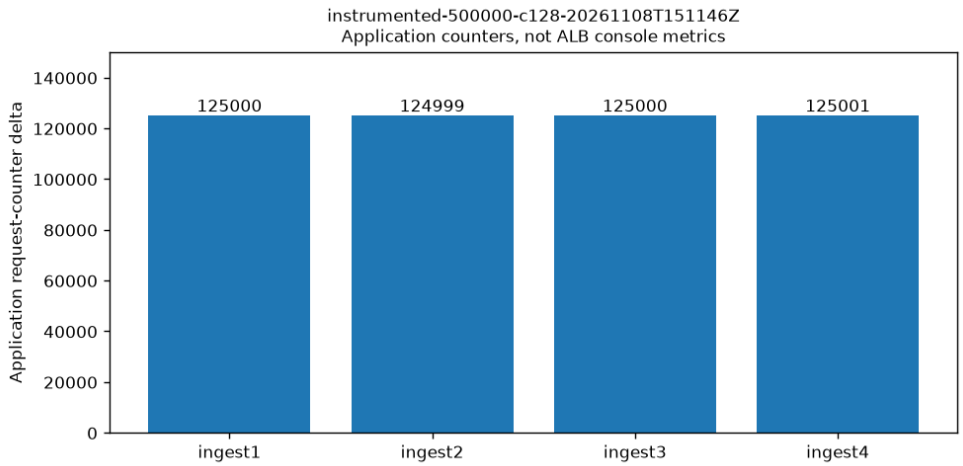
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	2,604.84 / 2,289.23 / 2,275.86
Ingestion / program runtime / drain seconds	218.414 / 223.529 / 1.253
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	48.42 / 78.05 / 241.01
Ready sampled mean / maximum	2.46 / 138
Unacked / Ready+Unacked maximum	1,242 / 1,380
Broker publish / delivery / ACK sampled mean per second	2,236.23 / 2,205.92 / 2,201.43
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	42.40
WebSocket updates / reconnections	918 / 0



Source: instrumented-500000-c128-20261108T151146Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 14 | resource use and distribution

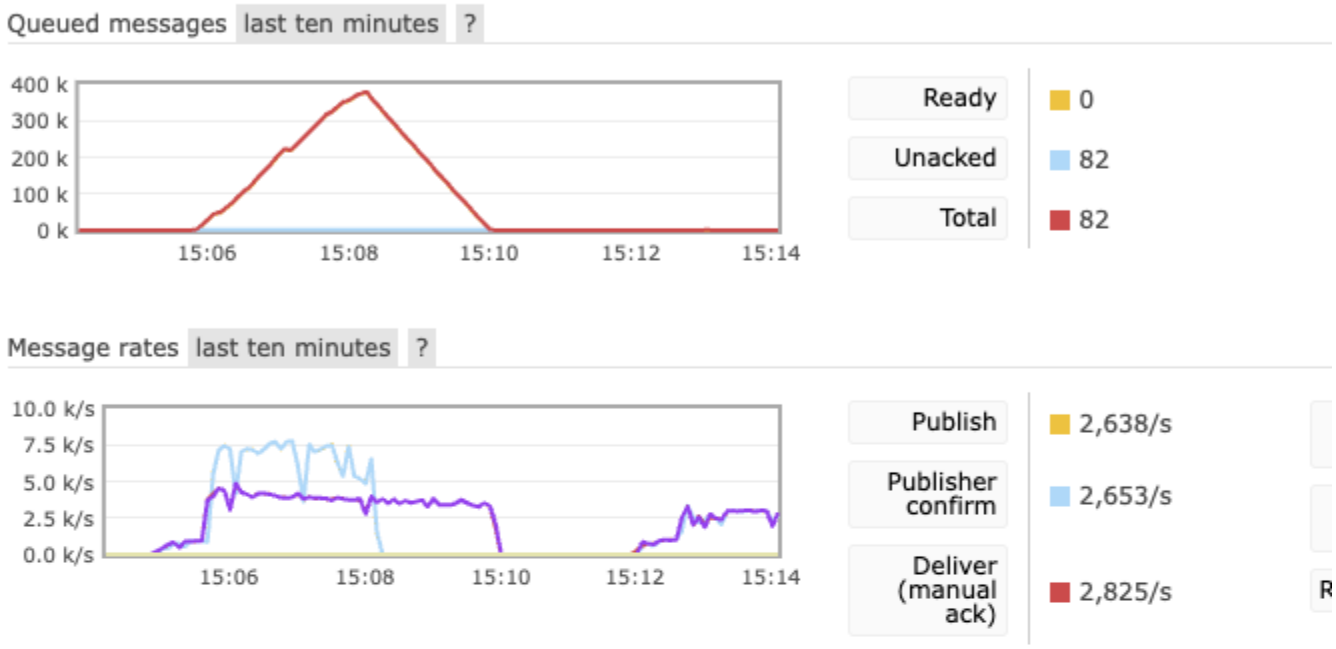




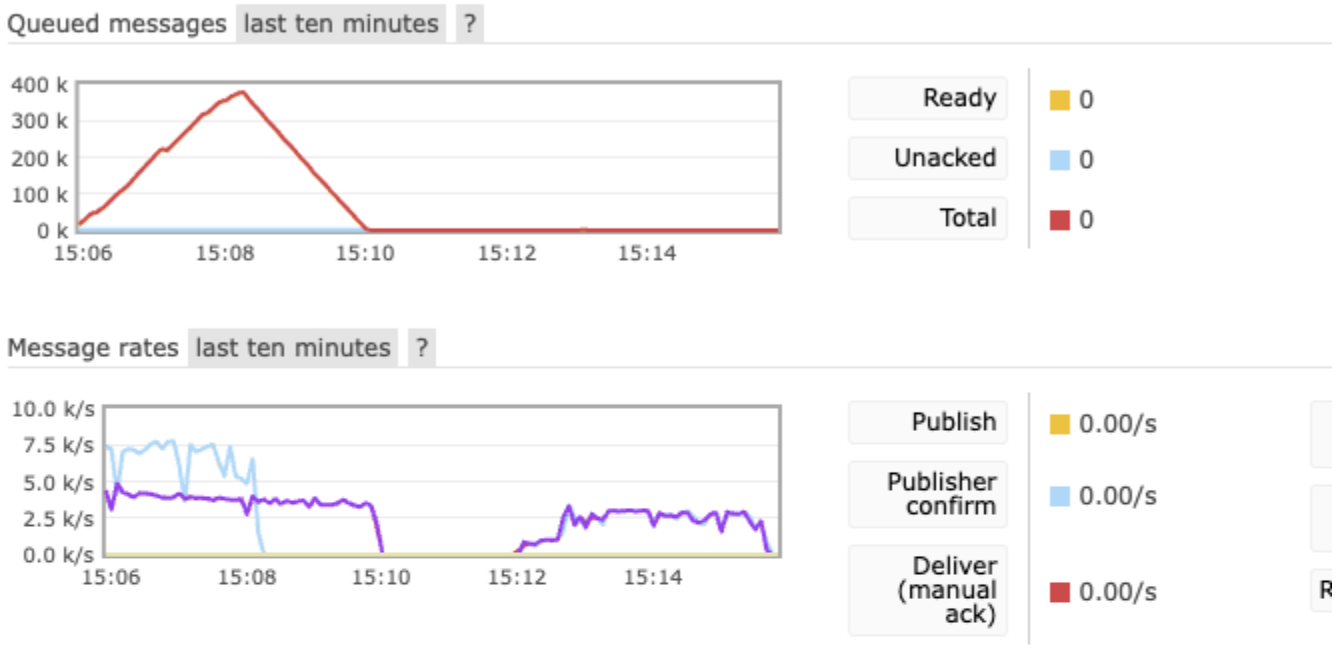
Collection gaps: consumer collection errors: {"timeout": 1}

Run 14 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:12:00.633818+00:00 to 2026-11-08T15:15:45.724134+00:00.



rabbitmq-20261108T151416395564Z.png | Original screenshot retained unchanged.



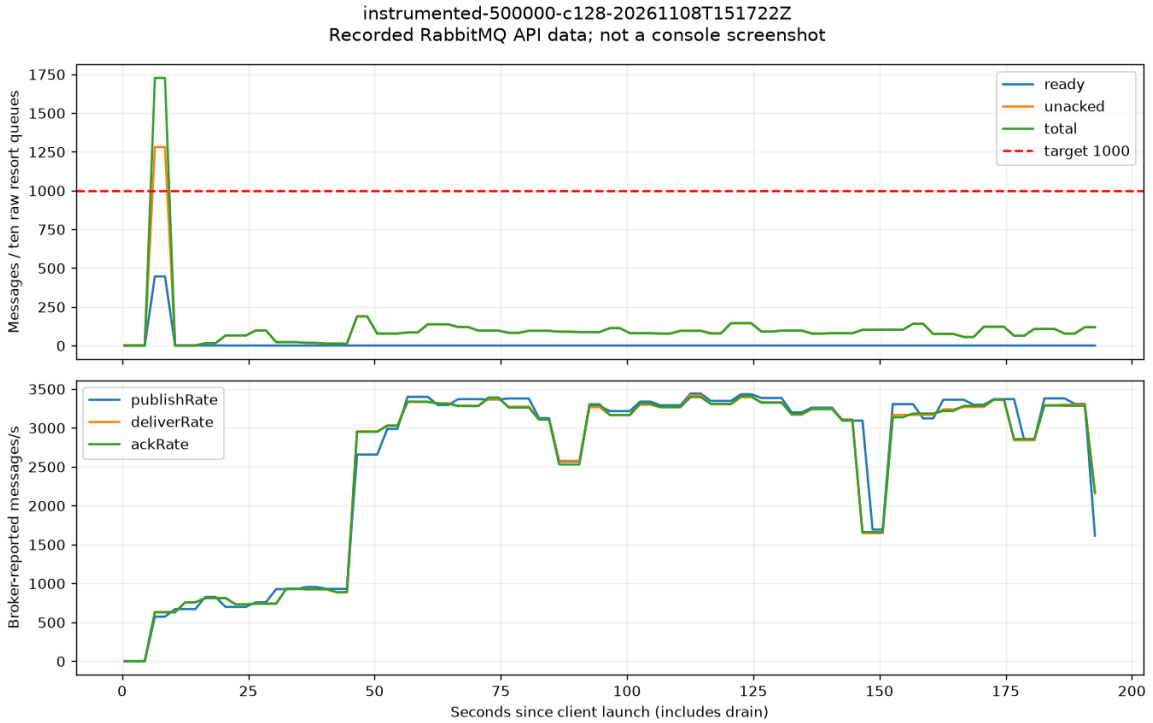
rabbitmq-20261108T151559637186Z-final.png | Original screenshot retained unchanged.

Run 15 | 4 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 10, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

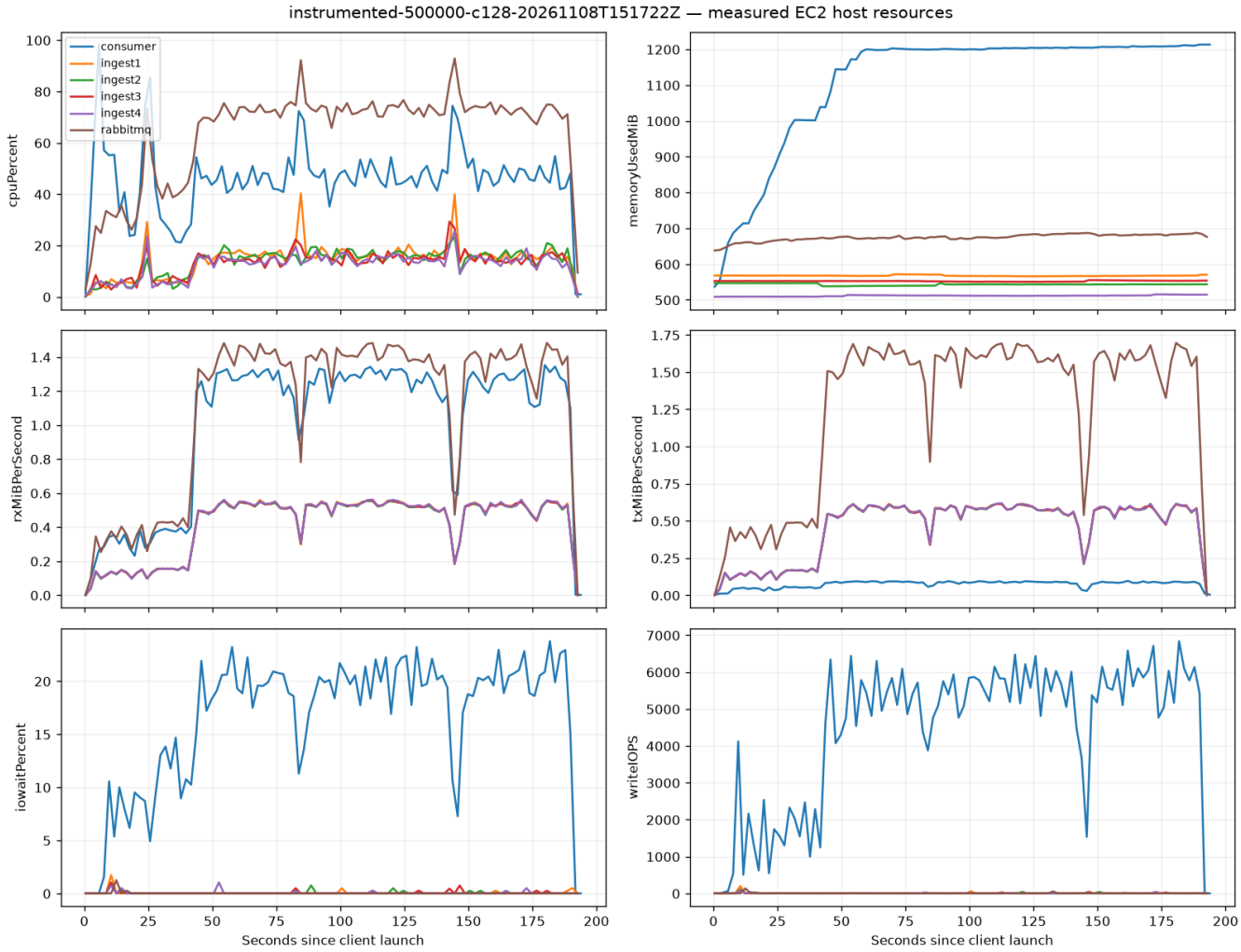
UTC 2026-11-08T15:17:38.725229+00:00 to 2026-11-08T15:20:52.587022+00:00

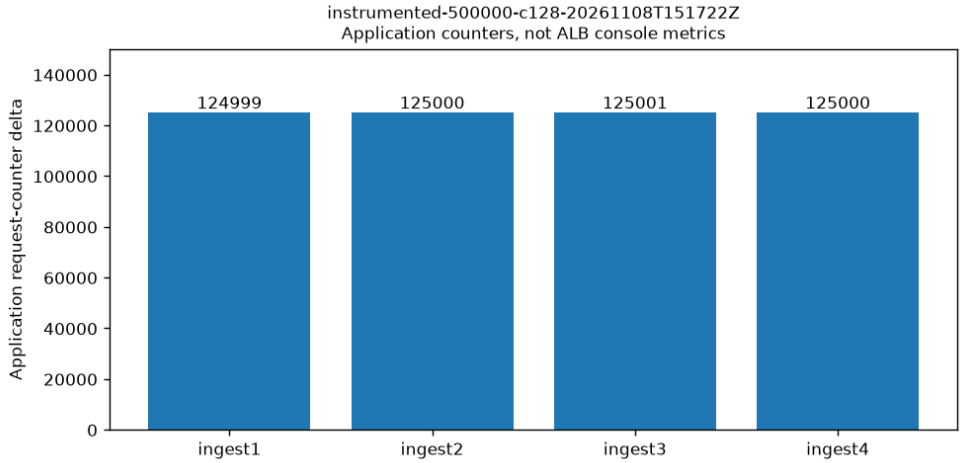
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	3,155.08 / 2,663.40 / 2,650.72
Ingestion / program runtime / drain seconds	187.730 / 192.907 / 0.800
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	40.43 / 55.58 / 92.10
Ready sampled mean / maximum	9.20 / 446
Unacked / Ready+Unacked maximum	1,280 / 1,726
Broker publish / delivery / ACK sampled mean per second	2,576.92 / 2,545.18 / 2,544.72
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	34.96
WebSocket updates / reconnections	917 / 0



Source: instrumented-500000-c128-20261108T151722Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 15 | resource use and distribution

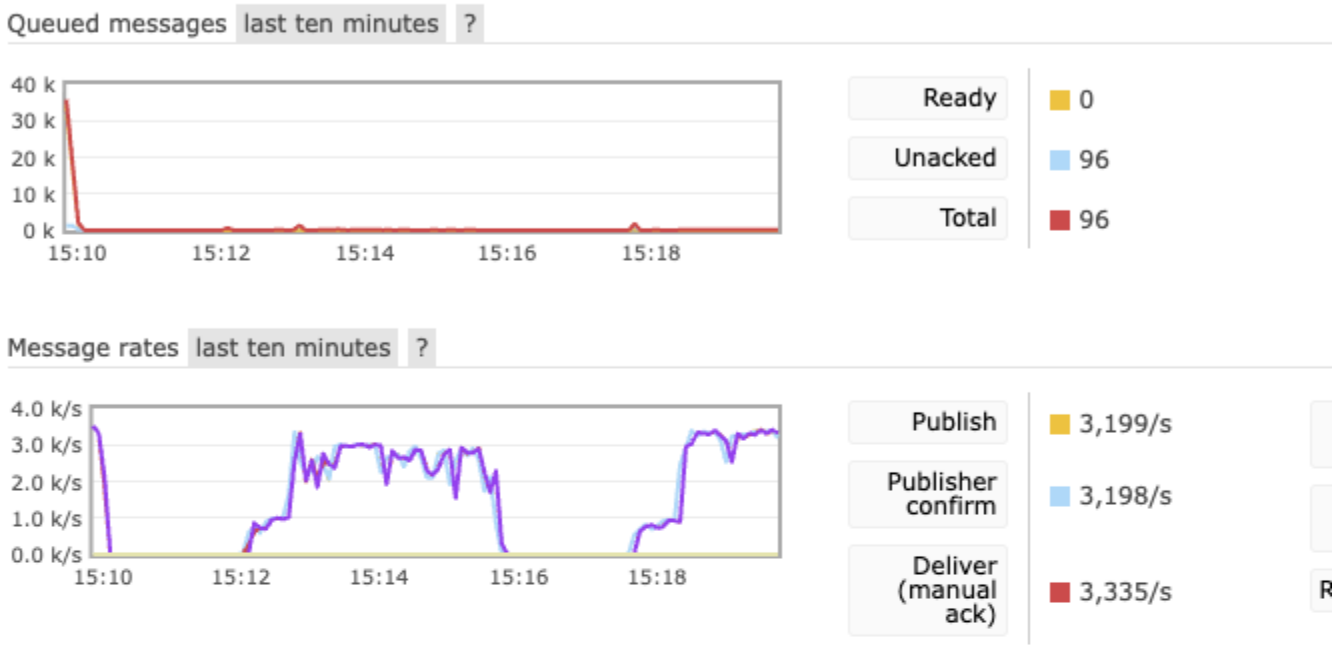




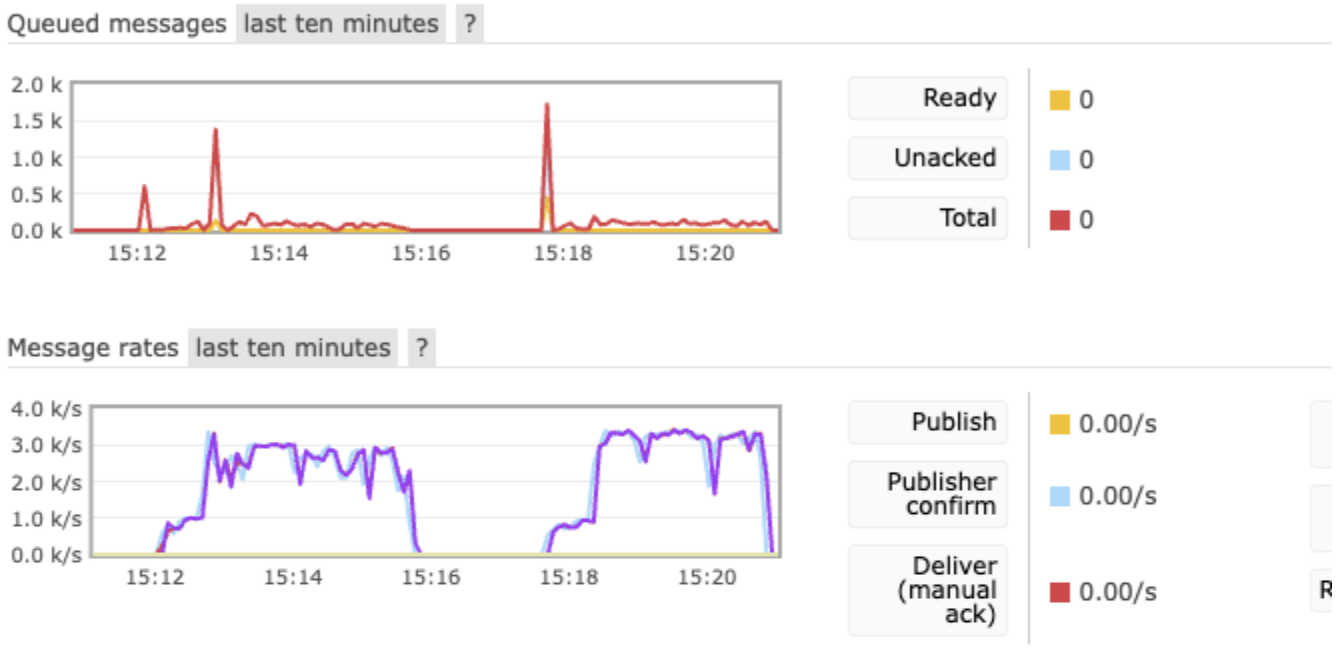
Collection gaps: None detected.

Run 15 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:17:38.725229+00:00 to 2026-11-08T15:20:52.587022+00:00.



rabbitmq-20261108T151954728312Z.png | Original screenshot retained unchanged.



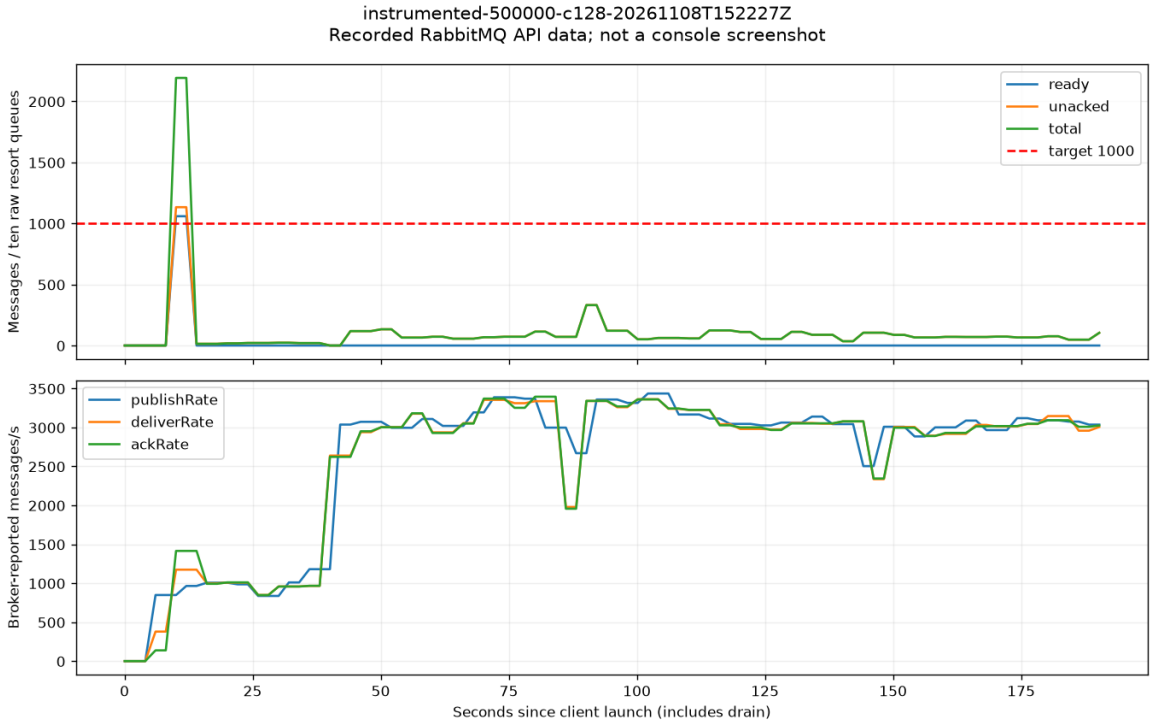
rabbitmq-20261108T152108146701Z-final.png | Original screenshot retained unchanged.

Run 16 | 4 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 15, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

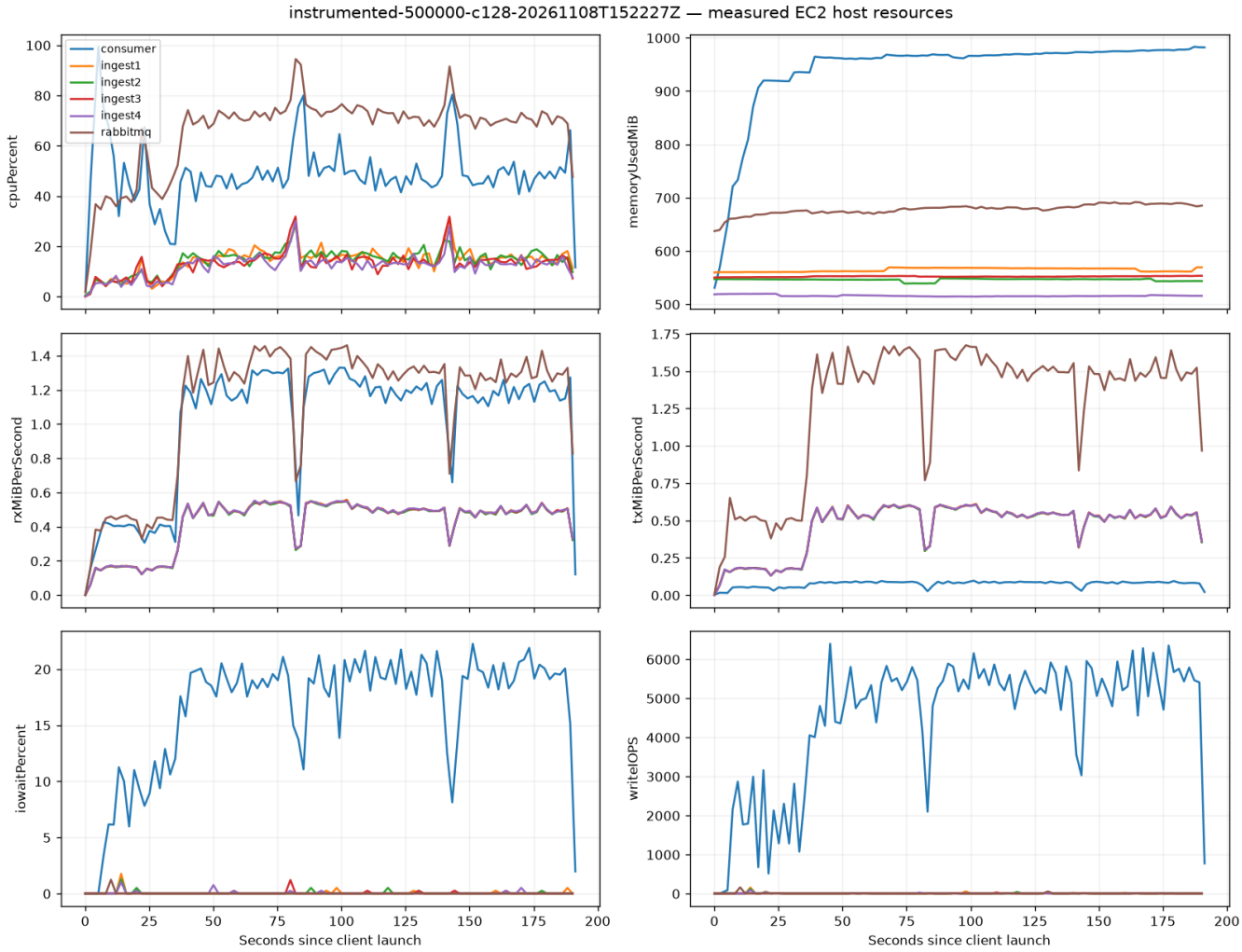
UTC 2026-11-08T15:22:40.634625+00:00 to 2026-11-08T15:25:52.373649+00:00

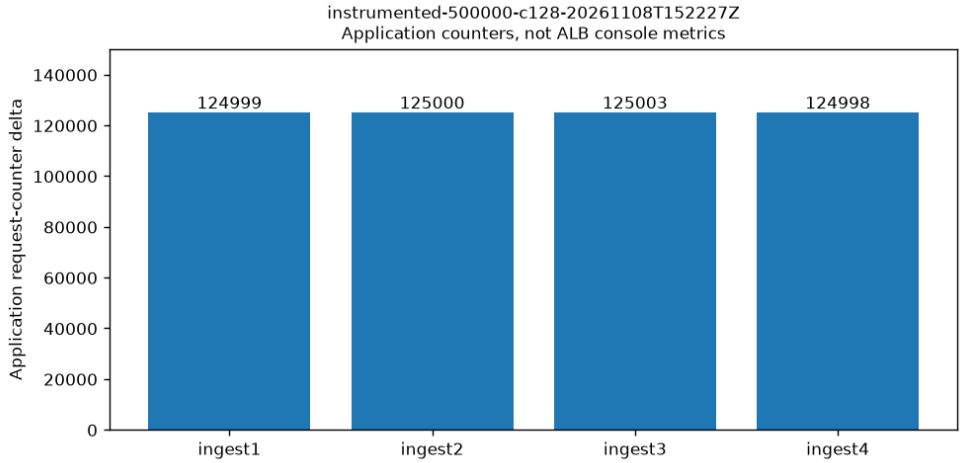
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	3,032.48 / 2,654.87 / 2,643.07
Ingestion / program runtime / drain seconds	188.333 / 190.874 / 0.816
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	41.66 / 59.26 / 93.09
Ready sampled mean / maximum	22.06 / 1,059
Unacked / Ready+Unacked maximum	1,133 / 2,192
Broker publish / delivery / ACK sampled mean per second	2,591.56 / 2,567.88 / 2,570.40
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	38.50
WebSocket updates / reconnections	915 / 0



Source: instrumented-500000-c128-20261108T152227Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 16 | resource use and distribution

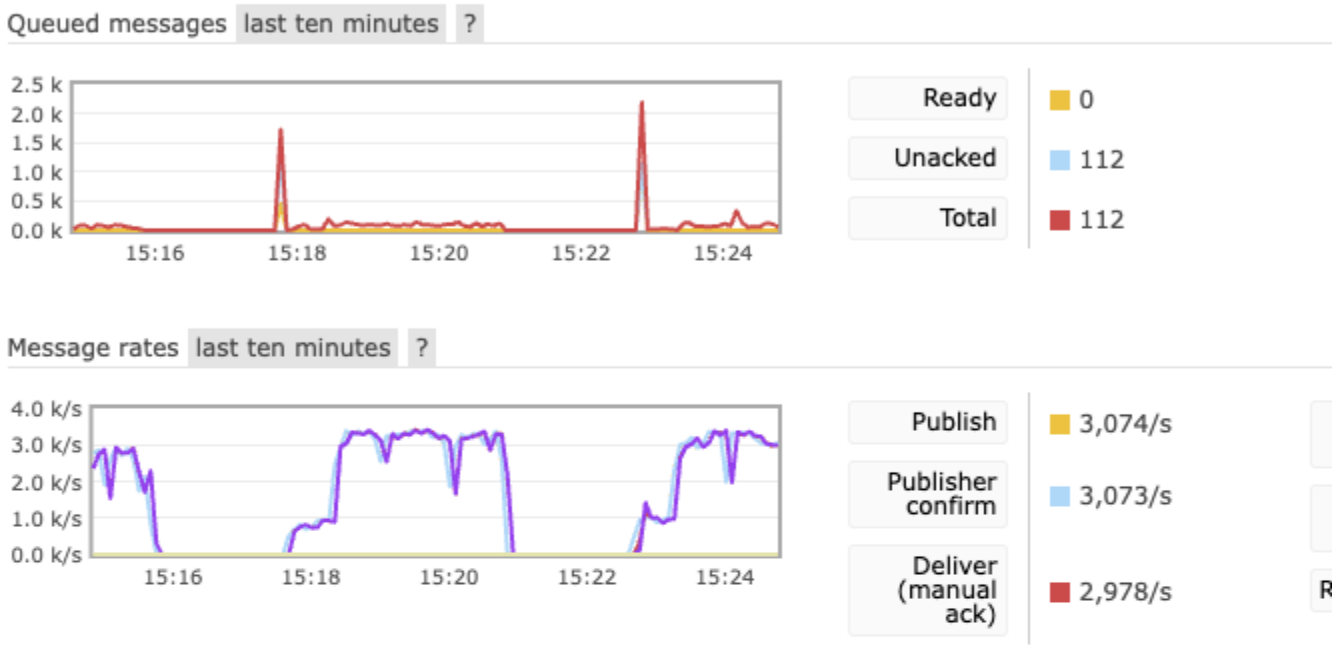




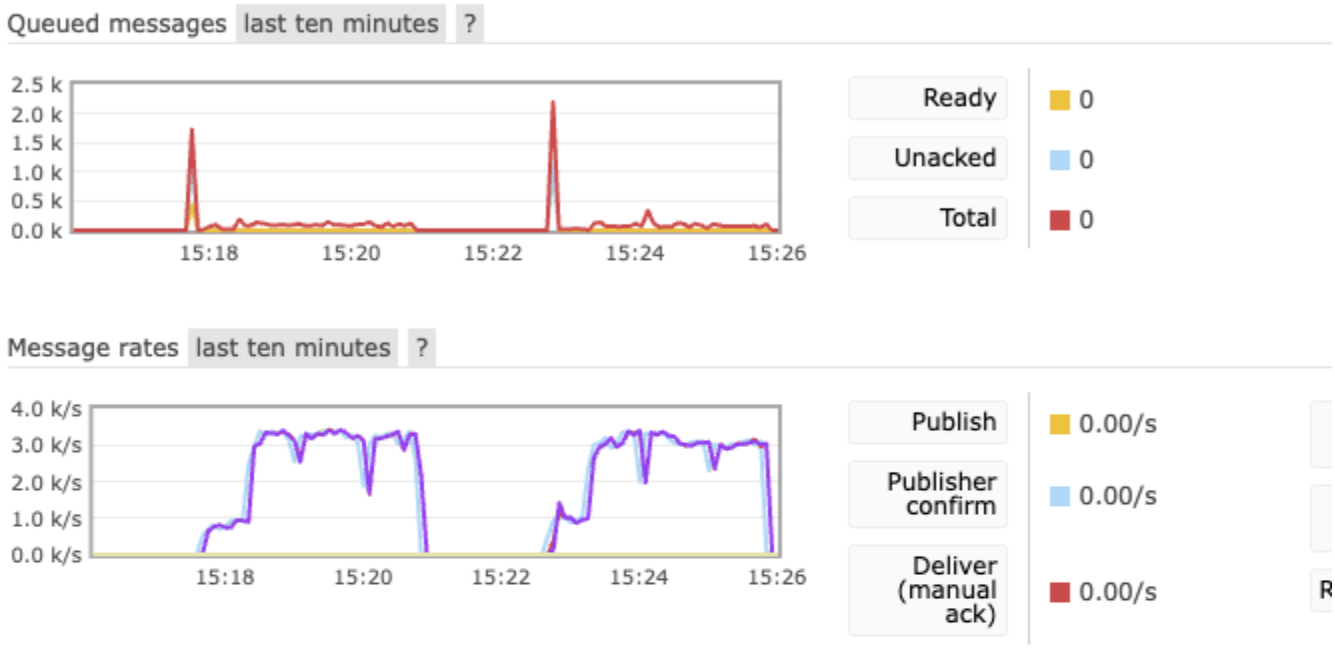
Collection gaps: None detected.

Run 16 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:22:40.634625+00:00 to 2026-11-08T15:25:52.373649+00:00.



rabbitmq-20261108T152456578021Z.png | Original screenshot retained unchanged.



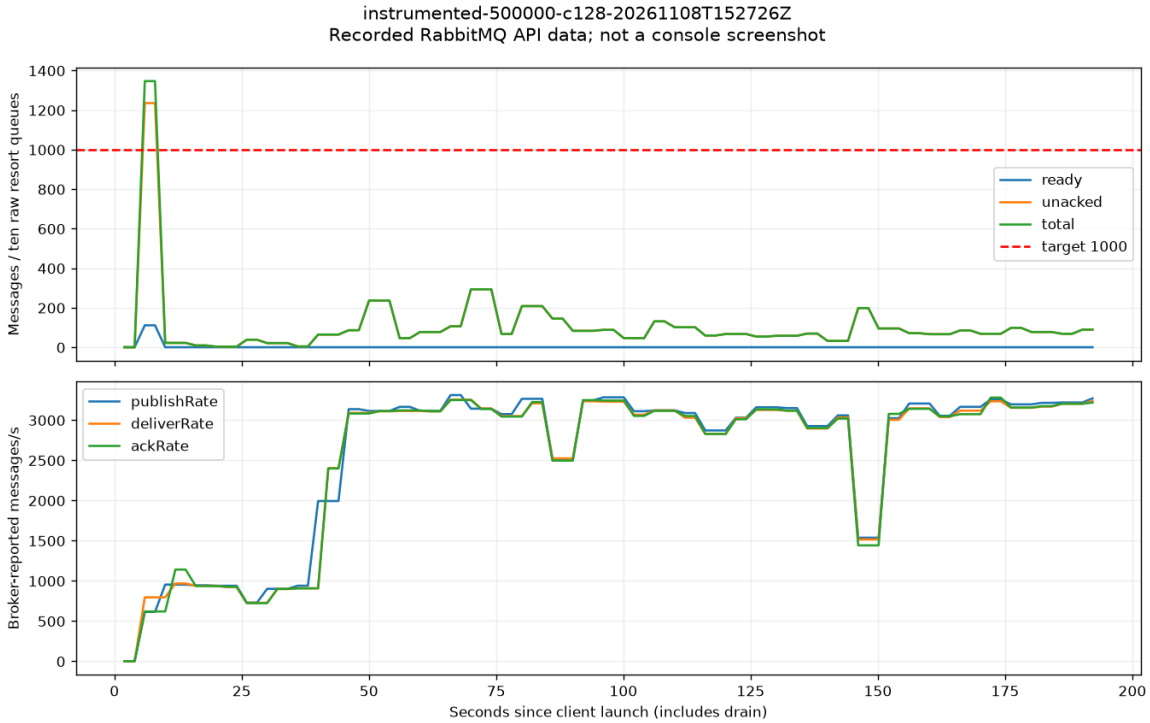
rabbitmq-20261108T152606268242Z-final.png | Original screenshot retained unchanged.

Run 17 | 4 ingest(s), 500,000 events

Client threads=128; Consumer={"configuredThreads": 20, "dynamicScaling": false, "prefetch": 128, "storeBatchSize": 64, "storeBatchWaitMs": 10, "ackBatchSize": 16}

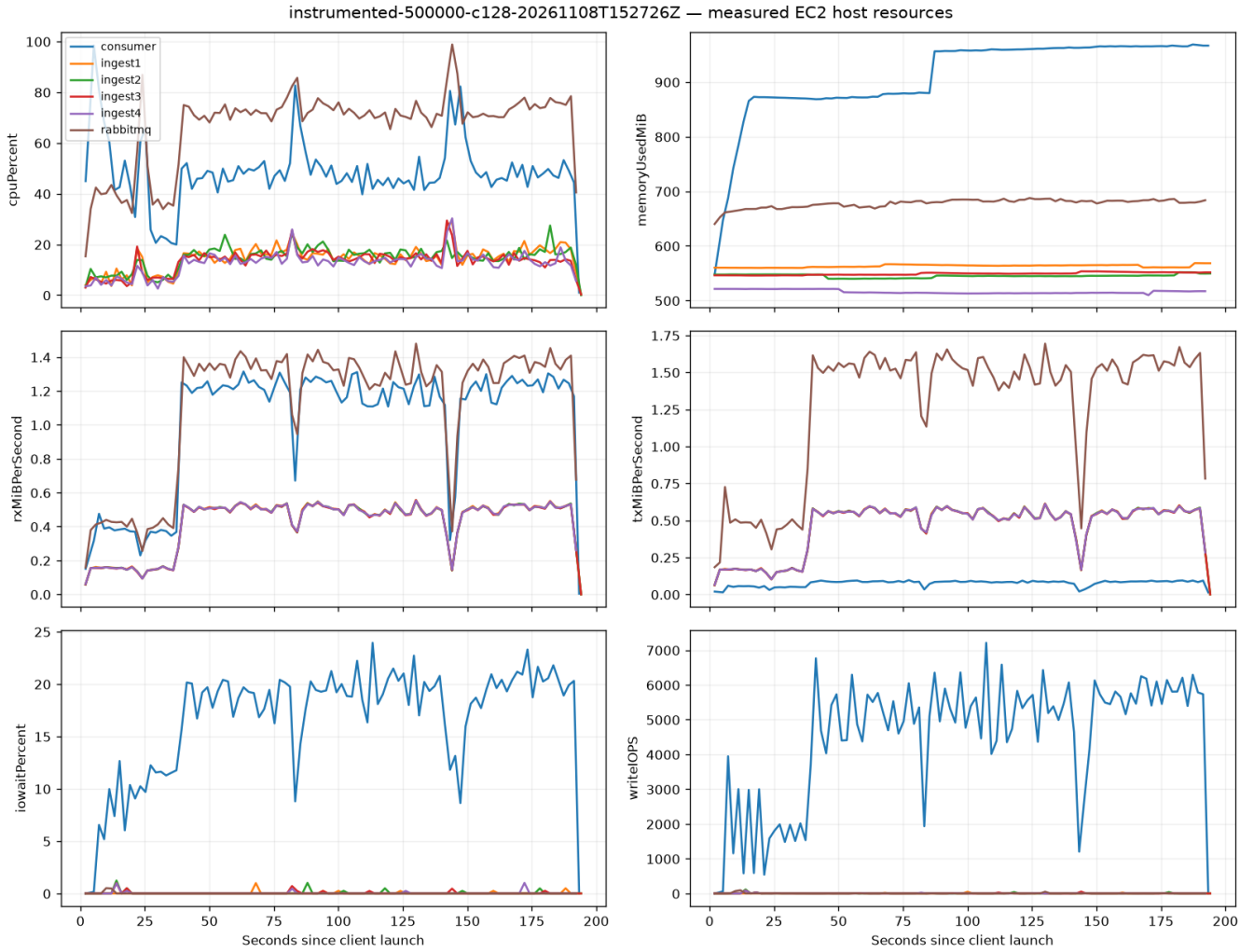
UTC 2026-11-08T15:27:39.783122+00:00 to 2026-11-08T15:30:53.922037+00:00

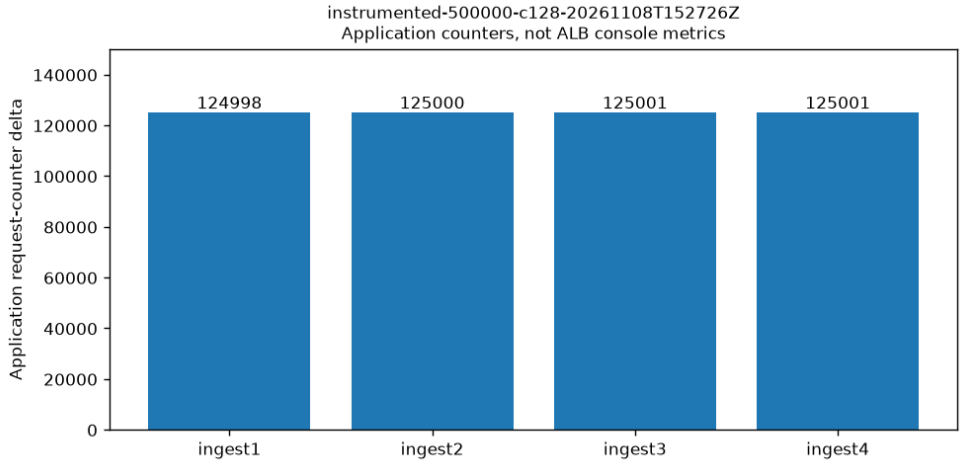
Measurement	Recorded result
Main HTTP / total ingestion / end-to-end events/s	3,036.31 / 2,631.02 / 2,620.63
Ingestion / program runtime / drain seconds	190.040 / 192.782 / 0.731
Successful / final failed / application retries	500,000 / 0 / 0
HTTP mean / p95 / p99 ms	41.73 / 58.24 / 99.31
Ready sampled mean / maximum	2.31 / 111
Unacked / Ready+Unacked maximum	1,235 / 1,346
Broker publish / delivery / ACK sampled mean per second	2,566.00 / 2,539.29 / 2,535.58
Additional processed / duplicate count	500000 / 0
Final Ready / in-flight / pending outbox	0 / 0 / 0
Mean run-specific Consumer lag ms	39.38
WebSocket updates / reconnections	915 / 0



Source: instrumented-500000-c128-20261108T152726Z; complete client-output.txt, queue-timeseries.csv and summary.json in the matching evidence directory.

Run 17 | resource use and distribution

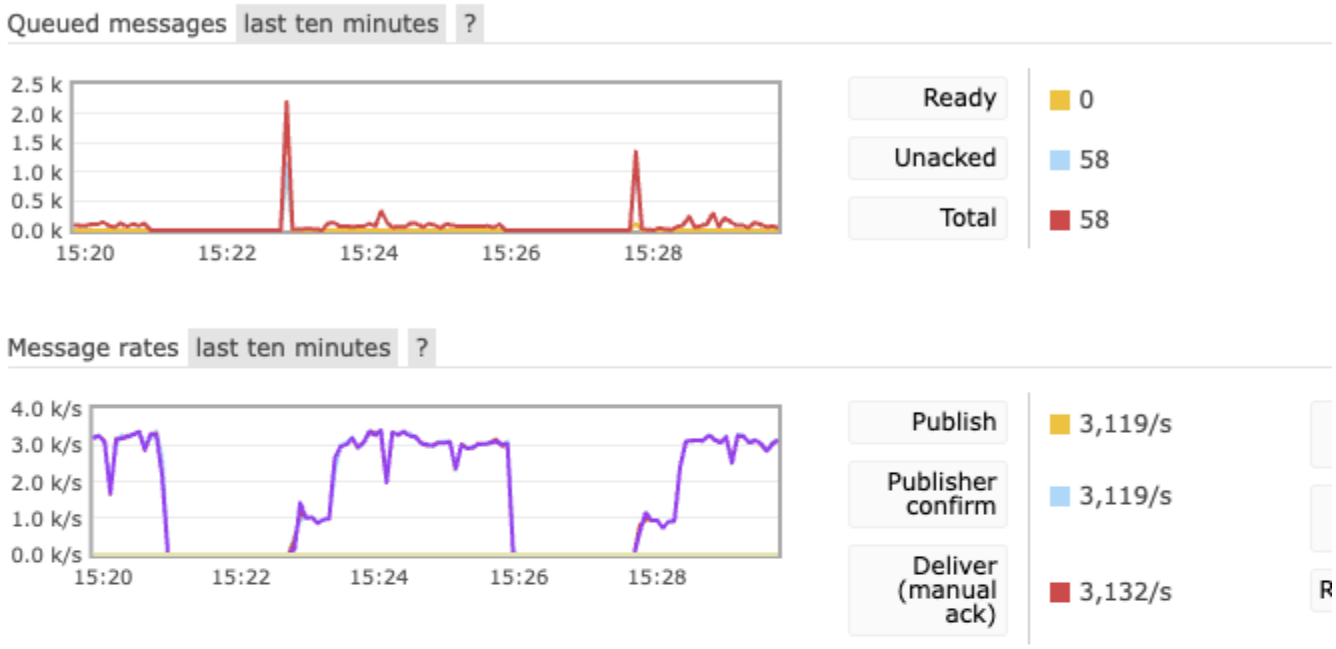




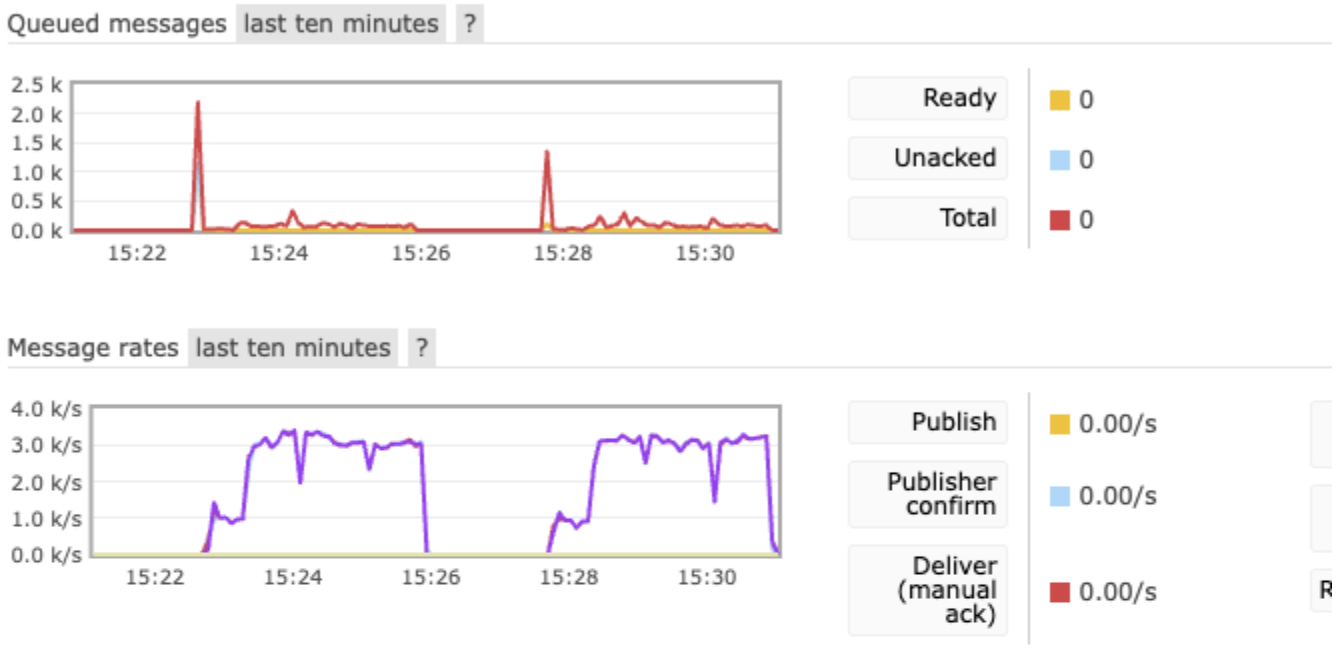
Collection gaps: None detected.

Run 17 | actual RabbitMQ console

Readable excerpts of the actual UI queue-depth and rate panels. Additional legend columns outside this viewport remain in the full original PNG files. Console history may include earlier tests; this run window is 2026-11-08T15:27:39.783122+00:00 to 2026-11-08T15:30:53.922037+00:00.



rabbitmq-20261108T152955823075Z.png | Original screenshot retained unchanged.



rabbitmq-20261108T153107468189Z-final.png | Original screenshot retained unchanged.

Functional tests, recovery and remaining evidence

alb.json

Recorded pass flag: True

distribution: {"ingest-2": 51, "ingest-4": 49, "ingest-1": 50, "ingest-3": 50}

stickyPassed: true

distributionTargetsPassed: true

Full result: results/a2-fresh/20261108T135433Z/verification/alb.json

pipeline.json

Recorded pass flag: True

demoLifts: false

uniqueEvents: 100

duplicateRequests: 20

serverRequestCounts: {"ingest-1": 27, "ingest-4": 25, "ingest-2": 25, "ingest-3": 23}

Full result: results/a2-fresh/20261108T135433Z/verification/pipeline.json

ec2-recovery.json

Recorded pass flag: True

brokerDownPublishStatus: 503

restoreErrors: []

initial: 3 unique events verified.

consumer restart, queued backlog and duplicate replay: 23 unique events verified.

broker restart and cross-ingest replay: 25 unique events verified.

Full result: results/a2-fresh/20261108T135433Z/verification/ec2-recovery.json

websocket-failover.json

Recorded pass flag: True

originalServer: "ingest-4"

replacementServer: "ingest-1"

reconnectSecondsFromStopStart: 7.199

restoredIngestHealthy: true

reconnectionMode: "Python test harness retries a new WebSocket through ALB"

Full result: results/a2-fresh/20261108T135433Z/verification/websocket-failover.json

Real, swipe-driven standings

The deterministic functional test sent 100 unique swipes and 20 duplicate requests using the official lift table. Both tested resorts finished at 50 processed unique events. The verifier compared the received WebSocket standings against independently computed expected totals.

Evidence	Resort / day	Processed	Skier	Vertical ft
Functional final snapshot	1 / 340	50	3	712
Functional final snapshot	2 / 340	50	18	605
Before stopping WebSocket owner	9 / 338	1	123	267
After reconnect and another lift-21 swipe	9 / 338	2	123	534

During the injected ingest failure, the Python harness observed disconnection and reconnected from ingest-4 to ingest-1 in 7.199 s measured from stop initiation. This demonstrates server/backplane recovery with harness-managed reconnection; it does not prove that the Java load client reconnects automatically.

Broker confirmation means broker acceptance, not WebSocket client receipt. The separate final-state and functional checks provide the client-observed evidence. Functional tests are correctness checks, not performance measurements.

Fresh local regression

{"tests": 61, "failures": 0, "errors": 0, "skipped": 0}; buildSuccess=True

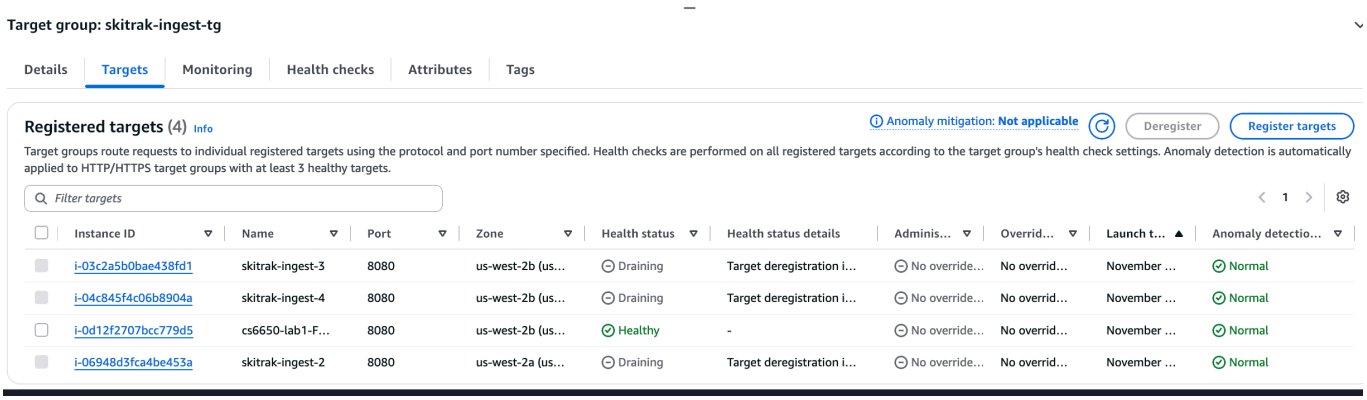
Initial restricted attempt could not bind localhost sockets. It is preserved separately; the complete rerun with local networking allowed passed without test-code changes.

Submission review items

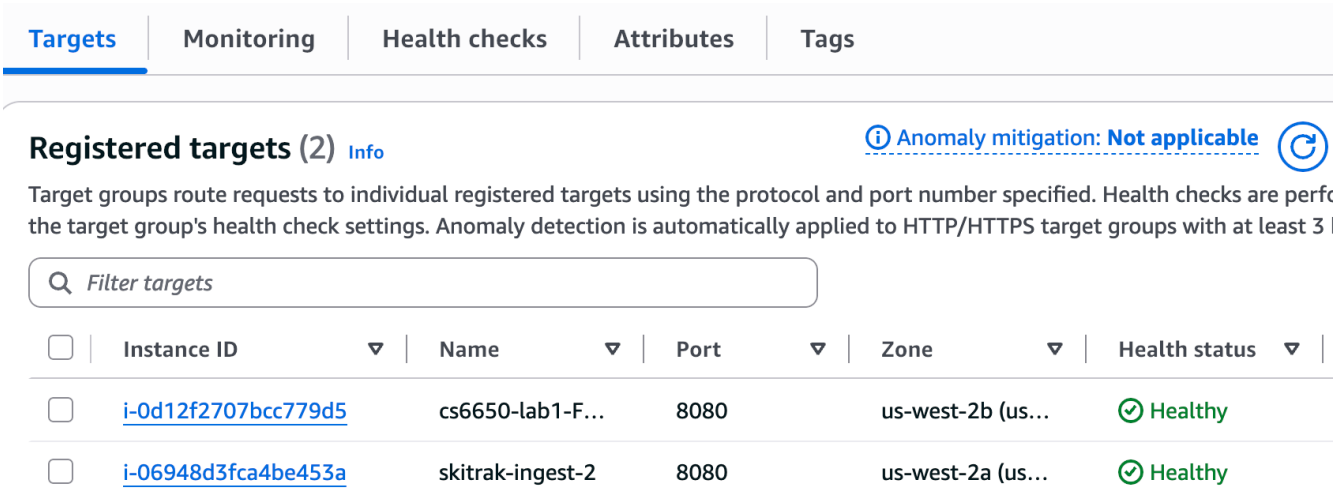
The new CloudWatch captures for the two- and four-ingest scaling runs are included. The single-ingest target image shows a draining transition, not a final single-target screenshot; fresh routing probes provide separate topology evidence. Consumer-sweep runs retain automated RabbitMQ and system evidence but do not have separate supplied CloudWatch captures. The student must review the AI-use reflection and verify that the A2 modules are published in the linked repository before submission.

ALB target-registration evidence

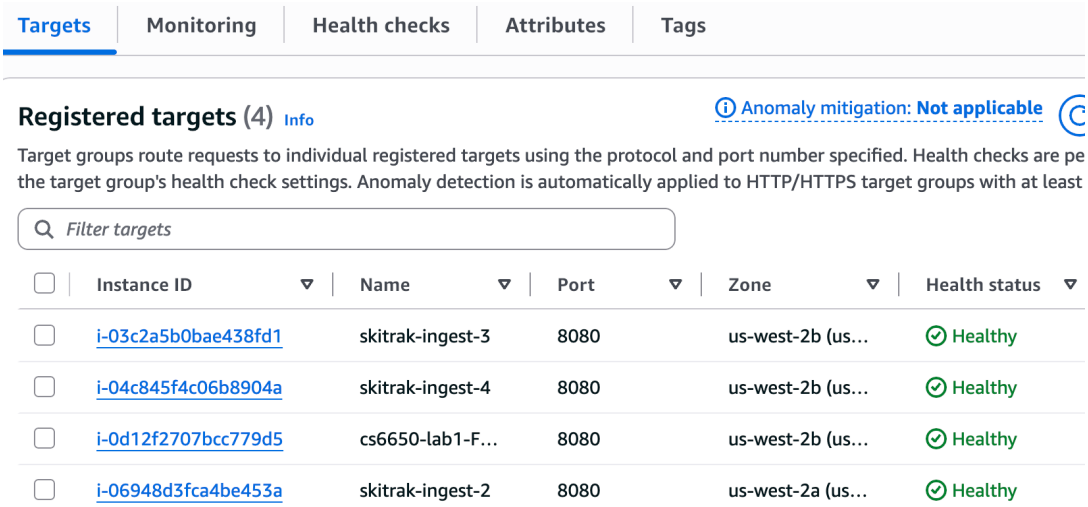
These are user-supplied captures of registered targets. Fresh-session health probes separately verify which server IDs actually responded. The single-ingest image records a deregistration transition, not a final one-target registration screenshot.



Transition toward single-ingest stage: three targets Draining, ingest1 Healthy.



Two registered targets, ingest1 and ingest2, both Healthy.



Four known registered ingest targets, all Healthy.

AI collaboration reflection

Factual collaboration record - student review required

Codex assisted with the channel pool and recovery ownership, official-table lookup, durable per-resort aggregation and deduplication, transactional outbox broadcasting, subscriber-aware topic bindings, deployment helpers, monitoring, verification scripts and this report. The student supplied the official lift table and instructor clarifications, configured AWS resources, collected console screenshots and requested independent repeats when results appeared questionable.

Suggestions accepted

1. Reuse a stable event UUID on retries, because an uncertain response must not create a second logical swipe. 2. Preserve ordered resort lanes and SQLite FULL durability while batching transactions, so lower commit overhead does not weaken atomic deduplication, totals and pending notifications.

Suggestions or conclusions corrected

1. Treating EBS IOPS as the sole explanation was too narrow; the evidence also permits database, synchronization, confirmation and broker limits. 2. One poor five-worker observation was insufficient to generalize; the student requested a repeat and a fresh formal matrix. 3. An early verifier claimed failover success without injecting failure; the revised evidence separates untested failover from a real stop/reconnect test.

A concrete measurement correction

ALB screenshots initially used a single AvailabilityZone metric instead of the whole LoadBalancer, and one replacement window included two runs. The corrected captures preserve LoadBalancer scope, UTC boundaries, Sum/Average and one-minute periods. Queue statistics use direct broker samples with missing samples retained as gaps; API timeouts are not converted to zeros.

The student should review this factual draft in their own words and confirm which implementation decisions they personally understand. No personal reasoning or transcript has been fabricated. The repository AI_USAGE.md must be reviewed separately before submission.

Original client output | 1 ingest(s)

500,000 events,512 client threads. Verbatim benchmark summary; startup and Top-N debug payloads remain in the complete original log. All other runs also retain their complete logs in the evidence directory.

instrumented-500000-c512-20261108T140951Z

```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 35.601 seconds

--- Main ---
Successful: 468000
Failed: 0
Runtime: 99.797 seconds
Main throughput: 4689.507472 swipes/sec

--- Total ingest ---
Successful: 500000
Failed: 0
Remaining in queue: 0
Runtime: 135.398 seconds
Throughput: 3692.81 swipes/sec

--- Throughput per resort ---
Resort 1: successful=49692, throughput=367.01 swipes/sec
Resort 2: successful=49912, throughput=368.63 swipes/sec
Resort 3: successful=49849, throughput=368.17 swipes/sec
Resort 4: successful=50106, throughput=370.06 swipes/sec
Resort 5: successful=49848, throughput=368.16 swipes/sec
Resort 6: successful=50012, throughput=369.37 swipes/sec
Resort 7: successful=50201, throughput=370.77 swipes/sec
Resort 8: successful=49661, throughput=366.78 swipes/sec
Resort 9: successful=50400, throughput=372.24 swipes/sec
Resort 10: successful=50319, throughput=371.64 swipes/sec

--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 500131
Application request retries: 131
TCP connect attempts: 512
TCP connections established: 512
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=512, pending=0, max=512
Sampled pool peaks (50 ms + attempt completion): leased=512, available=512, pending=0
Pipeline drain and final WebSocket verification: 19.110 seconds
End-to-end throughput: 3235.22 swipes/sec
HTTP attempts recorded: 500131
CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-1ingest-workers10/
instrumented-500000-c512-20261108T140951Z/results/swipes-1794147012485.csv
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-1ingest-
workers10/instrumented-500000-c512-20261108T140951Z/results/pushes-1794147012485.csv

--- WebSocket subscriptions ---
Resort 1: updates=92, connections=1, reconnections=0, avgConnectMs=100.368
Resort 2: updates=86, connections=1, reconnections=0, avgConnectMs=121.694
Resort 3: updates=91, connections=1, reconnections=0, avgConnectMs=121.251
Resort 4: updates=93, connections=1, reconnections=0, avgConnectMs=97.045
Resort 5: updates=96, connections=1, reconnections=0, avgConnectMs=94.182
Resort 6: updates=91, connections=1, reconnections=0, avgConnectMs=94.454
Resort 7: updates=89, connections=1, reconnections=0, avgConnectMs=90.429
Resort 8: updates=92, connections=1, reconnections=0, avgConnectMs=90.272
Resort 9: updates=91, connections=1, reconnections=0, avgConnectMs=92.716
Resort 10: updates=92, connections=1, reconnections=0, avgConnectMs=89.256
Total leaderboard updates: 913

--- HTTP attempt latency ---
Samples: 500131
Mean: 104.232 ms
Median: 71.693 ms
p95: 285.714 ms
p99: 469.871 ms
Min: 14.737 ms
Max: 3143.285 ms

--- WebSocket snapshot-to-receipt time difference ---
Initial snapshots excluded from summary (retained in CSV): 10
Negative update samples (clock offset): 700
Cross-host clock offset is included; these are not calibrated one-way network latencies.
Samples: 903
Mean: 61.292 ms
Median: -33.337 ms
p95: 1055.075 ms
p99: 1347.174 ms

Total program runtime: 157.901 seconds
```

Original client output | 2 ingest(s)

500,000 events,512 client threads. Verbatim benchmark summary; startup and Top-N debug payloads remain in the complete original log. All other runs also retain their complete logs in the evidence directory.

instrumented-500000-c512-20261108T143329Z

```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 40.877 seconds

--- Main ---
Successful: 468000
Failed: 0
Runtime: 76.278 seconds
Main throughput: 6135.415694 swipes/sec

--- Total ingest ---
Successful: 500000
Failed: 0
Remaining in queue: 0
Runtime: 117.156 seconds
Throughput: 4267.82 swipes/sec

--- Throughput per resort ---
Resort 1: successful=49692, throughput=424.15 swipes/sec
Resort 2: successful=49912, throughput=426.03 swipes/sec
Resort 3: successful=49849, throughput=425.49 swipes/sec
Resort 4: successful=50106, throughput=427.69 swipes/sec
Resort 5: successful=49848, throughput=425.48 swipes/sec
Resort 6: successful=50012, throughput=426.88 swipes/sec
Resort 7: successful=50201, throughput=428.50 swipes/sec
Resort 8: successful=49661, throughput=423.89 swipes/sec
Resort 9: successful=50400, throughput=430.20 swipes/sec
Resort 10: successful=50319, throughput=429.50 swipes/sec

--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 500055
Application request retries: 55
TCP connect attempts: 512
TCP connections established: 512
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=512, pending=0, max=512
Sampled pool peaks (50 ms + attempt completion): leased=512, available=512, pending=0
Pipeline drain and final WebSocket verification: 40.631 seconds
End-to-end throughput: 3168.37 swipes/sec
HTTP attempts recorded: 500055
CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-2ingest-workers10/
instrumented-500000-c512-20261108T143329Z/results/swipes-1794148427520.csv
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-2ingest-
workers10/instrumented-500000-c512-20261108T143329Z/results/pushes-1794148427520.csv

--- WebSocket subscriptions ---
Resort 1: updates=90, connections=1, reconnections=0, avgConnectMs=504.802
Resort 2: updates=86, connections=1, reconnections=0, avgConnectMs=454.539
Resort 3: updates=91, connections=1, reconnections=0, avgConnectMs=454.455
Resort 4: updates=90, connections=1, reconnections=0, avgConnectMs=496.045
Resort 5: updates=92, connections=1, reconnections=0, avgConnectMs=495.730
Resort 6: updates=90, connections=1, reconnections=0, avgConnectMs=495.173
Resort 7: updates=91, connections=1, reconnections=0, avgConnectMs=452.286
Resort 8: updates=94, connections=1, reconnections=0, avgConnectMs=450.970
Resort 9: updates=90, connections=1, reconnections=0, avgConnectMs=451.838
Resort 10: updates=89, connections=1, reconnections=0, avgConnectMs=442.577
Total leaderboard updates: 903

--- HTTP attempt latency ---
Samples: 500055
Mean: 80.520 ms
Median: 64.699 ms
p95: 172.687 ms
p99: 314.394 ms
Min: 14.813 ms
Max: 2790.499 ms

--- WebSocket snapshot-to-receipt time difference ---
Initial snapshots excluded from summary (retained in CSV): 10
Negative update samples (clock offset): 129
Cross-host clock offset is included; these are not calibrated one-way network latencies.
Samples: 893
Mean: 118.004 ms
Median: 11.298 ms
p95: 1222.129 ms
p99: 1373.851 ms

Total program runtime: 161.400 seconds
```

Original client output | 4 ingest(s)

500,000 events,512 client threads. Verbatim benchmark summary; startup and Top-N debug payloads remain in the complete original log. All other runs also retain their complete logs in the evidence directory.

instrumented-500000-c512-20261108T150014Z

```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 44.891 seconds

--- Main ---
Successful: 468000
Failed: 0
Runtime: 79.329 seconds
Main throughput: 5899.500084 swipes/sec

--- Total ingest ---
Successful: 500000
Failed: 0
Remaining in queue: 0
Runtime: 124.219 seconds
Throughput: 4025.14 swipes/sec

--- Throughput per resort ---
Resort 1: successful=49692, throughput=400.03 swipes/sec
Resort 2: successful=49912, throughput=401.81 swipes/sec
Resort 3: successful=49849, throughput=401.30 swipes/sec
Resort 4: successful=50106, throughput=403.37 swipes/sec
Resort 5: successful=49848, throughput=401.29 swipes/sec
Resort 6: successful=50012, throughput=402.61 swipes/sec
Resort 7: successful=50201, throughput=404.13 swipes/sec
Resort 8: successful=49661, throughput=399.78 swipes/sec
Resort 9: successful=50400, throughput=405.73 swipes/sec
Resort 10: successful=50319, throughput=405.08 swipes/sec

--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 500000
Application request retries: 0
TCP connect attempts: 512
TCP connections established: 512
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=512, pending=0, max=512
Sampled pool peaks (50 ms + attempt completion): leased=512, available=512, pending=0
Pipeline drain and final WebSocket verification: 38.567 seconds
End-to-end throughput: 3068.38 swipes/sec
HTTP attempts recorded: 500000
CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-4ingest-workers10/
instrumented-500000-c512-20261108T150014Z/results/swipes-1794150034354.csv
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-4ingest-
workers10/instrumented-500000-c512-20261108T150014Z/results/pushes-1794150034354.csv

--- WebSocket subscriptions ---
Resort 1: updates=91, connections=1, reconnections=0, avgConnectMs=102.433
Resort 2: updates=87, connections=1, reconnections=0, avgConnectMs=93.162
Resort 3: updates=90, connections=1, reconnections=0, avgConnectMs=95.165
Resort 4: updates=91, connections=1, reconnections=0, avgConnectMs=124.091
Resort 5: updates=92, connections=1, reconnections=0, avgConnectMs=90.134
Resort 6: updates=89, connections=1, reconnections=0, avgConnectMs=89.619
Resort 7: updates=94, connections=1, reconnections=0, avgConnectMs=89.342
Resort 8: updates=93, connections=1, reconnections=0, avgConnectMs=88.284
Resort 9: updates=91, connections=1, reconnections=0, avgConnectMs=88.281
Resort 10: updates=91, connections=1, reconnections=0, avgConnectMs=92.835
Total leaderboard updates: 909

--- HTTP attempt latency ---
Samples: 500000
Mean: 83.854 ms
Median: 67.170 ms
p95: 185.995 ms
p99: 331.415 ms
Min: 14.448 ms
Max: 2541.640 ms

--- WebSocket snapshot-to-receipt time difference ---
Initial snapshots excluded from summary (retained in CSV): 10
Negative update samples (clock offset): 707
Cross-host clock offset is included; these are not calibrated one-way network latencies.
Samples: 899
Mean: 66.934 ms
Median: -46.517 ms
p95: 1098.589 ms
p99: 1361.457 ms

Total program runtime: 167.748 seconds
```

Original client output | 4 ingest(s)

1,000,000 events,512 client threads. Verbatim benchmark summary; startup and Top-N debug payloads remain in the complete original log. All other runs also retain their complete logs in the evidence directory.

instrumented-1000000-c512-20261108T150442Z

```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 42.789 seconds

--- Main ---
Successful: 968000
Failed: 0
Runtime: 147.761 seconds
Main throughput: 6551.135492 swipes/sec

--- Total ingest ---
Successful: 1000000
Failed: 0
Remaining in queue: 0
Runtime: 190.550 seconds
Throughput: 5247.98 swipes/sec

--- Throughput per resort ---
Resort 1: successful=99486, throughput=522.10 swipes/sec
Resort 2: successful=99902, throughput=524.28 swipes/sec
Resort 3: successful=99816, throughput=523.83 swipes/sec
Resort 4: successful=100239, throughput=526.05 swipes/sec
Resort 5: successful=99785, throughput=523.67 swipes/sec
Resort 6: successful=100156, throughput=525.62 swipes/sec
Resort 7: successful=100317, throughput=526.46 swipes/sec
Resort 8: successful=99696, throughput=523.20 swipes/sec
Resort 9: successful=100366, throughput=526.72 swipes/sec
Resort 10: successful=100237, throughput=526.04 swipes/sec

--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 1000077
Application request retries: 77
TCP connect attempts: 512
TCP connections established: 512
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=512, pending=0, max=512
Sampled pool peaks (50 ms + attempt completion): leased=512, available=512, pending=0
Pipeline drain and final WebSocket verification: 110.256 seconds
End-to-end throughput: 3323.33 swipes/sec
HTTP attempts recorded: 1000077
CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-4ingest-workers10/
instrumented-1000000-c512-20261108T150442Z/results/swipes-1794150297853.csv
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/results/a2-fresh/20261108T135433Z/phase-4ingest-
workers10/instrumented-1000000-c512-20261108T150442Z/results/pushes-1794150297853.csv

--- WebSocket subscriptions ---
Resort 1: updates=106, connections=1, reconnections=0, avgConnectMs=124.687
Resort 2: updates=101, connections=1, reconnections=0, avgConnectMs=116.144
Resort 3: updates=101, connections=1, reconnections=0, avgConnectMs=137.585
Resort 4: updates=101, connections=1, reconnections=0, avgConnectMs=118.802
Resort 5: updates=94, connections=1, reconnections=0, avgConnectMs=114.711
Resort 6: updates=103, connections=1, reconnections=0, avgConnectMs=135.763
Resort 7: updates=106, connections=1, reconnections=0, avgConnectMs=136.970
Resort 8: updates=103, connections=1, reconnections=0, avgConnectMs=117.795
Resort 9: updates=105, connections=1, reconnections=0, avgConnectMs=116.991
Resort 10: updates=103, connections=1, reconnections=0, avgConnectMs=112.983
Total leaderboard updates: 1023

--- HTTP attempt latency ---
Samples: 1000077
Mean: 76.882 ms
Median: 64.924 ms
p95: 142.220 ms
p99: 312.635 ms
Min: 14.658 ms
Max: 3092.066 ms

--- WebSocket snapshot-to-receipt time difference ---
Initial snapshots excluded from summary (retained in CSV): 10
Negative update samples (clock offset): 774
Cross-host clock offset is included; these are not calibrated one-way network latencies.
Samples: 1013
Mean: 65.536 ms
Median: -13.433 ms
p95: 1084.033 ms
p99: 1273.302 ms

Total program runtime: 304.875 seconds
```